Time-Causal Scale Space from Hemigroup Axioms: Characterization of the Kernels

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September 27, 2026 (v1.1.0), doi:10.5281/zenodo.22992450
First released as v1.0.0, September 2, 2026 (doi:10.5281/zenodo.22259187). All versions: doi:10.5281/zenodo.22259186.

Abstract

We characterize the time-causal scale spaces: the families of smoothing operators through which an uncommitted observer can measure a temporal signal at all scales at once. One classical axiom is weakened: the cascade property is stated as a two-parameter hemigroup, $\Phi_{y,z}\Phi_{x,y} = \Phi_{x,z}$, retaining composition of measurements while dropping the tacit assumption that measurement stages are interchangeable. The admissible kernel families are then exactly the Laplace exponents of the self-decomposable laws, the marginals of self-similar additive (Sato) subordinators; reimposing interchangeability recovers the extremal stable densities together with a pure delay, exposing their heavy tails as the price of that assumption rather than of causality or covariance. The enlarged class contains members with all moments finite — foremost the Gamma family, implemented exactly, at integer shape, by cascades of first-order filter sections. Over the memory half-line, under a mild standing hypothesis, the scale evolution inverts into an evolution in present time, well posed in an explicit class: the memory line stores the Riemann–Liouville integrals of the past signal, the temporal N-jet reads that state instantaneously, and locality with the positive maximum principle singles out a degenerate transport member and a one-parameter Bessel family of media. Non-creation of temporal structure is settled in both directions: no nondegenerate member admits the scale-monotone form, while the endpoint form — no scale showing more sign changes than the signal — is attained exactly by the classified variation-diminishing members. The characterization and the signaling form are machine-checked in Lean 4, resting on one cited analytic fact.

1 Introduction

This paper characterizes the time-causal scale spaces: the families of smoothing operators through which an uncommitted observer can measure a temporal signal at all scales at once. A visual or auditory front end measures its input before anything is known about what the signal will contain, so it must hold records at every scale simultaneously; and in time it must do so causally — the future is unavailable — and out of its own state — the past cannot be revisited. The spatial form of this programme is Gaussian scale space; this paper settles the temporal form. Four questions organize everything that follows: which classical assumption has to give; what replaces the stable kernels once it does; what the enlarged class contains, and at what price; and which classical properties survive the enlargement, in what form.

The assumption. The axioms are those of causal, covariant measurement, with one change against the classical axiomatics. The semigroup form of the cascade property, $\Phi_\tau\Phi_{\tau'} = \Phi_{\tau+\tau'}$, formalizes "a measurement of a measurement is a measurement" together with a second, tacit assumption: that the composite depends only on the total *amount* of scale crossed — that measurement stages are interchangeable (§2.4). This paper keeps the first assumption and drops

the second, stating the cascade in its weakest defensible form, as a two-parameter *hemigroup*,

$$\Phi_{y,z}\,\Phi_{x,y} \;=\; \Phi_{x,z}, \qquad 0 \le x \le y \le z$$

(Figure 2). Every other axiom — causality, linearity, positivity, unit area, continuity, translation covariance, and *full continuous* scale covariance — is retained, restated only as the two-parameter setting requires: scale covariance with the relabelling of the scale axis left free (§2.3), and the extended point as the diagonal condition $\Phi_{x,x} = \mathrm{Id}$. One further difference is disclosed where it matters (Remarks 2.2 and 7.5): the kernels are no longer required to be continuous functions, which admits a degenerate pure-delay member. The derivation of [8] — hereafter *the semigroup theory* — is then rerun in the same order with the same tools: Laplace transforms, two functional equations, and the Bernstein-function calculus. A numbering convention, used throughout and self-contained: a theorem numbered with a prime — 3′ (the scale-Cauchy problem), 4′ (the signaling form), 5′ (locality) — is the hemigroup analogue of the correspondingly numbered theorem of the semigroup theory, and the text restates each recovered classical statement at the point where it recovers it, so the correspondence can be read without the 2005 paper at hand.

What replaces the stable family. Under these axioms the admissible kernel families are completely characterized (Theorem 7.3): in the canonical scale parametrization fixed in §6, they are exactly

$$\phi_x(t) \;=\; \mathcal{L}^{-1}\big[\, e^{-F(xs)} \,\big](t), \qquad F(s) = b_0\, s + \int_0^\infty \big(1 - e^{-st}\big)\, \frac{k(t)}{t}\, dt, \quad k \text{ nonincreasing},$$

the Laplace exponents of the *self-decomposable* laws — the delays T that shrink into themselves, $T \overset{d}{=} cT + R_c$ with R_c independent, for every $c \in (0,1)$ — equivalently, the kernels are the marginal laws of self-similar additive (Sato) subordinators. When stage-interchangeability is reimposed, the recovered sub-case is the extremal stable family together with the pure delay at its boundary (Corollary 7.4), and the heavy tails of the stable kernels are thereby exposed as the price of that tacit assumption, not of causality and covariance (Remark 8.18).

What the class contains, and at what price. The class contains members with all moments finite. The simplest of these, and the one the implementation half of the paper builds on, is the Gamma family

$$\phi_x(t) \;=\; \frac{t^{\gamma-1}\, e^{-t/x}}{\Gamma(\gamma)\, x^{\gamma}},$$

whose mean delay is finite, γx; whose tails are exponential; whose scale covariance is exact and continuous; and which is computed, for integer γ, by a cascade of first-order recursive filters. The price is stated before the consequences are: what the Gamma family gives up is locality of its memory-line medium. Its signaling operator is an integral operator, not a differential one. The trade-off between finite moments and a local medium is not an accident of the example; it is classified by the locality theorem (§10).

The classical constructions are recovered inside this class, and their limitations are located by it. The Gaussian scale space that answers the corresponding spatial question [13, 22] is unavailable in time: the Gaussian aperture is non-causal, and its causal truncations destroy the cascade structure. The known time-causal constructions each concede one desideratum. The scale-time of [19] applies Gaussian smoothing to a logarithmically remapped past, achieving causality but not time-recursivity. The Poisson scale space of [24] is causal and time-recursive but carries no dilation covariance of any span: its fixed elementary delay is a preferred time unit. The semigroup theory imposes full continuous covariance together with the semigroup cascade and obtains a complete answer, the kernels being exactly the extremal stable densities $\mathcal{L}^{-1}[e^{-\tau s^\alpha}]$, $0 < \alpha < 1$; but every member has infinite mean delay. The time-causal limit kernel of [23] restores finite moments through a recursive cascade of first-order integrators, with covariance established there under the discrete subgroup of rescalings $\sigma \in q^{\mathbb{Z}}$.

The memory line. The second half of the paper develops the theory over the memory line. The embodiment principle of the semigroup theory (§5 of that paper; §2 motivates it) — the observer must carry its usable past in its own state — appears as theorem content. Under a mild standing hypothesis on the exponent, the signaling form, Theorem 4′ (Theorem 9.17), inverts the evolution across scale into an evolution in present time over the memory half-line,

$$\partial_t u \;=\; A\,u, \qquad u(t,0) = f(t),$$

with the signal entering as boundary data at zero scale. That hypothesis is (H), Definition 9.1 in §9; every family worked in §8 satisfies it except the Gamma family at small shape. The operator A is constructed in §9 by harmonic analysis in the scale variable, and the problem is well posed. A is the unique inversion in its covariance class (Theorem 9.17(3)). The solution exists and is unique in an explicit class, with the classical period-one ambiguity closed by a Bohr–Mollerup-style rigidity (Corollary 9.21). And it depends continuously on the data, nonexpansively and uniformly over scales, and, under pointwise perturbation, on the exponent itself (Corollary 9.24). The construction comes with an exact account of what the memory holds. At each instant, the memory line stores the analytic family of Riemann–Liouville integrals of the past signal: the observer's embodiment of its past is fractional integration.

Two things follow. The temporal N-jet (§12), the causal analogue of the derivative-of-Gaussian receptive fields, is an instantaneous readout of that same state,

$$\partial_t^n u \;=\; A^n u :$$

differentiation is a re-indexing of the embodied record, for the Gamma family a binomial subtraction of coexisting filter states. No second memory is needed to differentiate, up to an order bound set by the same hypothesis (H). And the locality theorem, Theorem 5′ (Theorem 10.6), classifies the memory-line media: locality of the operator alone admits a ladder of every differential order, and the positive maximum principle closes the ladder at order two, where it leaves a degenerate transport member and the one-parameter Bessel family

$$A \;=\; \tfrac{1}{2}\,\partial_x^2 + \frac{\beta}{x}\,\partial_x,$$

unfolding the isolated heat case of the semigroup theory's Theorem 5, with the drift weight β buying any prescribed finite number of kernel moments.

The fourth question — which classical properties survive, and in what form — has its sharpest instance in the oldest axiom of the subject, and there it is settled in both directions (§8). Scale-monotone non-creation — every coarser record showing no more sign changes than every finer one — is *impossible*: an admissible family whose increments are all variation-diminishing is the pure delay (Proposition 8.15). What can exist is the endpoint form — no scale shows more sign changes than the signal itself — and it is classified and attained: the members with variation-diminishing kernels are exactly drift plus integer Thorin atoms of finite mean, the Gamma family foremost. The classical obstruction, that no causal convolution *semigroup* can be variation-diminishing at all, is thereby dissolved as far as it can be. Discrete covariance retains more: along a geometric ladder the limit kernel's rung increments are variation-diminishing, and there the counts are monotone from scale to scale (Remark 8.16). And along a feature's own delay path the field is non-enhancing at past-dominating extrema, with the echo of a preceding event as the sharp failure mode (Corollary A.21).

On the implementation side (§11) the hemigroup itself is the algorithm: the discrete cascade over any set of scale knots reproduces the continuum family exactly at the knots. For the Gamma family each increment is a pole–zero filter section whose zero moves with the finer scale, and the zeros are what the hemigroup adds. Finite Gamma cascades approximate the entire completely-monotone-kernel subclass from inside the axiom class, with rational cascade

realizations at integer weights. The whole architecture — cascade, jets, and the qualitative predictions — is exercised end to end in a numerical experiment (Example 12.7). Appendix A carries Theorem $3'$, the scale-Cauchy problem (Theorem A.19): a fractional relaxation equation across scale whose memory kernel is the function k of the representation itself. It also carries the Sonine conservation identity, by which the memory operator applied at each scale has an explicit right inverse, the potential kernel; a Volterra equation that derives the kernels from k by forward integration, with no Laplace inversion; and the pointwise non-enhancement corollary above.

One paragraph on the relation to the existing time-causal scale-space theory, since a complete time-causal, time-recursive construction already exists in the literature [23]. The relation is containment, not competition (Remark 13.1): the axioms without any covariance give the Bernstein hemigroups, which contain the Poisson scale space; covariance under the discrete subgroup $q^{\mathbb{Z}}$ gives the semi-selfdecomposable stratum; and full covariance gives the present class. The time-causal limit kernel itself has a completely monotone memory kernel, so it is a fully covariant member of the class characterized here, and of its variation-diminishing subclass — in retrospect, the first-discovered member: a construction that preceded the classification explaining it. Even the scale-time aperture is contained at the kernel level: pulled back to the time axis it is the lognormal density, a generalized gamma convolution by Thorin's theorem, hence an admissible Thorin-subclass kernel — only its log-domain composition structure lies outside the axioms (Remark 13.1).

The contributions, each with the section that carries it:

- the characterization of the time-causal scale spaces under the hemigroup axioms — exactly the self-decomposable Laplace exponents, with the gauge and exponent unique (Theorem 7.3, §7);
- continuous scale covariance for the whole class, with the discrete cascade exact at every set of scale knots (§11);
- the signaling form: the scale evolution inverted into an evolution in present time, with the unique covariant inversion and the exact account of what the memory holds (Theorem 9.17, §9);
- the classification of the local, positivity-respecting memory-line media — the Bessel family, closing the ladder at order two (Theorem 10.6, §10);
- non-creation settled in both directions: the scale-monotone form impossible for every non-degenerate member, the endpoint form classified and attained by the variation-diminishing members, and non-enhancement along the feature's delay path (§8, Appendix A);
- the temporal N-jet as an instantaneous readout of the embodied state (§12);
- machine-checked proofs of the central results, resting on one cited analytic fact (§1.1).

The paper is purely temporal; composition with a spatial scale space is orthogonal to everything done here. Its contribution is mathematical foundations together with an exact implementation route: no empirical claims about natural signals are made, and the one numerical experiment (§12) checks numbered results rather than applications. The signal model is L^1, where the transform apparatus lives; the pointwise extension of the smoothing to bounded measurable signals — the experiment's persistent step among them — is recorded in Remark 3.1. The organization follows the derivation. §2 states and motivates the axioms; §3 fixes notation and the Bernstein-function toolbox; §§4–6 derive the convolution representation, the infinite divisibility of the exponents, and the similarity form forced by covariance; §7 proves the characterization theorem; §8 gives examples and moment formulas; §9 the Mellin signaling form; §10

the locality theorem; §11 implementation; §12 the temporal jet; §13 concludes, with four open problems. Appendix A develops the memory-kernel and Sonine structure and the scale-Cauchy problem.

Two fast tracks, for two readers. The practitioner can take §2 for the axioms and the statement of Theorem 7.3, then go directly to §§11–12 — the exact cascade, the Thorin banks, the jets, and the numerical experiment — accepting the intermediate theory on faith. The reader here for the mathematics can take §§4–7 for the characterization, §9 for the Mellin solution theory, and §10 with Appendix A for the media and the scale-Cauchy problem, skipping the implementation sections entirely. §8 serves both.

1.1 What is machine-checked

The central results of this paper have been formalized in the Lean 4 proof assistant, on top of its mathematical library Mathlib, and this subsection says exactly which, resting on exactly what. It is self-contained: nothing below depends on reading the formal development.

What is proved. Both main theorems are machine-checked in full.

- Theorem 7.3, the characterization: a family $(\Phi_{x,y})$ satisfies (A1)–(A8) and (ND) if and only if it is convolution by the kernels of an exponent of the form (7.1) — the representation displayed at the start of this introduction — read through a gauge, an increasing reparametrization of the scale axis (§6); together with the uniqueness of the pair (gauge, exponent). All three parts are verified: the construction, the analysis, and the uniqueness.
- Theorem 9.17, the signaling form: the profiles $H(s\,\cdot)$ are eigenfunctions of the inversion operator A with eigenvalue s; the field solves the signaling problem, in its Laplace and its Mellin form; and A is the unique inversion in the covariant Mellin class.

Everything §§4–6 contribute to the characterization is verified with it: the convolution representation, the additivity and infinite divisibility of the exponents (Theorem 5.2), and the similarity form forced by covariance. In the appendix, the memory kernel, its transform, the Sonine conservation identity $\kappa^{(x)} * \ell^{(x)} = \mathrm{Leb}$, the existence, uniqueness and scaling of the potential kernel, the Volterra equation (V.1) with its uniqueness clause, the delay-derivative lemma, and the Phillips form of the generator with all five clauses of Lemma A.18 — the last interface-free — are verified, and so are the two equivalences of Proposition A.9(2). Of §9, every numbered result through the signaling theorem is, the Lévy-data reading (Lemma 9.3) and the log-growth Lemma 9.4 included. Of the well-posedness arc that follows it — Definition 9.19 through Corollary 9.24 — the rigidity Lemma 9.20 is verified, and the rest is paper-only. Of §10, the classification step (Lemma 10.2, both directions), the log-convexity of the moment function (Lemma 10.4), the vanishing $B(0) = 0$ (Lemma 10.3(1)) and the uniqueness of the log-convex solution of the moment recursion in the order-2 case (Lemma 10.5) are.

What it rests on. One analytic fact is taken from the literature rather than reproved. It is the *only* thing the verified results assume beyond the logical foundations of the proof assistant itself, and it is cited with a page reference checked against the source.

- *Self-decomposability, in the analysis direction.* If every dilation increment $F(b\,\cdot) - F(a\,\cdot)$ of an exponent is again an exponent, then its Lévy measure has a nonincreasing density $k(t)/t$ ([33], Prop. 5.17, p. 57). This is the implication (1) $\Rightarrow$ (3) of Lemma 7.1, and it is the only thing the analysis direction needs that is not proved here.

The constructive direction and the uniqueness of Theorem 7.3 use nothing beyond the proof assistant's foundations, and the analysis direction uses the fact above. That separation is

checked mechanically rather than claimed. The first release of this paper cited a second fact, the existence of a subordinator from its triple ([33], Thm. 5.2, p. 49), which turns an admissible exponent into actual kernels. It is now proved, by the compound-Poisson construction: the laws of the truncated Lévy measures are tight, and a weak limit point has the required transform.

Why this one and not more. The classical route to these results passes through the theory of completely monotone and Bernstein functions — the Bernstein–Widder theorem, closure of BF under pointwise limits, uniqueness of the Lévy–Khintchine triple, composition rules. Formalizing led us to replace five arguments by shorter ones that use the *representation* (7.1) directly, and those five are now the proofs printed in this paper; §13 names them. The surviving citation is the one place where the derivative-sign definition is irreducible: $B(s) = sF'(s)$ in Lemma 7.1 has no meaning until the representation is in hand, which is why the fact above is expected to be permanent. The same policy runs in the other direction: classical results of structural rather than analytic character are proved, not cited. Thus the convolution representation of §4, classically Wendel's theorem [41], is proved from scratch rather than admitted as a third interface; the citation would buy little in any case, since the sharpening the axioms force — a probability measure carried by the half-line — is most of the argument (Remark 4.4).

What is not formalized, and why. The scale-Cauchy problem (Theorem A.19) is not, though the obstruction is narrower than it was once thought to be: its proof is a scalar ODE per Laplace frequency and invokes no C_0-semigroup results — that theory, absent from Mathlib, is needed only by the fixed-scale reading (Proposition A.20), which is cited. What the theorem still needs is the distributional scale-evolution identity (Proposition A.3), which — like the exact-inversion corollary — requires distributions paired with measures, absent from Mathlib, together with the vector-valued fundamental-theorem step. Proposition A.9 is verified only in part. The two equivalences of its clause (2) are checked, with complete monotonicity read as a Laplace transform of a measure, since the formal development deliberately does not define the CM class by derivative signs. Its clause (1) and the second assertion of (2) rest on the theory of special and complete Bernstein functions, and are cited. §§8 and 11–12 are examples and implementation, formalized only where they were cheap.

§10 is a mixed case and worth separating out, because the obstruction there is narrower than it first appears. Its structural spine is verified: the classification step — Lemma 10.2, local operators are exactly the polynomial symbols, in both directions — together with the log-convexity of the moment function, the vanishing $B(0) = 0$, and the uniqueness of the log-convex solution of the moment recursion in the order-2 case. That last one deserves a word, because it removes a citation the theorem was stated to rest on: Theorem 10.6 applies the recursion $m(z+1) = Q(z)m(z)$ only where Q is linear, and there the Krull–Webster uniqueness theorem is the classical Bohr–Mollerup theorem, which is available; the order-n ladder of Proposition 10.7 has $n-1$ Gamma factors and does still need the general result. What remains cited in §10 are the analytic interfaces — Widder's singularity theorem (behind $z_* = \infty$), Courrège's order bound, Krull–Webster for the general ladder, and Bondesson's theory for the ladder's admissibility — and the Bessel identifications, which need K_ν. So the gap is four citations and one special function — not an absent theory.

Where to look. The development, its axiom ledger with page-verified citations for every cited interface, and a dependency graph in which each statement is colored by its verification status, are available at the repository accompanying this paper, `https://github.com/danielfagerstrom/hemigroup-causal-scale-space-kernels`. This version of the paper and the repository at its release tag `v1.1.0` are archived at `https://doi.org/10.5281/zenodo.22992450`, and every version under `https://doi.org/10.5281/zenodo.22259186`. The toolchain is pinned in-tree — Lean 4 `v4.31.0`, Mathlib `v4.31.0`, and the shared library

`ScaleSpaceCore` at its tag `v0.1.1` — so the verification is reproducible with one command, `lake build` in `Formalization/`. A second, `lake env lean CIAxiomGuard.lean`, prints the axiom usage of every named declaration and exits nonzero on any drift from the one-interface trust boundary above. The ledger records the twenty-one interfaces the paper cites, one of them since proved; the one above is the only one the verified results actually consume.

2 Temporal measurement

This section assembles the axiom system of the paper: a list of requirements on temporal measurement, each motivated in turn and collected at the end in Definition 2.1. The substantive change from [8] is one axiom, the cascade property of §2.4, and it is the subject of this paper; the two smaller differences are recorded in Remark 2.2.

It is impossible to measure the value of a signal at a point: a measurement must collect input from a region of non-vanishing extent — an *extended point* [11, 9]. What an observer holds is therefore a record of the signal at some resolution, never the signal itself, and a multi-scale front end maintains a family of such records, indexed by a scale parameter.

The front end is *uncommitted*: it is built before anything is known about what the signals will contain, so the family of records cannot be derived from a signal model. It must be derived from requirements on the measurement process itself. Those requirements are what this section states.

Two asymmetries distinguish time from space. The observer cannot access the future: measurement must be causal. And the observer cannot return to the past: whatever of the signal's history is to influence the present output must be carried in the observer's own state — the observer must *embody* its past. Causality is a condition on the measured values and becomes an axiom below (§2.2). Embodiment is a condition on how a measurement can be realized, and it is not an axiom: no requirement below mentions the observer's state. That the axioms nevertheless admit an embodied realization is a result of the theory: under a mild standing hypothesis on the family, stated in §9, that section constructs the state a covariant observer carries, and §12 shows that the temporal derivatives of the signal can be read off that state.

A temporal measurement is an operator that smooths the signal. Signals are integrable transients, $X = L^1(\mathbb{R}, dt)$, with positive cone X_+ and $\int f := \int_{\mathbb{R}} f(t)\,dt$; this and the other notation used informally in this section is collected in §3. Because a record is itself a signal, and is measured further in the cascade of §2.4, the measurement operators are indexed by ordered pairs of scales,

$$\Phi_{x,y} : X \to X, \qquad 0 \le x \le y < \infty,$$

with $\Phi_{x,y}$ carrying the record at scale x to the record at the coarser scale y, and the signal itself as the record at scale 0. In the classical theory a single scale index suffices; why two are needed here is explained in §2.4. The apertures — the measures realizing these operators — are not postulated: that the operators are convolutions is recovered from the axioms in §4, and which convolutions in §7.

One point of formulation is settled here, because the rest of the paper depends on it. The extended point says that no record is the signal and no measurement stage is the identity. It is more than a statement about what can be built: in the duality of state space and device space of [11], followed in [8] with the signal as the state, the device is the part of the measurement one designs, and designing it with an extended aperture is what makes the space of records differentiable when the space of signals is not — one reason to take the extended kernels, and not the pure delays, as the objects of the theory. The formal theory is nevertheless set on the *closed* half-line $[0, \infty)$ of scales, with the signal as the record at scale 0 and $\Phi_{x,x} = \mathrm{Id}$ on the diagonal. Neither is a realizable measurement; both enter as the limits the continuity requirement of §2.5 refers to, in the same way that the Dirac measure enters the extended point axiom of [8], $\lim_{\tau\to 0} \phi_\tau = \delta$ with ϕ_τ the kernel at scale τ: as a limit the family approaches and

never attains. Nothing is measured *at* scale 0; every statement below is about the family and its limits. The alternative — taking "a measurement of a measurement is a measurement" as a formal requirement, every stage strictly smoothing — leads to a discrete ladder of scales and would still need the diagonal in its closure to state continuity at all. It is not pursued. The same stance covers one degenerate member the axioms admit, the pure delay, exhibited after Definition 2.1 and kept separate in the text wherever a property fails on it.

2.1 Linearity, positivity, and mass

An uncommitted front end treats all intensity levels alike and prefers no signal feature over another. Superposition is the operational form of that indifference: a device that does not respect it responds to some combinations of inputs differently from the sum of its responses, and has thereby singled those combinations out before anything is known about the signals. So each $\Phi_{x,y}$ is linear. This is the assumption on which the convolution route of §4 rests; the nonlinear scale-space axiomatics [1], which give it up, are a different theory and outside the scope of this paper. Measurement is also stable — a small signal leaves a small record — which on X is boundedness of each $\Phi_{x,y}$ (A1).

An intensity signal is nonnegative, and a measurement of nonnegative input that reports negative values has created structure of its own — structure of the apparatus rather than of the signal. Measurement preserves the positive cone (A4).

In the spatial theory the corresponding normalization is *gray-level invariance*: smoothing an image may not change its average intensity, so the smoothing kernel has unit area and the smoothing operator is conservative — it redistributes intensity and neither creates nor absorbs any. Read on integrable transients the invariant quantity is the total mass $\int f$. This is (A5). A device that loses mass models absorption rather than measurement; the passive variant is recorded in Remark 2.2 and not pursued.

2.2 Causality

The record at time t may depend on the signal only through its restriction to $(-\infty, t]$: if the signal vanishes before some time t_0, so does every record of it (A3). This is the one requirement with no spatial counterpart, and it removes the Gaussian: the Gaussian aperture reads the future, and no truncation of it is compatible with the cascade structure discussed below — the characterization of §7 shows that every admissible kernel is infinitely divisible, which no truncated Gaussian is.

2.3 Covariance

A front end built before its signals has no preferred origin of time and no preferred unit of time.

No preferred origin: each $\Phi_{x,y}$ commutes with the translations

$$(\tau_a f)(t) = f(t-a), \qquad a \in \mathbb{R},$$

which is (A2).

No preferred unit: rescaling the time axis must carry the family into itself. The mass-preserving dilations are

$$(D_\sigma f)(t) = \sigma^{-1} f(t/\sigma), \qquad \sigma > 0.$$

One point requires care before the requirement can be stated. Nothing so far ties the scale index x to a unit of time: the scale axis is, at this stage, only an ordered parametrization of the records — which is finer, which is coarser — and the labels themselves have no metric meaning. A rescaling of the time axis can therefore not be asked to fix each operator; it can only be asked

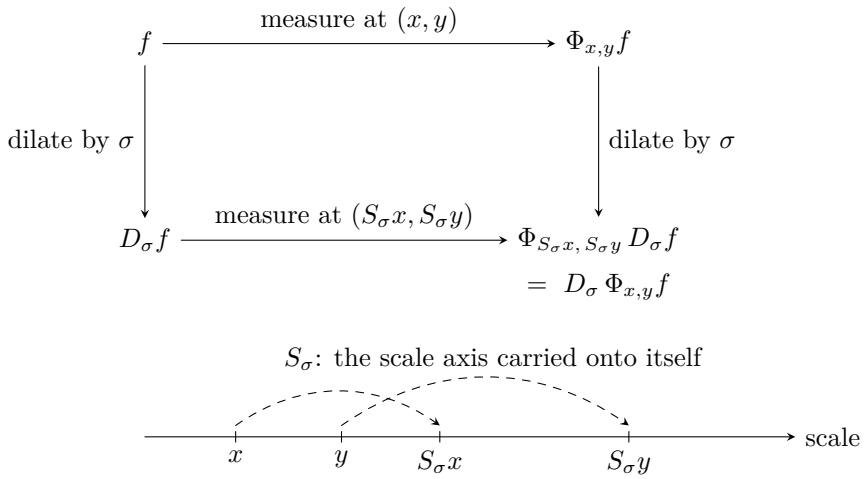


Figure 1: Scale covariance (A8). Dilating the signal by σ and then measuring is the same as measuring and then dilating, provided the measurement is taken at the transported scales $(S_\sigma x, S_\sigma y)$: the square commutes. The lower strip shows the relabelling. Nothing ties the scale labels to a unit of time at this stage, so (A8) can only ask that the family be mapped onto itself, with the labels rearranged by some increasing bijection S_σ; §6 shows that the axioms determine S_σ and thereby a canonical parametrization of the scale axis.

to map the family onto itself, with the scale labels rearranged. Formally: there is a family $(S_\sigma)_{\sigma>0}$ of increasing bijections of $[0, \infty)$ with

$$D_\sigma\, \Phi_{x,y} \;=\; \Phi_{S_\sigma x,\, S_\sigma y}\, D_\sigma \qquad \text{for all } 0 \le x \le y,\ \sigma > 0,$$

which is (A8) (Figure 1). The axioms, with the nondegeneracy of §2.5, determine each S_σ uniquely, and §6 recovers a canonical scale parametrization from the action. We demand *full* covariance, every $\sigma > 0$. The discrete form $\sigma \in q^{\mathbb{Z}}$, a fixed ratio between successive scales, is a genuinely weaker requirement with a theory of its own; §13 places it.

2.4 The cascade property

A record is itself a signal and can be measured further. A record at a coarser scale must then be computable from the record at any finer scale, without returning to the raw signal: a measurement of a measurement is a measurement, and the family closes under composition,

$$\Phi_{y,z}\, \Phi_{x,y} \;=\; \Phi_{x,z}, \qquad x \le y \le z, \qquad\qquad \Phi_{x,x} = \mathrm{Id},$$

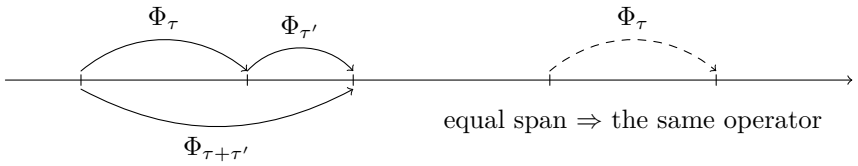
which is (A6).

This is the axiom the paper changes. In [8], as throughout the scale-space literature, the cascade is a one-parameter *semigroup*,

$$\Phi_\tau\, \Phi_{\tau'} \;=\; \Phi_{\tau+\tau'},$$

and this single equation asserts two things at once (Figure 2). The first is the closure just motivated: composites of measurements are measurements. The second is tacit: the composite depends only on the total *amount* of scale crossed. Equivalently, the elementary smoothing applied between fine scales equals the one applied between coarse scales — measurement stages are interchangeable. Nothing in the measurement picture supports the second assertion. A stage that carries a fine record to a slightly coarser one and a stage that does the same deep in the hierarchy act on different records, and there is no a priori reason the two operations should coincide.

(a) semigroup: increments indexed by an *amount* of scale



(b) hemigroup: increments indexed by *intervals* of the scale axis

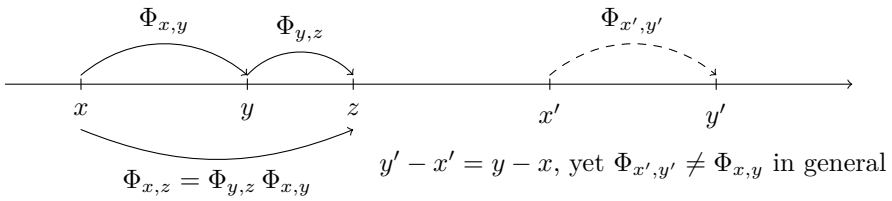


Figure 2: The two readings of the cascade property. (a) In the semigroup form the increment is indexed by an amount of scale: composition adds amounts, and increments of equal amount are the same operator wherever they act — the axis positions carry no labels because the operators cannot see them. (b) In the hemigroup form the increment is indexed by an interval of the scale axis: composition joins adjacent intervals, and increments of equal span at different positions need not coincide. The axioms retain what the two forms share — a measurement of a measurement is a measurement — and drop the rest.

The hemigroup form (A6) keeps the first assertion and drops the second, and nothing else: scale becomes an ordered interval structure rather than an additive quantity. Two consequences follow immediately. The family must be indexed by pairs, since an increment $\Phi_{x,y}$ is no longer determined by the difference $y-x$. And the scale axis acquires a gauge freedom — any increasing reparametrization of $[0,\infty)$ yields the same family relabelled — which is why (A8) was stated with the relabelling S_σ left free. §6 resolves the first: covariance determines the gauge up to normalization. And the semigroup case reappears when stage-interchangeability is reimposed: Corollary 7.4 in §7 returns the stable family of the semigroup theory together with the pure delay.

2.5 Continuity and nondegeneracy

Measurements at nearby scales must be comparable: the record depends continuously on the pair of scales, jointly, in the norm of X (A7). Together with the diagonal condition $\Phi_{x,x}=\mathrm{Id}$, continuity is the extended point in its two-parameter form: as $y\downarrow x$ the record at scale y recovers the record at scale x.

Finally, the scale axis carries no idle stretches: strictly coarser means strictly more smoothed, $\Phi_{x,y}\neq\mathrm{Id}$ for $x<y$ (ND). Nondegeneracy is a normalization rather than a restriction — a family that stalls on an interval of scales is the same family on a smaller scale axis.

2.6 The axiom system

Definition 2.1 (causal cascade measurement family)**.** A family of linear operators

$$\Phi_{x,y}: X\to X,\qquad 0\le x\le y<\infty,$$

is a *causal cascade measurement family* if it satisfies:

- **(A1) Boundedness.** Each $\Phi_{x,y}$ is a bounded operator on X.

- **(A2) Time-translation covariance.** $\Phi_{x,y}\,\tau_a = \tau_a\,\Phi_{x,y}$ for all $a \in \mathbb{R}$.
- **(A3) Causality.** If $f \in X$ vanishes a.e. on $(-\infty, t_0)$, then so does $\Phi_{x,y} f$.
- **(A4) Positivity.** $\Phi_{x,y} X_+ \subseteq X_+$.
- **(A5) Unit area.** $\int \Phi_{x,y} f = \int f$ for all $f \in X_+$.
- **(A6) Cascade (hemigroup).** $\Phi_{x,x} = \mathrm{Id}$, and $\Phi_{y,z}\,\Phi_{x,y} = \Phi_{x,z}$ for all $x \le y \le z$.
- **(A7) Continuity.** For every $f \in X$, the map $(x, y) \mapsto \Phi_{x,y} f$ is continuous from $\{0 \le x \le y\}$ to X.
- **(A8) Scale covariance.** There is a family $(S_\sigma)_{\sigma>0}$ of increasing bijections of $[0,\infty)$ such that
$$D_\sigma\,\Phi_{x,y} = \Phi_{S_\sigma x,\, S_\sigma y}\,D_\sigma \qquad \text{for all } 0 \le x \le y,\ \sigma > 0.$$
- **(ND) Nondegeneracy.** $\Phi_{x,y} \neq \mathrm{Id}$ whenever $x < y$.

Two members, at the two ends of what the axioms admit.
The *pure delay*,
$$\Phi_{x,y} = \tau_{y-x}, \qquad S_\sigma x = \sigma x,$$
satisfies every axiom at sight. It smooths nothing, and it is admitted for the reason the identity is: it is the boundary the class closes on (Remark 7.5).

The *Gamma family* of §1,
$$\Phi_{x,y} f = \rho_{x,y} * f, \qquad \hat{\rho}_{x,y}(s) = \Big(\frac{1+xs}{1+ys}\Big)^{\gamma},$$
with $\hat{\rho}$ the Laplace transform, satisfies (A6) because the transforms multiply, (A8) with $S_\sigma x = \sigma x$ because $\hat{\rho}_{\sigma x,\sigma y}(s) = \hat{\rho}_{x,y}(\sigma s)$, and (A7) because the transform is continuous in (x, y). That $\rho_{x,y}$ is a probability measure on $[0,\infty)$, which gives (A1)–(A5), is the one non-obvious point; it is the self-decomposability of the Gamma law (§8).

Between these two the theory has to fit; §7 says exactly what fits.

Remark 2.2 (provenance of the axioms). (A1)–(A5) and (A7) are as in [8]; the diagonal condition in (A6) together with (A7) is the two-parameter form of the *extended point* axiom $\lim_{\tau\to 0} \phi_\tau = \delta$, in the notation of that paper, ϕ_τ the kernel at scale τ.

The substantive change is (A6). In [8] the cascade reads $\Phi_\tau \Phi_{\tau'} = \Phi_{\tau+\tau'}$, which additionally assumes that the composite measurement depends only on the total amount of scale — that measurement stages are interchangeable. (A6) drops that assumption, and every result that follows is a consequence of dropping it. Two smaller differences: [8] also required the kernels to be continuous functions, which the pure delay violates, and dropping that requirement is what admits the drift term (Remark 7.5); and (ND) is a normalization not stated there.

(A8) is the two-parameter form of scaling covariance; note that an increasing bijection of $[0,\infty)$ automatically fixes 0. (ND) excludes idle stretches of the scale axis.

A *passive* variant replaces (A5) by $\int \Phi_{x,y} f \le \int f$ on X_+; all results below hold with sub-probability kernels and an extra killing term, and we do not pursue it.

3 Preliminaries and notation

Spaces. Time runs over $\mathbb{R}$. We write $X = L^1(\mathbb{R}, dt)$, real, with positive cone X_+ and $\int f := \int_{\mathbb{R}} f(t)\,dt$; its dual is $X^* = L^\infty(\mathbb{R})$, with pairing $\langle g, f\rangle = \int gf$. We write $M_+(\mathbb{R})$ for the finite positive Borel measures on $\mathbb{R}$, with total mass $\|\mu\| = \mu(\mathbb{R})$. For a function g on $(0,\infty)$, $g(0+)$

denotes its limit from the right at 0; and $\mathcal{D}'(\mathbb{R})$ denotes the space of distributions, the dual of $C_c^\infty(\mathbb{R})$, used only in Appendix A.

Actions (as in §2.3). Translations are $(\tau_a f)(t) = f(t-a)$, $a \in \mathbb{R}$. Dilations act on functions by $(D_\sigma f)(t) = \sigma^{-1} f(t/\sigma)$, $\sigma > 0$, which preserves mass, and on measures by $D_\sigma \mu :=$ the pushforward of μ under $t \mapsto \sigma t$.

Convolution. $(\mu * f)(t) = \int f(t-r)\,\mu(dr)$ for $\mu \in M_+(\mathbb{R})$, $f \in X$. Then $\|\mu * f\|_1 \le \|\mu\|\,\|f\|_1$, and dilation is compatible with convolution:

$$D_\sigma(\mu * f) = (D_\sigma \mu) * (D_\sigma f).$$

Remark 3.1 (the signal model; bounded signals). The working space is $X = L^1(\mathbb{R}, dt)$: it is where the Laplace transforms that carry the characterization live, and every statement of this paper is proved there. The smoothing itself asks for less. The kernels will turn out to be *probability* measures, and for μ a probability measure and f bounded and measurable the same formula defines the field pointwise,

$$(\mu * f)(t) \;=\; \mathbb{E}\big[f(t-T)\big], \qquad T \sim \mu,$$

with $\|\mu * f\|_\infty \le \|f\|_\infty$, and the cascade law survives by Fubini. What genuinely needs X is the analysis — the transforms, hence the characterization theorem and everything derived through it. What survives the extension unchanged is the measurement reading: the pointwise field, positivity and the sup bound, the sign-change statements of Proposition 8.13 (stated there for integrable *or bounded* arguments), and, in the experiment of §12, the persistent step — bounded, not integrable. We record the extension without developing it systematically.

Transforms. For $\mu \in M_+$ supported on $[0,\infty)$ the Laplace transform is

$$\hat\mu(s) = \int_{[0,\infty)} e^{-st}\,\mu(dt), \qquad s \ge 0,$$

and $\hat\mu(s) > 0$ for all $s \ge 0$ whenever $\mu \ne 0$, the integrand being positive.

Two classes of functions on $(0,\infty)$ carry the analysis: the completely monotone functions, which will be the transforms of the kernels, and the Bernstein functions, which will be their exponents.

Definition 3.2 (completely monotone)**.** A function $f : (0,\infty) \to \mathbb{R}$ is *completely monotone* (CM) if $f \in C^\infty$ and $(-1)^n f^{(n)} \ge 0$ for all $n \ge 0$.

Definition 3.3 (Bernstein function)**.** A function $g : (0,\infty) \to [0,\infty)$ is a *Bernstein function* if $g \in C^\infty$ and g' is completely monotone. We write $g \in \mathrm{BF}$, and $g \in \mathrm{BF}_0$ if moreover $g(0+) = 0$.

The toolbox that connects the two classes to Laplace transforms is classical, and it is cited rather than reproven: part (1) is [10], Vol. 2, § XIII.4, Theorems 1 and 1a, pp. 439–440; parts (2)–(4) are [33], Theorem 3.7, p. 27, Theorem 3.2, pp. 21–22, and Corollary 3.8, pp. 28–29, respectively.

Proposition 3.4 (the Bernstein-function toolbox)**.** *1. (Bernstein–Widder / Feller criterion.) f is the Laplace transform of a probability measure on $[0,\infty)$ if and only if f is completely monotone and $f(0+) = 1$. More generally, f is completely monotone if and only if $f(s) = \int_{[0,\infty)} e^{-st}\,F(dt)$ for a measure F on $[0,\infty)$, not necessarily finite, and F is then unique.*

2. For $g : (0,\infty) \to [0,\infty)$ the following are equivalent: $g \in \mathrm{BF}$; the composition $h \circ g$ is completely monotone for every completely monotone h; and $e^{-\tau g}$ is completely monotone for every $\tau > 0$.

3. *Every* $g \in \mathrm{BF}$ *is nonnegative, nondecreasing and concave, and has the Lévy–Khintchine representation*

$$g(s) = a + bs + \int_0^\infty \bigl(1 - e^{-st}\bigr)\,\nu(dt), \qquad a, b \geq 0, \quad \int_0^\infty (1 \wedge t)\,\nu(dt) < \infty,$$

with $g(0+) = a$. *The triple* (a, b, ν) *is unique.*

4. BF *is a convex cone, closed under pointwise limits: if* $g_n \in \mathrm{BF}$ *and* $g_n(s) \to g(s) < \infty$ *for every* $s > 0$, *then* $g \in \mathrm{BF}$. *Consequently* BF *is closed under mixtures* $s \mapsto \int g_u(s)\,m(du)$ *with* $m \geq 0$, *whenever the integral is finite.*

Which vocabulary does the work. Proposition 3.4 is used as a bridge and not as a toolkit. The object the arguments below manipulate is the *representation*, part (3) — a drift and a Lévy measure — and almost every proof in §§4–9 works with it directly. The increments of Theorem 5.2 are produced as representations, not as pointwise limits of Bernstein functions; the potential kernel of the appendix is constructed as a subordinator's potential measure, not represented from complete monotonicity; the Mellin calculus of §9 mentions neither class. Complete monotonicity and the derivative-sign definition of BF enter where the CM class is itself the subject (Proposition A.9(2), Proposition 11.6), in Lemma 7.1's implications (1) ⇒ (2) ⇒ (3), and in the classical proofs of §7, where the printed route and the machine-checked route differ (§1.1); Proposition 3.4 identifies the two vocabularies once so that no later proof has to.

The same class carries a second name, after its representation rather than its derivative signs; several statements below are given in both forms, and Proposition 3.4(3) is the identification.

Definition 3.5 (Lévy exponent)**.** A function $g : [0, \infty) \to [0, \infty]$ is a *Lévy exponent* if

$$g(s) = b_0\, s + \int_0^\infty \bigl(1 - e^{-st}\bigr)\,\nu(dt), \qquad b_0 \geq 0,$$

for a drift b_0 and a positive Borel measure ν on $(0, \infty)$ — with no killing term, so that $g(0) = 0$. We write $g \in \mathrm{LE}$, and call (b_0, ν) its *triple*.

The definition is valued in $[0, \infty]$ so that it carries no integrability hypothesis; for a measure ν on $(0, \infty)$, finiteness of g at a single $s > 0$ is equivalent to $\int_0^\infty (1 \wedge t)\,\nu(dt) < \infty$. On the finite members, by Proposition 3.4(3), $\mathrm{LE} = \mathrm{BF}_0$ as classes of functions, with the triple unique. Both names are kept because the two readings matter separately: the proofs of this paper argue mostly in the representation, and the machine-checked development (§1.1) argues *only* there — it never defines complete monotonicity at all — so a statement whose conclusion is given in LE is one the formalization can state verbatim.

Remark 3.6 (the circle of ideas). The classes just named are one circle of ideas, and the reader who sees the circle once does not need to memorize the toolbox. Start with complete monotonicity. The prototype completely monotone function is a single decaying exponential e^{-st}, whose derivatives alternate for the trivial reason that each differentiation pulls down a factor $-t \leq 0$. The sign conditions are linear in f and survive pointwise limits, so every positive mixture $\int e^{-st}\,F(dt)$ — every Laplace transform of a positive measure — is completely monotone for free. The Bernstein–Widder theorem, Proposition 3.4(1), is the remarkable converse: *every* completely monotone function is such a mixture, for a unique F. Geometrically: CM is the closed convex cone whose extreme rays are the exponentials, and the countably many derivative-sign conditions are exactly the visible shadow of positivity of the mixing measure. Complete monotonicity is to the Laplace transform what positive definiteness is to the Fourier transform — the intrinsic, transform-free description of "comes from a positive mass."

Bernstein functions are the same picture one level up, prompted by the question the cascade forces: for which $g \geq 0$ does $e^{-\tau g}$ remain a Laplace transform for *every* $\tau > 0$ — every τ,

because a cascade subdivides its exponents without limit (§5)? The answer is Proposition 3.4(2): exactly the Bernstein functions. And the cone BF has its own extreme rays, mirroring CM's exponentials: the elementary exponents $1 - e^{-st}$, the drift s, and the constants. That is what the Lévy–Khintchine representation, Proposition 3.4(3), *says* — the same cone described twice, once by sign conditions (checkable on a given function) and once by its extreme rays (structural). Definition 3.5 above chooses the extreme-ray description as primary, which is also the machine-checked development's choice.

Probability owns this cone through infinite divisibility. A law μ on $[0,\infty)$ divides, $\mu = \mu_n^{*n}$ for every n, exactly when $\hat{\mu}^{1/n}$ is again a transform for every n — that is, when $\hat{\mu} = e^{-g}$ with $e^{-g/n}$ completely monotone for all n, i.e. $g \in \mathrm{BF}_0$. And each extreme ray then acquires a probabilistic identity. Let delay events of size t occur at rate λ, independently across disjoint stretches of an interval of length τ: one infinitesimal slot contributes the transform factor $\big(1 - \lambda\, d\tau\big) + \lambda\, d\tau\, e^{-st} = 1 - \lambda\, d\tau\,(1 - e^{-st})$, and compounding the independent slots gives

$$\mathbb{E}\, e^{-sX_\tau} \;=\; \exp\big[-\lambda\tau\,(1 - e^{-st})\big]\,:$$

the extreme ray $1 - e^{-st}$ is the exponent of a Poisson stream of size-t jumps, and the Lévy–Khintchine integral is a superposition of independent such streams, one per jump size — with the drift as the deterministic limit and the killing constant a, excluded from LE by $g(0) = 0$, as loss of mass. Spread the divisions out as a flow and this is a *subordinator*: an increasing process with stationary independent increments and $\mathbb{E}\, e^{-sX_\tau} = e^{-\tau g(s)}$, whose triple describes the path — speed b_0, jumps of size in dt at rate $\nu(dt)$, the integrability $\int (1 \wedge t)\, \nu(dt) < \infty$ being exactly path-finiteness. The hemigroup of this paper relaxes precisely the stationarity: its exponent is $G(x, s)$ rather than $\tau g(s)$, its increments are independent but not stationary, and the classes $\mathrm{SD} \supset \mathrm{GGC}$ below are the strata of that relaxation under covariance.

The physical reading of the triple — drift as latency, ν as a catalogue of delays, the missing Gaussian term — is Remark 3.12 below. The extreme-ray pattern returns twice more, with the atoms reshaped by each added constraint: the Dickman exponents as the extreme rays of the self-decomposable cone (Proposition 7.8), and the Gamma exponents as the atoms of its Thorin subcone (Proposition 11.6).

Lemma 3.7 (vanishing lemma)**.** *If $g \in \mathrm{BF}_0$ and $g(s_0) = 0$ for some $s_0 > 0$, then $g \equiv 0$.*

Proof. g is nonnegative and concave by Proposition 3.4(3), with $g(0+) = g(s_0) = 0$. A nonnegative concave function vanishing at both ends of $[0, s_0]$ vanishes on it, so $g \equiv 0$ on $(0, s_0]$ and hence $g' \equiv 0$ on $(0, s_0)$. Now g' is nonincreasing, being the derivative of a concave function, and nonnegative, being completely monotone; a nonnegative nonincreasing function that vanishes on $(0, s_0)$ vanishes identically. So $g' \equiv 0$ on $(0, \infty)$, and with $g(0+) = 0$ this gives $g \equiv 0$. □

Two further classical facts are used repeatedly and are recorded here so that each later use can cite a statement; both are from [10], Vol. 2, § XIII.1: Theorem 2, p. 431, and Theorems 1 and 1a, pp. 430 and 432, the latter stated for arbitrary measures.

Proposition 3.8 (continuity theorem for Laplace transforms)**.** *Let μ_n, μ be probability measures on $[0,\infty)$. Then $\mu_n \to \mu$ weakly if and only if $\hat{\mu}_n(s) \to \hat{\mu}(s)$ for every $s > 0$.*

Proposition 3.9 (uniqueness of the Laplace transform)**.** *A measure on $[0,\infty)$ is determined by its Laplace transform: if $\hat{\mu}(s) = \hat{\rho}(s)$ for all s in some interval (a,∞), then $\mu = \rho$.*

The appendix compares measures — the memory kernel $\kappa^{(x)}$, the potential kernel $\ell^{(x)}$, and Lebesgue measure — none of which is finite, so it needs the following refinement, in which finiteness of the measures is replaced by convergence of the transform at a single point.

Proposition 3.10 (uniqueness without a finiteness assumption)**.** *Let μ, ρ be measures on $\mathbb{R}$ carried by $[0,\infty)$, not assumed finite, and let $s_0 \in \mathbb{R}$ be such that $\hat{\mu}(s_0) < \infty$. If $\hat{\mu}(s) = \hat{\rho}(s)$ for every $s \geq s_0$, then $\mu = \rho$.*

Corollary 3.11 (uniqueness for signed densities)**.** *Let V be a measurable real function on $\mathbb{R}$ vanishing a.e. on $(-\infty, 0)$, and let $s_0 \in \mathbb{R}$ be such that $\int_0^\infty e^{-s_0 t}\,|V(t)|\,dt < \infty$. If $\int_0^\infty e^{-st}\,V(t)\,dt = 0$ for every $s \geq s_0$, then $V = 0$ a.e.*

Proof. Write $V = V_+ - V_-$. The measures $V_+(t)\,dt$ and $V_-(t)\,dt$ are carried by $[0,\infty)$, their transforms are finite at s_0 — each is bounded by $\int e^{-s_0 t}|V|\,dt$ — and the hypothesis makes them equal for every $s \geq s_0$. By Proposition 3.10 the two measures coincide, so $V_+ = V_-$ a.e., i.e. $V = 0$ a.e. □

The causal vocabulary, used from §9 on and in the appendix, is the setting in which the family is read as an evolution in present time: a signal that starts at time 0, delayed and smoothed. A function $f \in X$ is *causal* if it vanishes a.e. on $(-\infty, 0)$; its transform $\hat{f}(s) = \int_0^\infty e^{-st} f(t)\,dt$ is then defined for $s > 0$ with $|\hat{f}(s)| \leq \|f\|_1$. We write

$$X_0 := L^1(\mathbb{R}_+),$$

identified with the causal elements of X; by (A3) every $\Phi_{x,y}$ restricts to X_0. On X_0 the translations $(\tau_r)_{r\geq 0}$ form the *delay semigroup*, $(\tau_r f)(t) = f(t-r)\,\mathbf{1}_{t\geq r}$: positive isometries, and strongly continuous because translation is continuous in L^1. The operator core used there is

$$\mathcal{D} \;:=\; \big\{\, f \in X_0 :\ f \text{ absolutely continuous},\ f' \in X_0,\ f(0) = 0 \,\big\}.$$

It is cut so that its members are exactly the primitives of their derivatives, $f = \int_0^{\cdot} f'$ with $f' \in X_0$, hence bounded, which is what widens the Mellin strips of §9. They also lie in X_0 themselves, which the other two conditions do not imply, a primitive of an L^1 function being integrable only when $\int_0^\infty f' = 0$.

Classes referred to later. Five named subclasses appear in §§10–11 and the appendix, always as the answer to "which laws have this property", never as tools; they are recorded here so that each use can point to a definition. Two are classes of functions on $(0,\infty)$:

- the *Stieltjes* functions $\mathcal{S}$,

$$f(s) = \frac{a}{s} + b + \int_{(0,\infty)} \frac{\mu(dt)}{s+t}, \qquad a, b \geq 0,$$

  a subclass of CM ([33], Def. 2.1, p. 16, and Thm. 2.2, p. 17);

- the *complete Bernstein* functions CBF $\subset$ BF: those whose Lévy measure has a completely monotone density ([33], Def. 6.1, p. 69).

Three are classes of laws on $[0,\infty)$:

- the *self-decomposable* laws SD, or class L: μ such that for every $b > 1$ there is a law ρ_b with

$$\hat{\mu}(s) = \hat{\mu}(s/b)\,\hat{\rho}_b(s)$$

  ([32], Def. 15.1, p. 90); they are infinitely divisible ([32], Prop. 15.5, p. 93), and their Laplace exponents are what §7 characterizes;

- the *generalized Gamma convolutions* GGC: the laws with Laplace transform

$$\exp\Big\{ -as - \int_{(0,\infty)} \log\big(1 + s/t\big)\,U(dt) \Big\}$$

  ([2], §3.1, p. 29) — the Thorin subclass of §7 — with GGC $\subset$ SD ([2], p. 30);

- the laws with *hyperbolically completely monotone* density, HCM: densities f with $f(uv)\,f(u/v)$ completely monotone in $v+v^{-1}$ for every $u>0$ ([2], §4.3, p. 55), with HCM $\subset$ GGC ([2], Thm. 4.3.1, p. 57, the regularity conditions removed on p. 68).

So HCM $\subset$ GGC $\subset$ SD.

Remark 3.12 (the delay dictionary). Every integral formula in this paper is an instance of one physical picture, worth fixing once. A causal kernel is the law of a *random delay*: the measurement $\Phi_{x,y}f$ reports the signal late by a random amount $T_{x,y}\sim\mu_{x,y}$, the transform $\hat\mu_{x,y}(s)=\mathbb{E}\,e^{-sT_{x,y}}$ is the fraction of a probe exponential surviving the stage, and the exponent $g=-\log\hat\mu$ is the attenuation it accumulates — additive along the cascade, as absorbances are additive through layered media (the optical reading is developed in Remark 4.3, where the analogy is made literal). In this reading the Lévy–Khintchine form

$$g(s)\;=\;b_0\,s\;+\;\int_0^\infty\bigl(1-e^{-st}\bigr)\,\nu(dt)$$

is an inventory of the two mechanisms by which delay can accumulate.

The drift b_0 is deterministic latency: a fixed transport lag, shifting every kernel bodily; its extreme case is the pure delay family $\mu_{0,x}=\delta_{b_0x}$.

The measure ν is a catalogue of random delay events: $\nu(dt)$ is the rate, per unit of scale, at which lag increments of size $\approx t$ occur. Each size contributes independently as a Poisson stream, and the integrand is that stream's cost: over an infinitesimal stretch of scale dx, a size-t event occurs with probability $\nu(dt)\,dx$ and multiplies the transform by e^{-st}, so the stage's factor is $1-\nu(dt)\,dx\,(1-e^{-st})$ and the exponent picks up $(1-e^{-st})\,\nu(dt)\,dx$. The integral is therefore literally a positive superposition — the catalogue lists which delays are available and how often. ("Event" here means an increment of the observer's accumulated *lag* as the aperture widens; the signal itself is never discontinuous.)

There is no diffusion term, and its absence is a theorem, not a choice. On the full line the Lévy–Khintchine formula carries a Gaussian term $\frac12\sigma^2s^2$; on the half-line it cannot appear, because a diffusing quantity oscillates and an accumulated delay only grows. The term that generates spatial scale space is thus structurally deleted from the time-causal axis — causality does not merely disfavor the Gaussian aperture, it removes its slot in the generator. Classifying time-causal scale spaces thereby becomes classifying delay catalogues. (Diffusion will reappear, but on the observer's memory line rather than the signal axis: the locality theorem of §10 closes that ladder at the Bessel operator, an honest second-order generator over the embodied record.)

The natural coordinate for catalogues is log-delay. Writing $t=e^\theta$ turns the density shape the characterization singles out into a profile per unit log-delay,

$$\nu(dt)=\frac{k(t)}{t}\,dt=\tilde k(\theta)\,d\theta,\qquad \tilde k(\theta):=k(e^\theta),$$

since dt/t is the invariant measure of the dilation group — so *dilation acts on catalogues by translation in θ*. The classification results of §7 then become shape conditions on one curve (Figure 3): full scale covariance forces $\tilde k$ nonincreasing — zooming out may only add delay intensity, which is self-decomposability; the semigroup subclass is the exactly scale-free profile $\tilde k(\theta)=c\,e^{-\alpha\theta}$, whose refusal to decay faster *is* the heavy tail. The extreme rays are unit steps — log-uniform bands of delays, the Dickman exponents — and every admissible catalogue is a unique stack of such steps; the completely monotone profiles form the Thorin subclass GGC above, whose finite moments correspond to a cliff-like cutoff of long delays.

Table 1 collects, for reference, the symbols that recur across sections, each with its defining formula and its reading in the delay vocabulary above, and Table 2 the four scale coordinates the paper distinguishes. Everything in them is defined where the middle column says; nothing is new here.

Table 1: The recurring symbols.

symbol	definition	delay reading
$\Phi_{x,y}$, $\mu_{x,y}$	increment operator and its kernel, $\Phi_{x,y}f = \mu_{x,y} * f$ (§4)	the stage delays the record by $T_{x,y} \sim \mu_{x,y}$
$g_{x,y}(s)$	increment exponent, $-\log \hat{\mu}_{x,y}(s)$ (§4)	attenuation of the probe e^{-st}; additive along the cascade
$G(x,s)$	accumulated exponent, $g_{0,x}(s)$ (§5)	optical depth from the boundary to scale x
$F(s)$	the family's exponent, (7.1); canonical gauge $G(x,s) = F(xs)$ (§7)	Laplace exponent of the unit delay T_1; one function per family
$k(t)$	memory kernel: Lévy density $k(t)/t$, k nonincreasing, (7.1)	size-weighted delay catalogue: $k(t) = t\times$ rate of size-t lag events
$\phi_x(t)$	kernel density, $\hat{\phi}_x(s) = e^{-F(xs)}$; exists for $k \not\equiv 0$ (§8, Prop. A.12)	density of the total delay $x\,T_1$
$H(s)$	the profile, $H := e^{-F}$, so $\hat{\mu}_{0,x}(s) = H(sx)$ (§9)	surviving fraction of the probe, in the dimensionless variable sx
$\tilde{H}(z)$, $m(z)$	Mellin data of the profile: $\tilde{H}(z) = \Gamma(z)\,m(z)$ with $m(z) := \mathbb{E}[T_1^{-z}]$, finite for $0 < \operatorname{Re} z < z_*$ (§9)	negative moments of the unit delay; (H) asks $z_* > 1$
$B(\theta)$	Mellin symbol of the inversion, $B(-z) = \tilde{H}(z+1)/\tilde{H}(z)$, $A = x^{-1}B(\theta)$ (§9)	the re-indexing that turns the memory record into evolution in present time
$B(s) = sF'(s)$	Bernstein function of Lemma 7.1 (§7) — a *different* B from the Mellin symbol above; its dilate $\varphi_x(s) := sF'(xs)$ is the per-scale symbol of Appendix A	transform of the memory measure below, up to the factor s
$\kappa^{(x)}$	memory measure, $b_0\,\delta_0(dt) + \frac{1}{x}k(\frac{t}{x})\,dt$, $\widehat{\kappa^{(x)}}(s) = F'(xs)$ (§7, App. A)	drift plus the catalogue dilated to scale x: what the medium remembers
$\ell^{(x)}$	potential kernel, $\widehat{\ell^{(x)}}(s) = 1/(sF'(xs))$ (App. A)	renewal measure of the delay stream; the memory operator's exact right inverse

Table 2: The scale coordinates.

coordinate	defined by	where used
original x	any increasing relabelling of $[0,\infty)$; only $G(x,s) = g_{0,x}(s)$ is meaningful	§§4–6, until the gauge is fixed
canonical $\tilde{x} = \chi(x)$	dilations act by multiplication, $G = F(\tilde{x}\,s)$; scale carries the dimension of time (Prop. 6.3)	the working gauge from §7 on, written plainly x
semigroup τ	the homogeneous member's own additive parameter, F-exponent τs^α: dilations act by $\tau \mapsto \sigma^\alpha\tau$, and $\chi(\tau) = \tau^{1/\alpha}$; coincides with the canonical gauge only at $\alpha = 1$ (§6)	the comparison with the semigroup theory
parabolic ξ	$\tilde{x} = \xi^2$ (Remark 6.4)	the Bessel family: its medium becomes $\frac{1}{2}\partial_\xi^2 + \frac{\beta}{\xi}\partial_\xi$ (§§9–10)

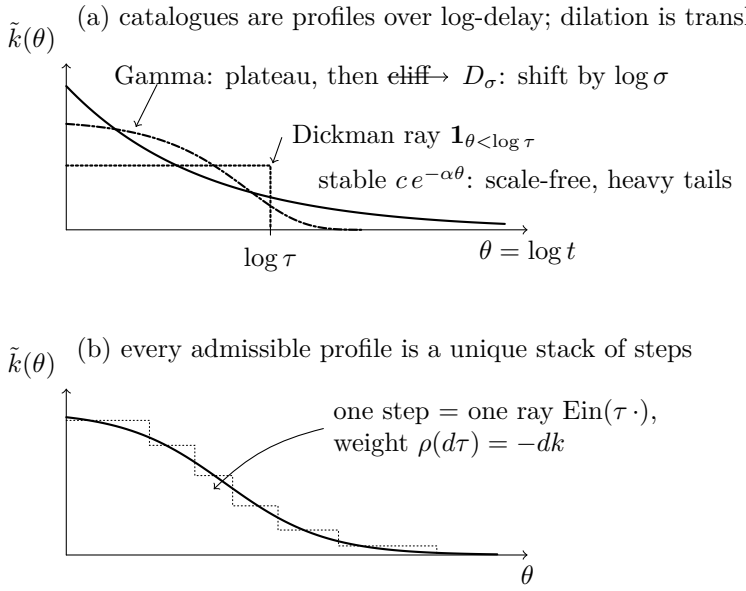


Figure 3: Delay catalogues in the log-delay coordinate $\theta = \log t$, where the Lévy measure $k(t)\,t^{-1}dt$ becomes the profile $\tilde{k}(\theta) = k(e^\theta)$ and dilation acts by translation. (a) Full scale covariance requires only that the profile be nonincreasing — zooming out may only add delay intensity (Lemma 7.1); within that freedom the named families are shapes: the stable family is the exactly scale-free exponential profile, whose slow decay is the heavy tail (Corollary 7.4); the Gamma family cuts long delays off at a cliff, buying finite moments; and the Dickman ray is a log-uniform band of delays with a hard cutoff. (b) The Dickman superposition (Lemma 7.7): every nonincreasing profile is a unique positive stack of unit steps, so every admissible family is a unique superposition of drift and dilated Dickman subordinators, with mixing measure $\rho = -dk$ (Proposition 7.8).

4 Convolution representation

The first step of the derivation uses no covariance and no cascade: the axioms (A1)–(A5) alone force the measurement operators to be convolutions by probability measures on $[0,\infty)$ — the apertures promised in §2, as measures; where they have densities, and which, is settled once covariance and (ND) are imposed (§7). The lemma is an equivalence, and its converse is what the constructive half of the characterization will later run backwards through. Continuity (A7) is added only for the second lemma, which extracts from the measures the one function the rest of the paper works with.

One device serves both lemmas. The exponential e^{-st} is unbounded on $\mathbb{R}$, so it is not a functional on X; the *clamped exponential* $w_s(t) := e^{-s\max(t,0)}$ is bounded, hence in X^*, and agrees with e^{-st} exactly where causal mass lives. For a measure μ carried by $[0,\infty)$ and $f = 1_{(0,1)}$, one has $\langle w_s, f(\cdot - r)\rangle = e^{-sr}\langle w_s, f\rangle$ for $r \geq 0$, so Fubini collapses the pairing to

$$\langle w_s, \mu * f\rangle = \langle w_s, f\rangle\,\hat{\mu}(s), \qquad \langle w_s, f\rangle \geq e^{-s} > 0 : \tag{1}$$

a single bounded functional reads off the Laplace transform of the kernel.

Lemma 4.1 (representation)**.** *Under (A1)–(A5), for each pair $0 \leq x \leq y$ there is a unique probability measure $\mu_{x,y}$ on $[0,\infty)$ with*

$$\Phi_{x,y} f = \mu_{x,y} * f, \qquad f \in X,$$

and conversely every such measure defines an operator satisfying (A1)–(A5).

Proof. Fix (x, y) with $0 \le x \le y$ and write $\Phi = \Phi_{x,y}$. The plan is classical: the representing measure ought to be Φ applied to a unit point mass, but δ_0 is not in L^1, so Φ is applied to an approximate identity instead, a limit measure is extracted by compactness, and the limit is then checked to represent Φ; the named steps follow that plan.

Convolution as a vector-valued integral. For $f \in L^1$ and $g \in X$,

$$f * g = \int_{\mathbb{R}} f(r)\, \tau_r g \, dr$$

as a Bochner integral *in* X: the integrand is continuous in r, because translation acts continuously on L^1, and is dominated by $|f(r)|\, \|g\|_1$. An X-valued integral commutes with every bounded operator, so (A1) and (A2) give the interchange identity

$$\Phi(f * g) = f * \Phi g, \qquad f \in L^1,\ g \in X,$$

which carries the existence argument; the remaining axioms enter where named.

The approximants. Let $\rho_\varepsilon = \varepsilon^{-1} 1_{(0,\varepsilon)}$ and $h_\varepsilon := \Phi\rho_\varepsilon$. By (A4) and (A5), $h_\varepsilon \ge 0$ with $\int h_\varepsilon = 1$; by (A3) it vanishes a.e. on $(-\infty, 0)$. So $\nu_\varepsilon := h_\varepsilon\, dt$ is a probability measure carried by $[0, \infty)$.

Tightness. By the interchange identity $\rho_1 * h_\varepsilon = \Phi(\rho_1 * \rho_\varepsilon)$, and $\rho_1 * \rho_\varepsilon \to \rho_1$ in L^1, so $\rho_1 * h_\varepsilon$ converges in X as $\varepsilon \downarrow 0$ — and a convergent sequence in L^1 has uniformly small tails. Since $\rho_1 \ge 0$ is carried by $(0, 1)$ and $h_\varepsilon \ge 0$, convolving can only move mass to the right:

$$\int_T^\infty h_\varepsilon \le \int_T^\infty \rho_1 * h_\varepsilon.$$

Along $\varepsilon_n = 1/(n+1)$ the right tails of ν_{ε_n} are therefore small uniformly in n, and (A3) kills the left tail outright, so $\{\nu_{\varepsilon_n}\}$ is tight, with compacts $[0, T]$.

The limit. By Prokhorov's theorem some subsequence $\nu_{\varepsilon_{n_k}}$ converges weakly to a probability measure μ; and $(-\infty, 0)$ being open, the portmanteau theorem bounds $\mu((-\infty, 0))$ by the lower limit of $\nu_{\varepsilon_{n_k}}((-\infty, 0)) = 0$, so $\operatorname{supp}\mu \subseteq [0, \infty)$. Both the mass and the support clause are thus part of the extraction rather than separate arguments.

Identification. Let $\Psi \in X^*$ and $f \in X$. The map $r \mapsto \Psi(\tau_r f)$ is continuous — again because translation acts continuously on L^1 — and bounded by $\|\Psi\|\, \|f\|_1$, so it is a legitimate test function for weak convergence. Reading $f * h_\varepsilon$ with h_ε in the density slot and applying Ψ,

$$\Psi(f * h_\varepsilon) = \int_{\mathbb{R}} \Psi(\tau_r f)\, \nu_\varepsilon(dr) \xrightarrow{k\to\infty} \int_{\mathbb{R}} \Psi(\tau_r f)\, \mu(dr) = \Psi(\mu * f),$$

the last equality because $\mu * f = \int \tau_r f \, \mu(dr)$ is again an X-valued Bochner integral. On the other hand $f * h_\varepsilon = \Phi(f * \rho_\varepsilon) \to \Phi f$ in X, by the interchange identity and (A1). Hence $\Psi(\Phi f) = \Psi(\mu * f)$ for every $\Psi \in X^*$, and so $\Phi f = \mu * f$. Note that this holds for *every* $f \in X$ at once: nothing in the display is pointwise, so no restriction to C_c and no density step is needed.

Uniqueness. Take $f = 1_{(0,1)}$ and pair $\mu * f$ against the clamped exponential $w_s(t) = e^{-s\max(t,0)}$: by the factorization (1), $\langle w_s, \mu * f\rangle = \langle w_s, f\rangle\, \hat\mu(s)$ with $\langle w_s, f\rangle > 0$. So Φ determines $\hat\mu$ on $[0, \infty)$, and a finite measure on $[0, \infty)$ is determined by its Laplace transform (Proposition 3.9). One test function suffices; no separating family is needed.

Converse. A probability measure μ on $[0, \infty)$ makes $f \mapsto \mu * f$ bounded with norm ≤ 1 by Young's inequality (A1), translation-commuting because convolution is (A2), support-preserving on $[t_0, \infty)$ because $\operatorname{supp}\mu \subseteq [0, \infty)$ (A3), positive (A4), and mass-preserving by Tonelli (A5). □

The rest of the paper works not with the measures but with one function of them, their *exponent* $g_{x,y}(s) := -\log \hat\mu_{x,y}(s)$ — the quantity the cascade will make additive (§5). That section's argument also needs the exponent to depend continuously on the pair of scales, so that a fine partition of $[x, y]$ gives a null array; this is what the second lemma supplies, from (A7).

Lemma 4.2 (continuity of transforms)**.** *Under (A1)–(A5) and (A7), the function*

$$g_{x,y}(s) := -\log \hat{\mu}_{x,y}(s) \in [0,\infty), \qquad s \geq 0,$$

is well defined, and $(x,y) \mapsto g_{x,y}(s)$ *is continuous on* $\{0 \leq x \leq y\}$ *for each* $s \geq 0$*; moreover* $g_{x,y}(0) = 0$ *and* $s \mapsto g_{x,y}(s)$ *is continuous on* $[0,\infty)$*.*

Proof. $\hat{\mu}(s) \in (0,1]$ for a probability measure on $[0,\infty)$ and $s \geq 0$: positive because the integrand is, and at most 1 because $e^{-st} \leq 1$ where the mass sits. So g is well defined and nonnegative, and $g_{x,y}(0) = -\log 1 = 0$. Continuity of $s \mapsto \hat{\mu}(s)$ on $[0,\infty)$ is dominated convergence with the constant bound 1 on the integrand. This is where causality is needed: e^{-st} is bounded on the support of μ only because that support lies in $[0,\infty)$.

For continuity in (x,y), take $f = 1_{(0,1)}$ and pair against the clamped exponential w_s; by (1),

$$\langle w_s, \Phi_{x,y} f \rangle = \langle w_s, \mu_{x,y} * f \rangle = \langle w_s, f \rangle \, \hat{\mu}_{x,y}(s).$$

The left side is a bounded functional applied to $\Phi_{x,y} f$, which is continuous in (x,y) on $\{0 \leq x \leq y\}$ by (A7); dividing by the constant $\langle w_s, f \rangle > 0$ gives continuity of $\hat{\mu}_{x,y}(s)$, and composing with $-\log$, legitimate because the transform never vanishes, gives it for $g_{x,y}(s)$. □

Remark 4.3 (transmittance and optical depth). The transform and the exponent deserve physical names, used from here on as a guide. Causal convolution acts on a probe exponential e^{-st} as multiplication — that is what (1) measures — and the multiplier $\hat{\mu}_{x,y}(s) \in (0,1]$ is the stage's *transmittance* at probe rate s: the fraction of the probe that survives. Transmittances multiply along the cascade, so their negative logarithms, the exponents $g_{x,y}(s)$, add: g is the stage's *absorbance*, and $G(x,s) = g_{0,x}(s)$ will be the *optical depth* of the medium between the surface $x = 0$ and depth x. Lemma 4.2 says the medium is physically sane: no layer is opaque, none amplifies, and the absorbance varies continuously with the layer and the probe.

The analogy is in fact literal, not visual. Beer–Lambert's law is, microscopically, a survival probability: a photon crossing a slab is absorbed at a constant rate per unit path length, so it survives a traverse of length L with probability $e^{-\alpha L}$. Here, absorb the traversing signal at rate s per unit of *accumulated delay* — dwell time playing the role of path length — and the transmittance is exactly the survival probability $\mathbb{E}\, e^{-sT_{x,y}}$, the slab's effective thickness being the random delay $T_{x,y}$. Equivalently, $e^{-sT} = \mathbb{P}(E > T)$ for an independent exponential deadline E of rate s: transmittance is the probability that the stage's delay beats a memoryless alarm clock, and transmittances multiply across independent stages *because* the exponential clock is memoryless — having survived one stage, it is statistically fresh for the next. So s is not a wavelength but an absorption rate (equivalently a discount rate, killing and discounting being the same computation), and varying s probes the medium with dyes of graded strength rather than light of graded color.

What plays the role of wavelength is the probe's *time constant* $1/s$: a small-s probe is long and patient, responding only to gross features — at $s = 0+$ it sees only total mass, which is $g(0) = 0$ being unit area. A large-s probe has died by time $1/s$ and sees only the immediate response. Sweeping s scans lateness structure from coarse to fine, as sweeping frequency scans oscillation structure. The normalized probes $s\, e^{-st}$ are, moreover, the impulse responses of first-order lag sections: the transform reads the kernels out through a bank of leaky integrators — the theory's own simplest member, and the hardware of §11.

In this language the next two sections say: absorbances of stacked layers add, so depth is measured from the surface (§5); and the medium is *self-similar* — descending the scale axis is indistinguishable from probing faster — so there is a coordinate in which one absorption spectrum describes it entirely (§6). The physical reading of the spectrum's parts — drift as latency, the Lévy measure as a catalogue of delays — is Remark 3.12.

Remark 4.4 (what the representation is, and what it is not). Lemma 4.1 is the Wendel step ([41]: the translation-invariant operators on L^1 of a group are exactly convolution by a measure), specialised to $\mathbb{R}$ and sharpened by (A3)–(A5) to a probability measure carried by $[0,\infty)$. It is proved here rather than cited; the only imported fact in this section is the uniqueness theorem for Laplace transforms, Proposition 3.9.

On the two routes. The classical argument regards $h_\varepsilon\,dt$ as a bounded family in $M(\mathbb{R}) = C_0(\mathbb{R})^*$ and extracts a weak-$*$ subnet by Banach–Alaoglu, recovering $\|\mu\| = 1$ from (A5) afterwards and reading the limit as a measure by Riesz–Markov. The proof given above instead makes the family *tight* — which costs one extra observation, that convolution with a density carried by $(0,1)$ only moves mass to the right — and then Prokhorov returns a probability measure carried by $[0,\infty)$ outright. The mass and support clauses are then part of the extraction rather than separate arguments, no dual space appears, and the weak-$*$ selection apparatus — Banach–Alaoglu and Riesz–Markov — is bypassed. The second route is the one the formalization follows, because it needs no dual space: the library has Banach–Alaoglu but not the identification of $M(\mathbb{R})$ with $C_0(\mathbb{R})^*$.

The identification step likewise has two forms. Testing against C_0 functions makes it pointwise, which forces the restriction to $f \in C_c$ and a density argument afterwards. Testing against $X^* = L^\infty$ makes it an identity in X, valid for every $f \in X$ at once, and the price is only that $f * g$ must be read as an X-valued Bochner integral — which the proof needs anyway, for the interchange identity $\Phi(f * g) = f * \Phi g$.

The multiplicative-group counterpart of this step, for the scale variable rather than time, is what §9 has to build by hand; there is no Wendel theorem to appeal to once the group is $(0,\infty)$ under multiplication and the operator is required only to be covariant.

5 The cascade: additivity and infinite divisibility

With the representation in hand, the cascade axiom (A6) becomes a statement about kernels: composition of operators is convolution of measures, so the exponent of §4 is *additive* over adjacent intervals of the scale axis — Beer–Lambert's law for the measurement medium: absorbances of stacked layers add (Remark 4.3). Set

$$G(x,s) := g_{0,x}(s),$$

the exponent accumulated from scale 0 to scale x — the optical depth from the surface. The section shows three things about it: that the exponent of any increment is a difference of its values, $g_{x,y} = G(y,\cdot) - G(x,\cdot)$, so that $G(\cdot,s)$ is nondecreasing; that every such increment is a Bernstein function, which is the hemigroup form of infinite divisibility; and that, under (ND), $G(\cdot,s)$ is strictly increasing — so that it is a coordinate on the scale axis, which is what §6 will use to straighten that axis. No covariance is used anywhere in this section: everything below holds for every family satisfying (A1)–(A7), and the stratification of §13 depends on exactly that.

Lemma 5.1 (additivity)**.** *Under (A1)–(A7), $\mu_{x,z} = \mu_{y,z} * \mu_{x,y}$ for $x \le y \le z$, hence*

$$g_{x,z}(s) = g_{x,y}(s) + g_{y,z}(s), \qquad g_{x,x} = 0.$$

Consequently, with $G(x,s) := g_{0,x}(s)$,

$$g_{x,y}(s) = G(y,s) - G(x,s),$$

where $G(0,\cdot) = 0$, $G(\cdot,s)$ is nondecreasing and continuous for each s, and $G(x,0) = 0$.

Proof. Immediate from (A6), Lemma 4.1 (under which composition of operators corresponds to convolution of measures), and Lemma 4.2. In detail: (A6) gives $\Phi_{y,z}\Phi_{x,y} = \Phi_{x,z}$, which by Lemma 4.1 reads $\mu_{y,z} * (\mu_{x,y} * f) = \mu_{x,z} * f$ for all $f \in X$, so $\mu_{x,z} = \mu_{y,z} * \mu_{x,y}$ by the uniqueness clause there. Taking Laplace transforms turns convolution into multiplication and hence $-\log$ into addition — legitimate because $\hat{\mu}(s) > 0$, which is the first clause of Lemma 4.2 — giving the additivity of g; $\Phi_{x,x} = \mathrm{Id}$ gives $\mu_{x,x} = \delta_0$ and so $g_{x,x} = 0$. Setting $x = 0$ defines G, and additivity gives $g_{x,y} = G(y,\cdot) - G(x,\cdot)$. Monotonicity in the first argument is the nonnegativity of $g_{x,y}$, continuity is Lemma 4.2, and $G(x,0) = 0$ is the corresponding clause there. □

The next step upgrades the increments from "exponents of probability measures" to Bernstein functions. In the semigroup case this is trivial, since there $\mu_\tau = \mu_{\tau/n}^{*n}$ exhibits every kernel as an n-fold convolution power and infinite divisibility is immediate. In the hemigroup case there is no such power: the increments over a partition need not be equal, or even comparable, and what replaces the argument is a null-array limit. On the half line it can be made elementary, and the proof below is the whole of it.

Theorem 5.2 (increments are Bernstein)**.** *Under (A1)–(A7), $g_{x,y} \in \mathrm{BF}_0$ for every $x \le y$. Equivalently, and this is the form the proof produces: $g_{x,y} \in \mathrm{LE}$, i.e. there are a drift $b_0(x,y) \ge 0$ and a Lévy measure $\nu_{x,y}$ on $(0,\infty)$ with*

$$g_{x,y}(s) = b_0(x,y)\, s + \int_0^\infty \bigl(1 - e^{-st}\bigr)\, \nu_{x,y}(dt).$$

Proof. Fix $x < y$ and the uniform partition $t_i = x + \frac{i}{n}(y-x)$, $i = 0, \dots, n$. Write $\mu_i := \mu_{t_i,t_{i+1}}$ and $g_i := g_{t_i,t_{i+1}}$, so that by Lemma 5.1

$$g_{x,y}(s) = \sum_{i=0}^{n-1} g_i(s) \qquad \text{for every } n.$$

Fix $s > 0$. Since $G(\cdot, s)$ is continuous on the compact interval $[x,y]$ it is uniformly continuous there, so $m_n := \max_i g_i(s) \to 0$ as $n \to \infty$. Using $0 \le u - (1 - e^{-u}) \le u^2/2$ for $u \ge 0$,

$$\Bigl| g_{x,y}(s) - \sum_i \bigl(1 - e^{-g_i(s)}\bigr) \Bigr| \;\le\; \tfrac{1}{2} \sum_i g_i(s)^2 \;\le\; \tfrac{m_n}{2}\, g_{x,y}(s) \;\longrightarrow\; 0.$$

On the other hand, with $\Pi_n := \sum_i \mu_i \in M_+([0,\infty))$, a finite measure,

$$\sum_i \bigl(1 - e^{-g_i(s)}\bigr) = \sum_i \bigl(1 - \hat{\mu}_i(s)\bigr) = \int_0^\infty \bigl(1 - e^{-st}\bigr)\, \Pi_n(dt),$$

which is a Bernstein function of s vanishing at $0+$ — it is in Lévy–Khintchine form with finite Lévy measure Π_n and no drift or killing term.

It remains to pass to the limit, and the point is to extract the *representation* rather than to invoke closure of BF under pointwise limits. The total masses $\Pi_n([0,\infty)) = n$ diverge, because mass piles up at the origin — and that pile *is* the drift. So weight the measures by $1 - e^{-t}$, which vanishes at the origin at exactly the rate that compensates, and substitute $v = 1 - e^{-t}$, carrying $[0,\infty]$ onto the compact $[0,1]$ with $\infty \mapsto 1$. Weighting μ_i by $1 - e^{-t}$ and integrating gives $1 - \hat{\mu}_i(1)$, so the weighted measures have total mass

$$\sum_i \bigl(1 - \hat{\mu}_i(1)\bigr) \;=\; \sum_i \bigl(1 - e^{-g_i(1)}\bigr) \;\le\; \sum_i g_i(1) \;=\; g_{x,y}(1),$$

using $1 - e^{-u} \le u$ for $u \ge 0$ and Lemma 5.1 at $s = 1$; this bound is uniform in n. So the weighted measures live on the fixed compact $[0,1]$ with uniformly bounded mass, and a subsequence converges weakly, with no further tightness argument needed. Integrating against

$$k_s(v) \;=\; \frac{1 - (1-v)^s}{v},$$

which has a removable singularity at $v = 0$ with value s, gives

$$g_{x,y}(s) \;=\; b_0(x,y)\,s + \int_0^\infty \big(1 - e^{-st}\big)\,\nu_{x,y}(dt),$$

where the limit measure ρ on $[0,1]$ is read in three pieces. The atom at $v = 0$ gives the drift $b_0(x,y) := \rho(\{0\})$, because $k_s(0) = s$. On the interior $(0,1)$, the substitution $v \mapsto t = -\log(1-v)$ carries ρ back to a measure $\nu_{x,y}$ on $(0,\infty)$: the weight $1/v$ in k_s undoes the weighting by $1-e^{-t}$, and $1-(1-v)^s = 1-e^{-st}$, so $k_s(v)\,d\rho(v)$ on $(0,1)$ reads as $(1-e^{-st})\,d\nu_{x,y}(t)$ on $(0,\infty)$; finiteness of $\int_0^\infty (1\wedge t)\,\nu_{x,y}(dt)$ follows from the finiteness of ρ, since $1-e^{-t} \geq (1-e^{-1})(1\wedge t)$. The atom at $v = 1$ would be a killing term, and is excluded by $g_{x,y}(0) = 0$ (Lemma 4.2). So $g_{x,y} \in \mathrm{BF}_0$ by the representation itself. The case $x = y$ is $g_{x,x} = 0 \in \mathrm{BF}_0$. □

In particular every kernel is infinitely divisible: for each n, $e^{-g_{x,y}/n}$ is completely monotone with value 1 at 0+ (Proposition 3.4(2) and (1)), hence the Laplace transform of a probability measure on $[0,\infty)$, whose n-fold convolution is $\mu_{x,y}$.

Corollary 5.3 (strict monotonicity)**.** *Assume in addition (ND). Then for every $s > 0$ and $x < y$,*

$$G(y,s) - G(x,s) = g_{x,y}(s) > 0,$$

i.e. $G(\cdot, s)$ is strictly increasing for every $s > 0$.

Proof. By (ND), $\Phi_{x,y} \neq \mathrm{Id}$, so $\mu_{x,y} \neq \delta_0$ by the uniqueness clause of Lemma 4.1, and therefore $g_{x,y} \not\equiv 0$. By Theorem 5.2, $g_{x,y} \in \mathrm{BF}_0$; by Lemma 3.7 a member of BF_0 vanishing at a single point of $(0,\infty)$ vanishes identically, so $g_{x,y}(s) > 0$ for all $s > 0$. The displayed identity is Lemma 5.1. □

The corollary is the formal content of "the scale axis carries no idle stretches" (§2.5): for each $s > 0$ the accumulated exponent $G(\cdot, s)$ is injective. That injectivity is what makes the relabelling S_σ of (A8) rigid and the canonical gauge exist, in §6.

6 Scale covariance: the similarity form

Covariance now turns the free relabelling S_σ of (A8) into a coordinate: the action is rigid, its orbit through 1 parametrizes the scale axis, and in the resulting gauge the accumulated exponent is a dilate of a single function, $G(x,s) = F(\chi(x)s)$ — so that the exponent of any kernel is a difference of two dilates of F, which is the form §7 characterizes. In the optical reading of Remark 4.3, the section says: the covariance equation makes the medium *self-similar* — descending the scale axis is indistinguishable from probing faster — and the rigidity lemma says a self-similar medium has no marked layer. The gauge is the coordinate in which the self-similarity becomes homogeneity, one absorption spectrum F read through the dimensionless product $\chi(x)\,s$. The first lemma only translates (A8) into the language of kernels and exponents.

Lemma 6.1 (Laplace form of covariance)**.** *Under (A1)–(A5), the following are equivalent:*

1. *axiom (A8);*
2. *$D_\sigma\,\mu_{x,y} = \mu_{S_\sigma x,\, S_\sigma y}$ for all $0 \leq x \leq y$, $\sigma > 0$;*
3. *$g_{S_\sigma x,\, S_\sigma y}(s) = g_{x,y}(\sigma s)$ for all $0 \leq x \leq y$, $\sigma, s > 0$.*

In particular, under (A8),

$$G(S_\sigma x,\, s) = G(x,\, \sigma s) \qquad \text{for all } x \geq 0,\ \sigma, s > 0. \tag{6.1}$$

Proof. The compatibility $D_\sigma(\mu * f) = (D_\sigma\mu) * (D_\sigma f)$ turns (A8) into the statement that $D_\sigma\mu_{x,y}$ and $\mu_{S_\sigma x, S_\sigma y}$ represent the same operator, and the uniqueness clause of Lemma 4.1 makes that an identity of measures; this is (1)⇔(2). For (2)⇔(3), $\widehat{D_\sigma\mu}(s) = \hat{\mu}(\sigma s)$, so taking $-\log$ gives $g_{S_\sigma x, S_\sigma y}(s) = g_{x,y}(\sigma s)$. Equation (6.1) is the case $x = 0$ of that identity together with $S_\sigma 0 = 0$, which holds because an increasing bijection of $[0,\infty)$ fixes 0; here $G(x,s) = g_{0,x}(s)$, which needs only Lemma 4.1. □

What the translated axiom buys is rigidity. Nothing in (A8) says that the relabellings S_σ are unique, compose, or move every point; the next lemma says all three, and its engine is one fact — for fixed s the accumulated exponent $G(\cdot, s)$ is injective (Corollary 5.3) — so that (6.1) reads S_σ off G. Clause (1) is the claim made in §2.3, that the axioms with (ND) determine each S_σ.

Lemma 6.2 (rigidity of the action)**.** *Assume (A1)–(A8) and (ND). Then:*

1. *For each σ, S_σ is uniquely determined by (6.1).*
2. *$S_\sigma S_\tau = S_{\sigma\tau}$ for all $\sigma, \tau > 0$, and $S_1 = \mathrm{Id}$.*
3. *$\sigma \mapsto S_\sigma x$ is continuous and strictly increasing for each $x > 0$.*
4. *The action has no fixed point in $(0,\infty)$: if $S_\sigma x^* = x^*$ for all σ, then $x^* = 0$.*

Proof. (1) Uniqueness. If S and S' both satisfy (6.1) then $G(S_\sigma x, s) = G(x, \sigma s) = G(S'_\sigma x, s)$ for every $s > 0$. By Corollary 5.3, $G(\cdot, s)$ is strictly increasing, hence injective, so $S_\sigma x = S'_\sigma x$.

(2) The group law. Applying (6.1) twice,

$$G(S_\sigma S_\tau x, s) = G(S_\tau x, \sigma s) = G(x, \sigma\tau s) = G(S_{\sigma\tau} x, s),$$

and injectivity of $G(\cdot, s)$ gives $S_\sigma S_\tau = S_{\sigma\tau}$. Taking $\sigma = 1$ in (6.1) gives $G(S_1 x, s) = G(x, s)$, whence $S_1 = \mathrm{Id}$.

(3) Continuity and strict monotonicity in σ. Fix $x > 0$ and any $s_0 > 0$. By (6.1), $G(S_\sigma x, s_0) = G(x, \sigma s_0) = g_{0,x}(\sigma s_0)$.

The map $\sigma \mapsto g_{0,x}(\sigma s_0)$ is continuous, and it is *strictly* increasing: $g_{0,x}$ is a nonzero member of BF_0 by Corollary 5.3, and a nonzero member of BF_0 is strictly increasing, because its derivative is completely monotone and not identically zero, hence strictly positive on $(0,\infty)$ — a completely monotone function vanishing at one point vanishes beyond it, and vanishing on an interval forces the representing measure, and so the function, to be zero.

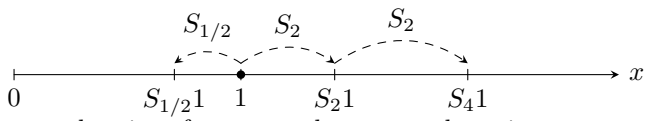
The map $G(\cdot, s_0)$ is continuous (Lemma 5.1) and strictly increasing (Corollary 5.3), so it has a continuous strictly increasing inverse on its range. Composing, $S_\sigma x = G(\cdot, s_0)^{-1}\big(g_{0,x}(\sigma s_0)\big)$ is continuous and strictly increasing in σ.

(4) No interior fixed point. Suppose $S_\sigma x^* = x^*$ for all $\sigma > 0$. Then (6.1) gives $G(x^*, s) = G(x^*, \sigma s)$ for all $\sigma, s > 0$, so $G(x^*, \cdot)$ is constant on $(0,\infty)$; its value is $G(x^*, 0+) = 0$ by Lemma 5.1. Hence $g_{0,x^*} \equiv 0$, i.e. $\mu_{0,x^*} = \delta_0$ and $\Phi_{0,x^*} = \mathrm{Id}$. By (ND) this forces $x^* = 0$. □

Three of the four clauses are injectivity of $G(\cdot, s)$ and nothing else; (3) is the one that needs more, namely that $g_{0,x}$ is strictly increasing in s. Clause (4) is where (ND) bites in this section: a fixed point x^* of the action would be a scale at which $\Phi_{0,x^*} = \mathrm{Id}$, an idle stretch. Without (ND) the orbit below could stall there and there would be no coordinate at all. §2.5 called (ND) a normalization; this is what the normalization is for.

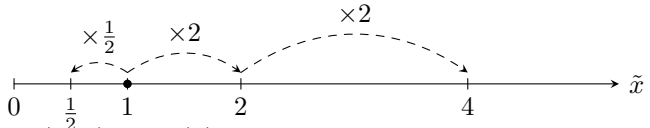
The coordinate itself is the orbit. Fix the point 1 on the scale axis and follow it under the action: $\sigma \mapsto S_\sigma 1$ is continuous, strictly increasing, and — having no fixed point to stall at — sweeps the whole axis. So every scale $x > 0$ is reached by exactly one σ, and labelling x by that σ straightens the action into multiplication (Figure 4).

(a) the scale axis as given: the orbit of 1 under S_σ



equal ratios of σ, unequal steps on the axis

(b) the canonical gauge $\tilde{x} = \chi(x)$: the same action is multiplication



$\chi(S_\sigma x) = \sigma\,\chi(x)$: equal ratios of σ, equal ratios on the axis

Figure 4: The orbit coordinate. (a) On the scale axis as the family presents it, the action S_σ of (A8) is some increasing bijection; the orbit of the point 1, $\sigma \mapsto S_\sigma 1$, is continuous, strictly increasing and has no fixed point, so it sweeps the whole axis (Lemma 6.2, Proposition 6.3). Equal ratios of σ are unequal steps on the axis. (b) Labelling each point by the σ that reaches it, $\tilde{x} = \chi(x)$, the same action becomes multiplication by σ: the canonical gauge. In this gauge every accumulated exponent is a dilate of one function, $G(x, s) = F(\tilde{x}s)$.

Proposition 6.3 (orbit coordinate; the gauge)**.** *Assume (A1)–(A8) and (ND). Then* $c(\sigma) := S_\sigma 1$ *is a continuous strictly increasing bijection of* $(0,\infty)$ *onto* $(0,\infty)$ *with* $c(1) = 1$*. Consequently*

$$\chi := c^{-1} : (0,\infty) \to (0,\infty), \qquad \chi(0) := 0,$$

is an increasing bijection — the canonical gauge — *satisfying* $\chi(S_\sigma x) = \sigma\,\chi(x)$*, and*

$$G(x,s) = F\big(\chi(x)\,s\big), \qquad F := G(1,\cdot) \in \mathrm{BF}_0,\ F \not\equiv 0. \tag{6.2}$$

Equivalently $F \in \mathrm{LE}$*, with a drift* $b_0 \geq 0$ *and a Lévy measure* ν *on* $(0,\infty)$*.*

Proof. *c is continuous* by Lemma 6.2(3), and $c(1) = S_1 1 = 1$ by Lemma 6.2(2).

c is injective. Suppose $c(\sigma) = c(\tau)$ with $\rho := \sigma/\tau \neq 1$. The group law 6.2(2) gives $S_\rho 1 = 1$, so by (6.1)

$$F(s) = G(1,s) = G(S_\rho 1, s) = G(1,\rho s) = F(\rho s) \qquad \text{for all } s > 0.$$

Iterating along $\rho^{\pm n}$ gives $F(s) = F(\rho^n s)$ for every $n \in \mathbb{Z}$. Replacing ρ by ρ^{-1} if necessary we may take $\rho < 1$, and then $\rho^n s \to 0$, so $F(s) = F(0+) = 0$ for every s by Lemma 5.1. Thus $F \equiv 0$, which contradicts Corollary 5.3 under (ND).

c is increasing. Being continuous and injective on the connected set $(0,\infty)$, c is strictly monotone, so it suffices to exclude the decreasing case. Suppose c were decreasing and fix $\sigma > 1$, so that $S_\sigma 1 < 1$. Since $G(\cdot, s)$ is strictly increasing (Corollary 5.3),

$$G(S_\sigma 1, s) < G(1,s) = F(s) \qquad \text{for every } s > 0.$$

But (6.1) evaluates the left-hand side as $G(1,\sigma s) = F(\sigma s)$, and F is nondecreasing, being a Bernstein function, so $F(\sigma s) \geq F(s)$. The two are incompatible, so c is increasing.

c is onto $(0,\infty)$. Its limits as $\sigma \to 0+$ and $\sigma \to \infty$ exist in $[0,\infty]$ by monotonicity, and each is a fixed point of the action or a boundary point; by Lemma 6.2(4) there is no fixed point in $(0,\infty)$, so the limits are 0 and ∞. With continuity, c is a bijection onto $(0,\infty)$.

The gauge relation. By the group law, $S_\sigma x = S_\sigma S_{\chi(x)} 1 = S_{\sigma\chi(x)} 1$, so $\chi(S_\sigma x) = \sigma\chi(x)$.

The similarity form. For $x > 0$, writing $x = S_{\chi(x)}1$ and using (6.1),

$$G(x,s) = G\big(S_{\chi(x)}1,\ s\big) = G\big(1,\ \chi(x)\,s\big) = F\big(\chi(x)\,s\big),$$

and $F = g_{0,1} \in \mathrm{BF}_0$ by Theorem 5.2, with $F \not\equiv 0$ by (ND) through Corollary 5.3. The case $x = 0$ is $G(0,\cdot) = 0 = F(0)$. □

In the semigroup case the straightening is the classical reparametrization: in the semigroup's own additive parameter τ the accumulated exponent is $G(\tau, s) = \tau s^\alpha$, so the action is $S_\sigma \tau = \sigma^\alpha \tau$ and the gauge is $\chi(\tau) = \tau^{1/\alpha}$; in the straightened coordinate $x = \tau^{1/\alpha}$ the action is $S_\sigma x = \sigma x$ and $G = F(xs)$ with $F(s) = s^\alpha$ (Corollary 7.4; Remark A.23 makes the identification $\tau = x^\alpha$ explicit), the two coordinates coinciding only for the pure delay $\alpha = 1$. The proposition says that the hemigroup case differs from this only by the generality of the reparametrization and the freedom in F.

Remark 6.4 (gauge freedom). In the hemigroup setting the scale coordinate carries no intrinsic parametrization: replacing x by any increasing bijection of $[0,\infty)$ yields an equivalent family. Proposition 6.3 fixes the *canonical gauge* $\tilde{x} := \chi(x)$, in which S_σ acts by multiplication and $G(x,s) = F(\tilde{x}\,s)$. In it, scale carries the dimension of time: s is dual to time, and $\tilde{x}s$ must be dimensionless for $F(\tilde{x}s)$ to make sense. The canonical gauge is the working gauge of the rest of the paper.

The other homogeneous gauges are $c\,\tilde{x}^{\,\alpha}$, $c, \alpha > 0$ (the *α-gauge*). One of them recurs, for the Bessel family of §8–10: the *parabolic gauge* ξ, defined by

$$\tilde{x} = \xi^2,$$

under which that family's memory-line evolution is exactly the Bessel operator $\frac{1}{2}\,\partial_\xi^2 + \frac{\beta}{\xi}\,\partial_\xi$ — the heat-equation form; the factor $\frac{1}{2}$ arises from the substitution itself, as for Brownian motion at $a = \frac{1}{2}$. It is named at each use and never written $\tilde{x}$.

7 Self-decomposability and the characterization theorem

The two preceding sections reduce the family to one function. By Lemma 5.1 the exponent of a kernel is a difference of accumulated exponents, $g_{x,y} = G(y,\cdot) - G(x,\cdot)$, and by Proposition 6.3 the accumulated exponent is a dilate of one function, $G(x,s) = F(\chi(x)s)$. So every kernel is read off F by stretching its argument and subtracting:

$$g_{x,y}(s) \;=\; F(bs) - F(as), \qquad a = \chi(x) \le b = \chi(y),$$

and since χ is onto, (a,b) ranges over all pairs $0 \le a \le b$. Theorem 5.2 requires each of these differences to be a Bernstein function vanishing at the origin. That is a condition on F alone — not that F is Bernstein, which it is, but that *every stretched difference* $F(b\,\cdot) - F(a\,\cdot)$ is — and it is exactly the condition this section solves.

It has a probabilistic name. Let T_1 be the random delay whose law is the kernel at canonical scale 1, so that $\mathbb{E}\,e^{-sT_1} = e^{-F(s)}$; in the canonical gauge the kernel at scale x is then the law of xT_1. That $F(s) - F(cs)$ is a Laplace exponent for $c \in (0,1)$ says that the law of T_1 is the convolution of the law of cT_1 with some probability law: there is a random variable R_c, independent of T_1, with

$$T_1 \;\stackrel{d}{=}\; c\,T_1 + R_c \qquad \text{for every } c \in (0,1).$$

A law with this property — shrink it by any factor, and what was removed is itself a random variable independent of the shrunk copy — is called *self-decomposable*, or of class L ([32], Def. 15.1, p. 90; §3). It is the hemigroup form of infinite divisibility: in the semigroup case the

kernel at scale τ is the n-fold convolution power of the kernel at τ/n, all stages identical; here the stages differ, but covariance turns a shrinking of the scale axis into a shrinking of the delay, and the cascade requires what remains, the passage from scale c to scale 1, to be a measurement in its own right — that is R_c. The lemma below is the analytic form of this statement: the classical characterization of self-decomposable laws by their Lévy measure ([32], Cor. 15.11, p. 95), which is what makes F concrete.

Lemma 7.1 (self-decomposable exponents)**.** *For $F \in \mathrm{BF}_0$ the following are equivalent:*

1. *$F(b\,\cdot) - F(a\,\cdot) \in \mathrm{BF}_0$ for all $0 < a \le b$;*
2. *$B(s) := s\,F'(s)$ is a Bernstein function;*
3. *F has the representation*

$$F(s) = b_0\, s + \int_0^\infty \bigl(1 - e^{-st}\bigr)\,\frac{k(t)}{t}\,dt, \qquad b_0 \ge 0, \tag{7.1}$$

with $k : (0,\infty) \to [0,\infty)$ nonincreasing, $\int_0^1 k < \infty$ and $\int_1^\infty k(t)/t\,dt < \infty$.

Proof. (1) $\Rightarrow$ (2). For $h > 0$ the difference quotient $s \mapsto \bigl(F(e^h s) - F(s)\bigr)/h$ lies in BF, being a positive multiple of an element of BF_0. As $h \downarrow 0$ it converges pointwise to $sF'(s)$, since F is smooth on $(0,\infty)$; the limit is therefore Bernstein by Proposition 3.4(4).

(2) $\Leftrightarrow$ (3). Write $F(s) = b_0 s + \int (1 - e^{-st})\,\nu(dt)$, with no killing term because $F(0+) = 0$ (Proposition 3.4(3)), so that

$$B(s) = sF'(s) = b_0 s + \int_0^\infty s\,t\,e^{-st}\,\nu(dt).$$

Note $B(0+) = 0$, since $0 \le sF'(s) \le F(s)$ by concavity.

Suppose $\nu(dt) = k(t)\,t^{-1}dt$ with k nonincreasing. Write $k(t) = \rho\bigl((t,\infty)\bigr)$ for the positive measure $\rho = -dk$; right-continuity of k may be assumed, and $k(\infty) = 0$ is forced by $\int^\infty k(t)t^{-1}dt < \infty$. Tonelli then gives

$$\int_0^\infty s\,e^{-st}\,k(t)\,dt = \int_0^\infty \rho(du) \int_0^u s\,e^{-st}\,dt = \int_0^\infty \bigl(1 - e^{-su}\bigr)\,\rho(du),$$

so $B(s) = b_0 s + \int (1 - e^{-su})\rho(du) \in \mathrm{BF}_0$.

Conversely, if $B \in \mathrm{BF}$ then $B \in \mathrm{BF}_0$ by the remark above, so $B(s) = \beta s + \int (1 - e^{-su})\rho(du)$. Running the same computation backwards and invoking uniqueness of the Lévy–Khintchine triple (Proposition 3.4(3)) identifies $\beta = b_0$ and $\nu(dt) = k(t)t^{-1}dt$ with $k(t) = \rho((t,\infty))$ nonincreasing.

(3) $\Rightarrow$ (1). Write $F(cs) = b_0 cs + \int_0^\infty \bigl(1 - e^{-cst}\bigr)\,k(t)\,\frac{dt}{t}$ for $c > 0$ and substitute $u = ct$. The measure dt/t is invariant under dilations, so

$$F(cs) = b_0 c\, s + \int_0^\infty \bigl(1 - e^{-su}\bigr)\,k(u/c)\,\frac{du}{u},$$

i.e. dilating the argument by c dilates the density and rescales the drift, and does nothing else. Subtracting the identity at $c = a$ from the one at $c = b$,

$$F(bs) - F(as) = b_0(b-a)\,s + \int_0^\infty \bigl(1 - e^{-su}\bigr)\,\bigl(k(u/b) - k(u/a)\bigr)\,\frac{du}{u},$$

which lies in BF_0: the drift $b_0(b-a)$ is nonnegative since $a \le b$, the density is nonnegative since $u/b \le u/a$ for $u > 0$ and k is nonincreasing, and both integrals are finite, each being bounded by $F(bs) < \infty$. Vanishing at $0+$ is monotone convergence. □

The three conditions are not equally deep, and the asymmetry organizes what follows. The implication (3) ⇒ (1) is elementary — a change of variables in the representation, using no property of the CM class — and it is all that the constructive direction of Theorem 7.3 consumes. The legs (1) ⇒ (2) ⇒ (3) are different in kind: $B(s) = sF'(s)$ has no meaning in the representation vocabulary until the representation is in hand, and both legs spend the classical theory — closure of BF under pointwise limits, and uniqueness of the Lévy–Khintchine triple. This is why the constructive direction of the main theorem is cheap and the analytic one is not. The printed proofs take the classical route: the constructive direction of the theorem reaches its measures by Bernstein–Widder, and the analysis direction by the legs (1) ⇒ (2) ⇒ (3). The machine-checked proofs reach the constructive direction with no cited fact, the subordinator being built by a compound-Poisson construction, and the analysis direction by the one cited fact of §1.1; it is that asymmetry which §1.1 certifies.

Condition (2) has a reading that the section's tail supplies. By Lemma 7.10 below, F' is the drift plus the Laplace transform of the nonincreasing function k, so $B(s) = sF'(s)$ is s times that transform; and since $\partial_x F(xs) = sF'(xs)$, it is the symbol of the evolution of the family across scale. Condition (2) says that this symbol is itself a Bernstein function — the evolution across scale is, at each scale, again a subordination.

Remark 7.2 (the probabilistic reading). Probabilistically, (7.1) together with the similarity structure below says that the kernels are the marginal laws of a *self-similar additive process*, a Sato subordinator: an increasing process $(T_{\tilde{x}})_{\tilde{x}\geq 0}$ with independent — but not stationary — increments, satisfying $T_{\sigma\tilde{x}} \overset{d}{=} \sigma\, T_{\tilde{x}}$.

Theorem 7.3 (Main theorem)**.** *A family $(\Phi_{x,y})_{0\leq x\leq y}$ satisfies (A1)–(A8) and (ND)* ***iff*** *there exist an increasing bijection χ of $[0,\infty)$ — the gauge — and a function F of the form (7.1) with $F \not\equiv 0$ such that*

$$\Phi_{x,y} f = \mu_{x,y} * f, \qquad \hat{\mu}_{x,y}(s) = \exp\Big[-\big(F(\chi(y)\, s) - F(\chi(x)\, s)\big)\Big].$$

In particular, in the canonical gauge the scale-space kernels are

$$\phi_x(t) \;=\; \mathcal{L}^{-1}\Big[\, e^{-F(x\, s)} \,\Big](t), \qquad F \text{ as in } (7.1),$$

and the pair (χ, F) is unique up to the normalization $\chi(1) = 1$.

Read against §2: the apertures are the marginal laws $\mu_{0,x}$ of one Sato subordinator, the kernel at canonical scale x being the law of xT_1, and the whole family is generated by one law and the gauge. What the axioms leave free is exactly the pair (b_0, k) in (7.1): a deterministic front, and a nonincreasing density.

Proof. (⇒) Lemmas 4.1, 4.2 and 5.1, Theorem 5.2, Corollary 5.3, Lemmas 6.1 and 6.2, and Proposition 6.3 together give the representation with $F = G(1,\cdot) \in \mathrm{BF}_0$ and $F \not\equiv 0$. Every increment $F(\chi(y)s) - F(\chi(x)s)$ lies in BF_0 by Theorem 5.2, and χ is onto $(0,\infty)$ by Proposition 6.3, so condition (1) of Lemma 7.1 holds for every pair $0 < a \leq b$, and the lemma yields (7.1).

(⇐) Let F be of the form (7.1) with $F \not\equiv 0$; work in the canonical gauge. For $0 \leq a \leq b$ the function $g_{a,b}(s) := F(bs) - F(as)$ lies in BF_0 by Lemma 7.1, (3) ⇒ (1) (for $a = 0$ the increment is $F(b\,\cdot)$ itself, which is in BF_0 by hypothesis), so $e^{-g_{a,b}}$ is completely monotone with value 1 at 0+ (Proposition 3.4(2)), and by the Bernstein–Widder criterion (Proposition 3.4(1)) it is the Laplace transform of a unique probability measure $\mu_{a,b}$ on $[0,\infty)$. Define $\Phi_{a,b} f := \mu_{a,b} * f$. Then: (A1)–(A5) hold by the converse direction of Lemma 4.1; (A6) holds because exponents add, $g_{a,b} + g_{b,c} = g_{a,c}$, and Laplace transforms determine measures (Proposition 3.9); (A8) holds with $S_\sigma x = \sigma x$, since $\hat{\mu}_{\sigma a,\sigma b}(s) = \hat{\mu}_{a,b}(\sigma s)$; and (ND) holds because F is strictly increasing, by Lemma 3.7 applied to the increments of F as in Corollary 5.3.

For (A7): if $(a_n, b_n) \to (a, b)$ then $\hat{\mu}_{a_n,b_n} \to \hat{\mu}_{a,b}$ pointwise, by continuity of F, so $\mu_{a_n,b_n} \to \mu_{a,b}$ weakly by the continuity theorem (Proposition 3.8). Weak convergence of probability measures implies $\mu_{a_n,b_n} * f \to \mu_{a,b} * f$ in $L^1(\mathbb{R}, dt)$ for every $f \in X$: writing $\mu * f = \int \tau_r f \, \mu(dr)$ as an X-valued Bochner integral, the map $r \mapsto \tau_r f$ is bounded and continuous, and an $\varepsilon/3$ argument on a compact carrying mass $1-\varepsilon$ concludes, first for compactly supported continuous f and then for all of X by density, the error not growing because each $\Phi_{a,b}$ is a contraction.

Finally, for an arbitrary gauge χ set $\Phi'_{x,y} := \Phi_{\chi(x),\chi(y)}$: (A1)–(A6) and (ND) hold verbatim, (A7) because an increasing bijection of $[0, \infty)$ is continuous, and (A8) with $S'_\sigma = \chi^{-1}(\sigma\, \chi(\cdot))$.

Uniqueness. The exponents $g_{0,x}$ are determined by the family (Proposition 3.9, with exp injective), and $\chi(1) = 1$ recovers $F = g_{0,1}$ (without the normalization the pair is determined up to $(\chi, F) \mapsto (c\,\chi,\ F(c^{-1}\,\cdot))$, $c > 0$, which is the freedom the normalization removes). For each x, $\chi(x)$ is then the unique c with $g_{0,x} = F(c\,\cdot)$: such a c exists by the representation, and if $F(cs) = F(c's)$ for all s with $\rho := c/c' \neq 1$, then iterating $F(s) = F(\rho^{\pm n} s)$ towards the origin and using $F(0+) = 0$ gives $F \equiv 0$, a contradiction. □

Corollary 7.4 (recovering the semigroup case)**.** *Assume in addition that the family is one-parameter: $\Phi_{x,y}$ depends only on $y - x$. Then, after the normalization $g_{0,1}(1) = 1$,*

$$F(s) = s^\alpha \quad \textit{for some } 0 < \alpha \leq 1,$$

i.e. the kernels are exactly the extremal stable densities of [8], Thm. 2, together with the boundary case $\alpha = 1$ — the pure delay $\mu_{0,x} = \delta_x$.

Proof. Homogeneity gives $g_{x,y} = (y - x)\, g$ with $g := g_{0,1}$, and (6.1) becomes $(S_\sigma y - S_\sigma x)\, g(s) = (y - x)\, g(\sigma s)$. Hence S_σ is affine, and fixing 0 it is linear: $S_\sigma x = c(\sigma)x$ with $c(\sigma)g(s) = g(\sigma s)$.

Then c is a continuous multiplicative homomorphism of $(0, \infty)$, so $c(\sigma) = \sigma^\alpha$ by Cauchy's functional equation, as in [8], Lemma 2. Now $g(\sigma s) = \sigma^\alpha g(s)$ forces $g(s) = g(1)s^\alpha$, and the normalization makes $g(1) = 1$. Membership of BF_0 restricts α to $(0, 1]$. The value $\alpha = 1$ gives $sF'(s) = s \in \mathrm{BF}$, so Lemma 7.1 admits it and it is not excluded by the present axioms. □

Remark 7.5 (the boundary case $\alpha = 1$ and the drift term). [8] excluded $\alpha = 1$ through the pointwise-continuity requirement on the kernel functions, which the deterministic delay $\delta(t - x)$ violates. Under Definition 2.1 the delay family — and more generally the drift term $b_0 s$ in (7.1), which shifts every kernel by $b_0 x$ — is admissible.

If that is undesired it is removed by one additional axiom: absolute continuity of the kernels for $x > 0$, or the requirement $\lim_{s\to\infty} F(s)/s = 0$, which says there is no deterministic front.

Lemma 7.6 (the admissible exponents form a convex cone)**.** *If F_1, F_2 are admissible in the sense of Theorem 7.3 — of the form (7.1) with drift $b_0 \geq 0$ and $k \geq 0$ nonincreasing — then so are $F_1 + F_2$, with data $(b_0^1 + b_0^2,\ k_1 + k_2)$, and $c\,F_1$ for $c \geq 0$, with data $(c\,b_0^1,\ c\,k_1)$. So the admissible exponents form a convex cone in the pointwise order, and the correspondence with the pairs (b_0, k) is linear.*

Proof. Each of the six defining conditions is stable: nonnegativity and monotonicity of k are preserved by sums and by nonnegative multiples, the drift stays nonnegative, and the exponent of a sum is the sum of the exponents by additivity of the integral — so finiteness is inherited. Linearity of $(b_0, k) \mapsto F$ is the same computation read forwards. □

Lemma 7.7 (Dickman superposition)**.** *Write* $\mathrm{Ein}(z) := \int_0^z \big(1 - e^{-u}\big)\, u^{-1} du$.

1. (The rays.) *For $\tau > 0$ the density $k = \mathbf{1}_{(0,\tau)}$ is admissible, with drift 0, and its exponent is $s \mapsto \mathrm{Ein}(\tau s)$.*

2. (The superposition.) *Every admissible F, with drift b_0 and density k, satisfies*

$$F(s) \;=\; b_0\, s \;+\; \int_{(0,\infty)} \mathrm{Ein}(\tau s)\, \rho(d\tau) \qquad (s \geq 0),$$

where ρ is any positive measure on $(0,\infty)$ with $\rho\big((t,\infty)\big) = k(t)$ for almost every $t > 0$; such a ρ exists.

Proof. (1) is the substitution $u = st$ in $\int_0^\tau (1 - e^{-st})\, t^{-1} dt$; admissibility is the integrability criterion of Proposition 8.1, both halves being immediate for a bounded, compactly supported k.

(2) k is nonincreasing and nonnegative and $k(\infty) = 0$, the last forced by $\int^\infty k(t)\, t^{-1} dt < \infty$; so a positive measure with tail k exists. Then Tonelli over the region $\{0 < t < \tau\}$ gives

$$\int_0^\infty \big(1 - e^{-st}\big)\, \frac{k(t)}{t}\, dt = \int_{(0,\infty)} \rho(d\tau) \int_0^\tau \big(1 - e^{-st}\big)\, \frac{dt}{t} = \int_{(0,\infty)} \mathrm{Ein}(\tau s)\, \rho(d\tau),$$

the inner integral being $\mathrm{Ein}(\tau s)$ by (1). □

Proposition 7.8 (Choquet structure of the admissible cone)**.** *The superposition map of Lemma 7.7,*

$$(b_0, \rho) \;\mapsto\; F, \qquad F(s) = b_0\, s + \int_{(0,\infty)} \mathrm{Ein}(\tau s)\, \rho(d\tau),$$

is a linear bijection *from $\big\{(b_0,\rho) : b_0 \geq 0,\ \rho$ a positive Borel measure on $(0,\infty)$, $\int \big[\tau \wedge (1 + \log_+ \tau)\big]\, \rho(d\tau) < \infty\big\}$ onto the cone of Lemma 7.6, with inverse $\rho = -dk$, $k(t) = \rho\big((t,\infty)\big)$. Consequently the extreme rays of the cone are the pure delay $s \mapsto s$ together with the one-parameter family $\mathrm{Ein}(\tau\,\cdot)$, $\tau > 0$ — the Laplace exponents, with Lévy density $t^{-1}\mathbf{1}_{(0,\tau)}(t)$, of the* dilated Dickman subordinators *([3], (1.1)–(1.2), p. 1) — and every admissible family is a* unique *superposition of Dickman rays and drift.*

Proof. By Theorem 7.3 an admissible F is exactly one of the form (7.1) with $b_0 \geq 0$ and k nonincreasing, and k may be taken right-continuous with $k(\infty) = 0$, the latter forced by $\int^\infty k(t)\, t^{-1} dt < \infty$. So $\rho := -dk$ is a positive measure on $(0,\infty)$ with $k(t) = \rho((t,\infty))$, and Tonelli gives

$$\int_0^\infty \big(1 - e^{-st}\big)\, \frac{k(t)}{t}\, dt = \int_{(0,\infty)} \rho(d\tau) \int_0^\tau \big(1 - e^{-st}\big)\, \frac{dt}{t} = \int_{(0,\infty)} \mathrm{Ein}(\tau s)\, \rho(d\tau),$$

which is the displayed representation. Since $\mathrm{Ein}(z) \leq z$ and $\mathrm{Ein}(z) = \log z + \gamma_E + o(1)$, γ_E Euler's constant, as $z \to \infty$, the integral is finite for one $s > 0$ iff it is finite for all, iff $\int [\tau \wedge (1 + \log_+ \tau)]\, \rho(d\tau) < \infty$; so the stated domain is exactly the set of pairs the representation accepts. The map is linear in (b_0, ρ) by construction and surjective by the above. It is injective because the Lévy–Khintchine triple of F is unique (Proposition 3.4(3)): the triple determines b_0 and the Lévy measure $k(t)t^{-1}dt$, hence k up to a null set, hence $\rho = -dk$.

A linear bijection carries extreme rays to extreme rays, so it suffices to identify them in the domain, which is the product of the ray $[0,\infty)$ with a cone of positive measures. Its extreme rays are $(1,0)$ and $(0,\delta_\tau)$, $\tau > 0$: any (b_0,ρ) decomposes uniquely as $b_0\,(1,0) + \int (0,\delta_\tau)\, \rho(d\tau)$, and a pair is extreme only if it admits no nontrivial such splitting, which forces $\rho = 0$ or ρ a single atom with $b_0 = 0$. Under the map, $(1,0) \mapsto s$, the pure delay; and $(0,\delta_\tau) \mapsto \mathrm{Ein}(\tau s)$, whose $k(t) = \delta_\tau((t,\infty)) = \mathbf{1}_{(0,\tau)}(t)$ gives the Lévy density $t^{-1}\mathbf{1}_{(0,\tau)}(t)$. □

Remark 7.9 (the three corners are slices, not extremes). None of the distinguished families of this paper is extreme; each is a slice cut by a different principle — the stable family by dilation symmetry (Corollary 7.4, Proposition 8.5), the Gamma family as the atoms of the Thorin

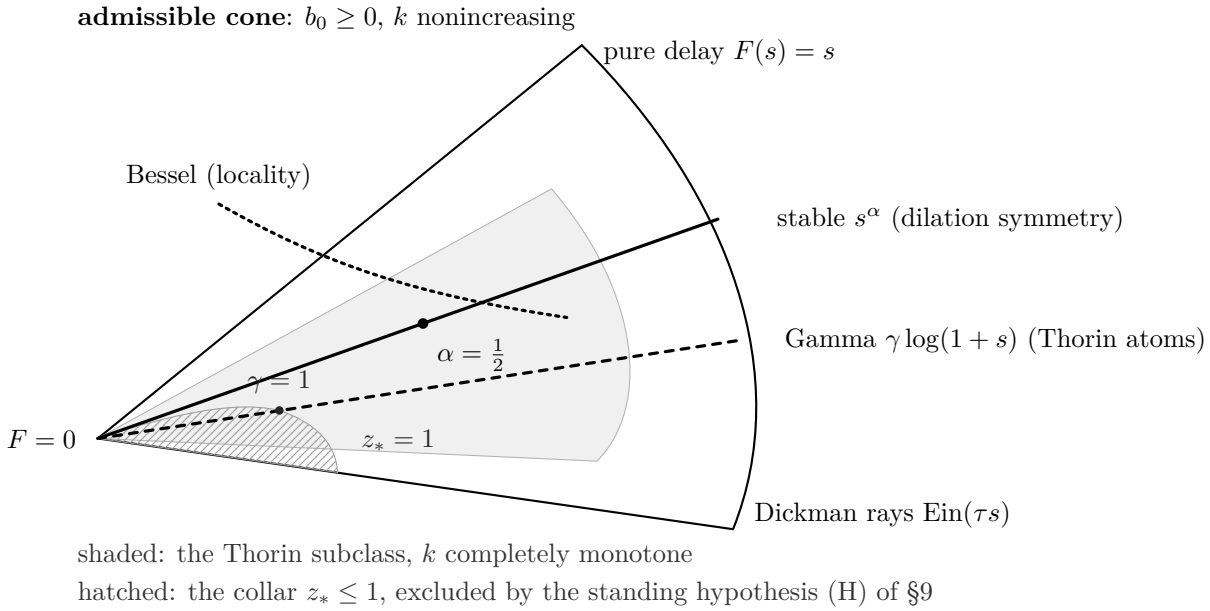


Figure 5: The admissible cone. Theorem 7.3 identifies the exponents F with pairs (b_0, k), a drift and a nonincreasing density; these form a convex cone (Lemma 7.6) whose extreme rays are the Dickman rays $\mathrm{Ein}(\tau\,\cdot)$ and the pure delay (Proposition 7.8). Inside it, the exponents with completely monotone k form the Thorin subclass. None of the three distinguished families of the paper is extreme; each is a slice cut by a different principle — the stable exponents s^α by dilation symmetry (Corollary 7.4), the Gamma family as the atoms of the Thorin representation, the Bessel family by locality (§10) — and the stable and Bessel slices meet at $\alpha = \frac{1}{2}$, the heat-signaling member, which is the one point the semigroup axioms allowed on the memory line. The hatched collar at the apex is excluded by the standing hypothesis (H) of §9 (Definition 9.1). Unlike the Thorin condition, which selects rays and so cuts a subcone, (H) cuts *across* rays: z_* is homogeneous of degree one along them (Lemma 9.4), so a ray of finite z_* enters (H) beyond the threshold multiple $1/z_*$, and the excluded set $\{z_* \leq 1\}$ is a collar around the apex — pinching to nothing on every exponent with drift and on the stable and Bessel slices, where $z_* = \infty$, reaching the unit point of each Dickman ray, where $z_* = k(0^+) = 1$, and crossing the Gamma ray at $\gamma = 1$. The drawing is schematic: the cone is infinite-dimensional.

representation (Proposition 8.8), the Bessel family by locality (Proposition 8.17, Theorem 10.6). Since all three have completely monotone k, the closed convex cone the three corners generate together is exactly the Thorin subclass — strictly smaller than the whole class, the difference being the memory kernels that are monotone but not completely monotone (plateaus, kinks, hard cutoffs). The Dickman rays are the paradigm of that difference: their kernels solve the delay equation obtained from the Volterra identity of Appendix A with the step kernel $k = \mathbf{1}_{[0,\tau)}$, and carry derivative singularities at the delays $\tau, 2\tau, \ldots$ — echo signatures unattainable within the Thorin subclass.

Two bounds hold throughout the cone off the drift ray. If $k \not\equiv 0$, every kernel $\mu_{0,x}$, $x > 0$, is a nondegenerate self-decomposable law, hence absolutely continuous ([32], Thm. 27.13) and *unimodal*, by Yamazato's theorem on class L laws ([43], Thm. 1, p. 523); on the drift ray $k \equiv 0$ the kernel is the point mass $\delta_{b_0 x}$. The axioms never produce multimodal kernels. And every kernel with $k \not\equiv 0$ is infinitely divisible and carried by $[0, \infty)$, so its tail is never lighter than $e^{-c\,t \log t}$ ([32], Thm. 26.1, p. 168): no truncation of the Gaussian is an admissible kernel among these, which is the claim made in §2; the drift ray is the point mass above, whose tail is of course lighter than any such bound.

The representation (7.1) attaches to every admissible family a nonincreasing function k,

and one identity fixes its analytic role. It is recorded here because the moment formulas of §8 already use it; the reading it opens — k as the memory kernel of the evolution across scale — is developed in Appendix A.

Lemma 7.10 (the kernel behind F')**.** *Let F be as in (7.1). Then*

$$F'(s) \;=\; b_0 \;+\; \int_0^\infty e^{-st}\, k(t)\, dt, \qquad s > 0.$$

Proof. Differentiating the integrand of (7.1) in s gives $e^{-st}k(t)$, which for $s \geq s_0 > 0$ is dominated by $e^{-s_0 t}k(t)$. That dominating function is integrable near 0 because $\int_0^1 k < \infty$, and at infinity because k is bounded there, being nonincreasing; so differentiation under the integral sign is justified. □

Thus F' is, up to the drift, the Laplace transform of k. Appendix A reads this as the statement that k — in L^1-normalized dilation to the current scale, $\kappa^{(x)}(dt) := b_0\,\delta_0(dt) + \frac{1}{x}\,k(\frac{t}{x})\,dt$ — is the memory kernel of the evolution across scale. The monotonicity of k, which entered abstractly as condition (3) of Lemma 7.1, is precisely the admissibility of $\kappa^{(x)}$ as a Lévy tail. In the extremal stable case $k(t) = \alpha t^{-\alpha}/\Gamma(1-\alpha)$, and the operator $\partial_t(\kappa^{(x)} * \cdot)$ is the Riemann–Liouville derivative $D^\alpha_{t,+}$ of [8], up to the scale factor $\alpha x^{\alpha-1}$ that Remark A.23 absorbs into the semigroup's own parametrization; in the Gamma family $k(t) = \gamma e^{-t}$, a bounded kernel.

8 Examples and moments

Throughout this section we work in the canonical gauge, so that the kernel at scale x is $\mu_{0,x} = \mathcal{L}^{-1}[e^{-F(xs)}]$, with density ϕ_x where it has one, and T_x denotes a random variable with law $\mu_{0,x}$ — the delay at scale x, so that $T_x \overset{d}{=} xT_1$. The section does three things. It states once the one criterion by which every family below is admissible, and its converse, which makes the two integrability conditions of (7.1) characteristic of the class rather than an artifact of an estimate. It gives the mean delay and the moment criterion, the two quantities by which the families are then compared. And it works the three families the rest of the paper uses, each the worked case of one of the slices in Figure 5: the stable family, cut by dilation symmetry, which is the semigroup theory; the Gamma family, the atoms of the Thorin representation, which is what §11 implements; and the Bessel family, cut by locality, which is the subject of §§9–10.

Proposition 8.1 (admissibility criterion for a memory kernel)**.** *Let $b_0 \geq 0$ and let $k : (0,\infty) \to [0,\infty)$ be nonincreasing with*

$$\int_0^1 k(t)\, dt < \infty \qquad \text{and} \qquad \int_1^\infty \frac{k(t)}{t}\, dt < \infty.$$

Then $F(s) = b_0 s + \int_0^\infty (1 - e^{-st})\, k(t)\, \frac{dt}{t}$ is finite for every $s \geq 0$.

Consequently F is of the form (7.1), and if $F \not\equiv 0$ Theorem 7.3 applies to it.

Proof. Split the integral at $t = 1$. On $(0,1]$ use $1 - e^{-st} \leq st$, which is $1 + x \leq e^x$ at $x = -st$; the integrand is then at most $s\,k(t)$, integrable by the first hypothesis. On $(1,\infty)$ use $1 - e^{-st} \leq 1$; the integrand is at most $k(t)/t$, integrable by the second. The drift term is finite for every real s. □

Lemma 8.2 (the criterion is characteristic)**.** *Let $b_0 \geq 0$ and let $k : (0,\infty) \to [0,\infty)$ be nonincreasing. If the $[0,\infty]$-valued exponent (7.1) at $s = 1$,*

$$b_0 + \int_0^\infty (1 - e^{-t})\, k(t)\, \frac{dt}{t},$$

is finite, then $\int_0^1 k(t)\, dt < \infty$ and $\int_1^\infty k(t) t^{-1} dt < \infty$.

Proof. Take $s = 1$ in (7.1), so that $\int_0^\infty (1 - e^{-t})k(t)t^{-1}dt < \infty$. For $t > 0$, $1 - e^{-t} \geq t/(1 + t)$, so the integrand is at least $k(t)/(1 + t)$. On $(0, 1]$ that is at least $k(t)/2$, and on $(1, \infty)$ it is at least $k(t)/(2t)$; both claims follow. □

In particular every admissible F carries both integrability facts already, so nothing below assumes regularity of k beyond the definition.

The first quantity to compare families by is the mean delay.

Proposition 8.3 (the mean delay and the influence curve)**.** $\mathbb{E}\,T_x = x\,F'(0+) = x\big(b_0 + \int_0^\infty k(t)\,dt\big)$*, finite if and only if* k *is integrable at infinity.*

Proof. Differentiating the transform at the origin, $\mathbb{E}\,T_x = -\partial_s e^{-F(xs)}\big|_{s=0+} = xF'(0+)$, and monotone convergence in (7.1) gives $F'(0+) = b_0 + \int_0^\infty k(t)\,dt$, finite exactly when k is integrable at infinity. □

The *influence curve* of [8], §8, p. 106, is the distance $t_0 - t_m$ between the time t_0 at which a measurement is made and the time t_m with the largest influence on it, as a function of scale. There t_m had to be the mode of the kernel — whose scaling is Proposition 8.7 — the mean of the stable kernels being infinite (*ibid.*, p. 102). Whenever the mean is finite it can be taken as $t_0 - t_m$ instead, and the proposition says that in the canonical gauge the influence curve is then exactly linear, $t_0 - t_m = \mathbb{E}\,T_x = F'(0+)\,x$. The condition "$k$ integrable at infinity" is the case $n = 1$ of the general criterion, which runs through one cited equivalence.

Proposition 8.4 (the moment criterion)**.** $\mathbb{E}\,T_x^n < \infty$ *if and only if* $\int_1^\infty t^{\,n-1}k(t)\,dt < \infty$.

Proof. The Lévy measure of T_x is $\nu(dt) = k(t)t^{-1}dt$ up to the scaling by x, so $\int_{t>1} t^n\,\nu(dt) = \int_1^\infty t^{\,n-1}k(t)\,dt$; the moment criterion for infinitely divisible laws ([32], Thm. 25.3) equates finiteness of that integral with finiteness of $\mathbb{E}\,T_x^n$. □

The stable family is the semigroup theory seen from inside the cone: $F(s) = s^\alpha$ is the only exponent, up to scale, that is homogeneous, and its k is a pure power.

Proposition 8.5 (the extremal stable family: the exponent)**.** *For* $0 < \alpha < 1$ *the exponent* $F(s) = s^\alpha$ *is of the form (7.1), with Lévy density factor*

$$k(t) = \frac{\alpha}{\Gamma(1-\alpha)}\,t^{-\alpha},$$

which is nonincreasing.

Proof. $s^\alpha = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^\infty (1 - e^{-st})\,t^{-\alpha-1}\,dt$ is the standard integral representation, which is (7.1) with $b_0 = 0$ and $k(t)/t = \alpha t^{-\alpha-1}/\Gamma(1-\alpha)$; k is a negative power, hence nonincreasing, and both integrability conditions of Proposition 8.1 hold ($t^{-\alpha}$ integrable at 0, $t^{-\alpha-1}$ integrable at infinity), so F is finite and of the form (7.1). □

Proposition 8.6 (the extremal stable family: every moment is infinite)**.** *With* F *as in Proposition 8.5,* $\mathbb{E}\,T_x^n = \infty$ *for every* $n \geq 1$ *and every* $x > 0$*, so the delay of the 2005 stable kernels cannot be measured by any moment.*

Proof. By Proposition 8.3 the mean delay is $\mathbb{E}\,T_x = x\big(b_0 + \int_0^\infty k\big)$, finite exactly when k is integrable at infinity; here $b_0 = 0$ and $k(t) = \alpha t^{-\alpha}/\Gamma(1-\alpha)$, which is not integrable at infinity for $\alpha < 1$. So $\mathbb{E}\,T_x = \infty$. Every higher moment follows without a further criterion: $t \leq 1 + t^{\,n}$ for $t \geq 0$ and $n \geq 1$, so on the probability measure $\mu_{0,x}$ we get $\mathbb{E}\,T_x \leq 1 + \mathbb{E}\,T_x^n$, and a divergent mean forces $\mathbb{E}\,T_x^n = \infty$. □

Proposition 8.7 (the extremal stable family: the mode)**.** *With F as in Proposition 8.5, T_x is unimodal, and its mode is x times the mode of T_1 in the canonical gauge; in the parametrization of the one-parameter family of Corollary 7.4, where $\chi(x) = x^{1/\alpha}$, it scales as $x^{1/\alpha}$.*

Proof. By Remark 7.9, T_x is unimodal, since $k \not\equiv 0$ for the stable family. In the canonical gauge $\mu_{0,x}$ is the law of xT_1, so its mode is x times the mode of T_1. The self-similarity is the case $\chi(x) = x^{1/\alpha}$ of Corollary 7.4, and the mode of $x^{1/\alpha}T_1$ is $x^{1/\alpha}$ times the mode of T_1. □

The Gamma family is the one the introduction started from. Its k is an exponential — bounded, with every moment finite — and its kernels are the Gamma densities, which is what makes it implementable by a cascade of first-order filters.

Proposition 8.8 (the Gamma family: the exponent)**.** *For $\gamma > 0$ the exponent $F(s) = \gamma \log(1+s)$ is of the form (7.1) with $k(t) = \gamma\, e^{-t}$, which is nonincreasing.*

Proof. The Frullani integral gives $\log(1+s) = \int_0^\infty (1 - e^{-st})\, e^{-t}\, t^{-1}\, dt$, so F has the form (7.1) with $b_0 = 0$ and $k(t) = \gamma e^{-t}$, nonincreasing, and both integrability conditions of Proposition 8.1 hold (e^{-t} bounded near 0, integrable at infinity), so F is finite and of the form (7.1); hence F is admissible by Lemma 7.1. □

Proposition 8.9 (the Gamma family: the transforms and the mean)**.** *With F as in Proposition 8.8, the kernels and the increment measures have transforms*

$$e^{-F(xs)} = (1+xs)^{-\gamma}, \qquad e^{-g_{a,b}(s)} = \left(\tfrac{1+as}{1+bs}\right)^{\gamma},$$

and the mean delay is $\mathbb{E}\,T_x = \gamma x$.

Proof. $F(s) = \gamma \log(1+s)$ by Proposition 8.8, so $e^{-F(xs)} = (1+xs)^{-\gamma}$; and $g_{a,b} = F(b\cdot) - F(a\cdot)$, so the increment transform is the ratio $e^{-F(bs)}/e^{-F(as)}$, which is the stated one. For the mean, Proposition 8.3 gives $\mathbb{E}\,T_x = x\big(b_0 + \int_0^\infty k\big)$ with $b_0 = 0$ and $k(t) = \gamma e^{-t}$, whose integral is γ. □

Proposition 8.10 (the Gamma family: the kernel density)**.** *With F as in Proposition 8.8 and $\gamma > 0$,*

$$\phi_x(t) = \frac{t^{\gamma-1} e^{-t/x}}{\Gamma(\gamma)\, x^{\gamma}},$$

the Gamma density of fixed shape γ and scale x.

Proof. $(1+xs)^{-\gamma}$ is the Laplace transform of the stated Gamma density — the Gamma integral $\int_0^\infty t^{\gamma-1} e^{-(1/x+s)t}\, dt = \Gamma(\gamma)/(1/x+s)^{\gamma}$ — so Proposition 3.9 identifies ϕ_x. □

Remark 8.11 (the Gamma family as a Sato subordinator). The increment measures $\mu_{a,b}$ of this family are the increments of the Gamma-type Sato subordinator. This is Remark 7.2 read at the Gamma exponent.

Proposition 8.12 (the Gamma family: the higher moments)**.** *With F as in Proposition 8.8, every moment of T_x is finite, and $\operatorname{Var} T_x = \gamma x^2$.*

Proof. By Proposition 8.10 the law of T_x is the Gamma law of shape γ and rate $1/x$, so $\mathbb{E}\,T_x^n = \Gamma(\gamma+n)x^n/\Gamma(\gamma)$, finite for every n. In particular $\mathbb{E}\,T_x = \gamma x$ and $\mathbb{E}\,T_x^2 = \gamma(\gamma+1)x^2$, so $\operatorname{Var} T_x = \gamma(\gamma+1)x^2 - \gamma^2 x^2 = \gamma x^2$. □

Proposition 8.13 (non-creation from the signal; the Pólya frequency members)**.** *Write $S^-(g)$ for the number of sign changes of a function g on $\mathbb{R}$.*

1. *For integer γ the Gamma kernels are one-sided Pólya frequency densities, and convolution with them is variation-diminishing: $S^-(\mu_{0,x} * g) \le S^-(g)$ for every $x > 0$ and every integrable or bounded g, sign changes counted in the essential (almost-everywhere) sense. In particular, for the field $u(\cdot, x) := \mu_{0,x} * f$,*

$$S^-\big(u(\cdot,x) - c\big) \;\le\; S^-\big(f - c\big) \qquad \text{for every level } c \in \mathbb{R},$$

so the field has, at every scale, no more zero-crossings than the signal; and if f is the primitive of a function $f' \in X_0$ — that is, $f \in \mathcal{D}$ — then, counting local extrema as essential sign changes of the derivative,

$$S^-\big(\partial_t u(\cdot,x)\big) \;\le\; S^-\big(f'\big) :$$

no more local extrema either. For non-integer γ the kernels are not Pólya frequency densities.

2. *An admissible family (Theorem 7.3) has Pólya frequency kernels if and only if*

$$F(s) \;=\; b_0\, s \;+\; \sum_i \log\Big(1 + \frac{s}{\tau_i}\Big), \qquad \tau_i > 0, \quad \sum_i \frac{1}{\tau_i} < \infty$$

— drift together with integer-weighted Thorin atoms, the summability condition being finiteness of the mean delay, $\mathbb{E}\,T_1 = b_0 + \sum_i \tau_i^{-1}$.

Proof. Two classical facts carry the proof. A density supported on $[0,\infty)$ is a Pólya frequency density if and only if the reciprocal of its Laplace transform is an entire function of the form $Ce^{\delta s}\prod_i(1+\lambda_i s)$ with $C > 0$, $\delta \ge 0$, $\lambda_i \ge 0$ and $\sum_i \lambda_i < \infty$ — Schoenberg's representation theorem ([15], Ch. 7, Thm 3.2(b), p. 345; the one-sided form, Remark 3.1, p. 346). And convolution with a kernel is variation-diminishing if and only if the kernel is totally positive ([15], Ch. 5, Thm 4.2), which for a translation kernel is the Pólya frequency property.

(1) For the Gamma kernel, $1/\widehat{\mu}_{0,x}(s) = (1+xs)^\gamma$: for integer γ this is a polynomial with the single real negative root $-1/x$, of the displayed form with $\delta = 0$ and γ equal factors, so the kernels are Pólya frequency densities and convolution with them diminishes essential sign changes of integrable and of bounded functions. The level form follows applied to $g = f - c$, since $\mu_{0,x} * (f - c) = u(\cdot,x) - c$ by unit mass, and the zero-crossing clause is the level $c = 0$. For the extremum clause, $f \in \mathcal{D}$ gives $f = \mathbf{1}_{[0,\infty)} * f'$, hence $u(\cdot,x) = \mathbf{1}_{[0,\infty)} * (\mu_{0,x} * f')$ and $\partial_t u(\cdot,x) = \mu_{0,x} * f'$ a.e.; the variation-diminishing bound applied to $g = f'$ gives the display. (The count is read from the derivative and not from the levels: no single level need meet every extremum — two oscillations can occupy disjoint value ranges — so the level bounds alone do not bound the number of extrema.) For non-integer γ, $(1+xs)^\gamma$ has a branch point at $s = -1/x$ and is not entire, so by the same representation theorem the kernel is not a Pólya frequency density.

(2) The kernels are supported on $[0,\infty)$ with transform $e^{-F(xs)}$, so the family is Pólya frequency exactly when $e^{F(xs)}$ has the displayed form — for one x or, equivalently, for all: replacing s by xs rescales δ and the λ_i by x and preserves the form. Matching at $x = 1$: $F(0+) = 0$ forces $C = 1$; the exponential factor is the drift, $\delta = b_0$; and the linear factors are atoms, $\lambda_i = 1/\tau_i$, giving the display, the condition $\sum_i \lambda_i < \infty$ being $\sum_i \tau_i^{-1} < \infty$, i.e. $F'(0+) = b_0 + \sum_i \tau_i^{-1} < \infty$. Every such F is admissible: it is of the form (7.1) with the nonincreasing memory kernel $k(t) = \sum_i e^{-\tau_i t}$ — a Frullani integral per atom — and $F(1) \le b_0 + \sum_i \tau_i^{-1} < \infty$, so Theorem 7.3 applies. □

Remark 8.14 (causal non-creation: what survives, and what cannot). Non-creation is the oldest axiom of the subject: Witkin's founding observation was that Gaussian convolution creates no

new extrema in one dimension, and non-creation of local extrema or zero-crossings is the one-dimensional entry in the standard catalogue of scale-space axioms ([22], §2). Proposition 8.13 places the recommended family inside the class that carries its endpoint form. Three glosses are in order. First, within the *semigroup* axioms, non-creation and time-causality are incompatible: any continuous-parameter convolution semigroup of Pólya frequency densities is Gaussian ([15], Ch. 7, Thm 5.2, p. 354), and the Gaussian is two-sided — so no time-causal semigroup scale space is variation-diminishing at all. The hemigroup dissolves the obstruction for the endpoint form: the Gamma family is a continuous-parameter *hemigroup* of one-sided Pólya frequency densities, one at every scale. The class of clause (2) is moreover closed under weak limits — atoms of integral mass cannot split in the limit ([2], p. 36) — so the Pólya frequency members form a closed subclass of the cone, not a boundary accident. Second, the property belongs to the accumulated kernels, not to the increments: the increment transforms carry zeros, which by the same Schoenberg representation place them outside the Pólya frequency class, so non-creation holds from the signal to every scale but not stage-to-stage along a cascade — the same zeros that buy continuous covariance in §11 spend the increments' variation-diminishing property; and this is forced, not an accident of the Gamma family: Proposition 8.15 below shows that no admissible family except the pure delay has all its interior increments variation-diminishing. Third, the pointwise form of non-enhancement — the value at a fixed-time extremum must not grow — fails for *every* causal family, because the kernels delay what they smooth. A peak decays while traveling down the delay line, and at a fixed time just past it the value rises as the delayed mass arrives. Counting, not pointwise decay, is the causal form of non-creation. A pointwise form that *does* survive — along the feature's own delay path rather than at fixed time — is Corollary A.21 in Appendix A: at extrema dominating their past, the tracked value is nonincreasing in scale, for every admissible family.

Proposition 8.15 (scale-monotone variation diminution is impossible)**.** *Let* $(\Phi_{x,y})$ *be an admissible family (Theorem 7.3), in the canonical gauge, and suppose that* every *interior increment operator* $\Phi_{x,y}$*,* $0 < x < y$*, is variation-diminishing. Then* $F(s) = b_0 s$*: the family is the pure delay. In particular no nondegenerate admissible family has the scale-monotone form of non-creation, in which every coarser record has no more sign changes than every finer one; the endpoint form of Proposition 8.13 is the strongest variation-diminishing property an admissible family can carry, and the integer-atom members of its clause (2) attain it.*

Proof. Step 1: the increments' Lévy tails are integer sums of exponentials. Fix $0 < x < y$ and write $g(s) := F(ys) - F(xs)$, the increment exponent, a Bernstein function with drift $b_0(y-x)$ and Lévy tail $h(t) := k(t/y) - k(t/x) \geq 0$, by substitution in (7.1). For $\varepsilon > 0$ let G_ε be the Gamma density of shape 2 at scale ε, with transform $(1+\varepsilon s)^{-2}$ — a one-sided Pólya frequency density, by the argument of Proposition 8.13(1). The measure $\nu_\varepsilon := \mu_{x,y} * G_\varepsilon$ is absolutely continuous, and its convolution operator, the composition of two variation-diminishing operators, is variation-diminishing; so its density is a one-sided Pólya frequency density ([15], Ch. 5, Thm 4.2, as in Proposition 8.13), and the Schoenberg representation ([15], Ch. 7, Thm 3.2(b), p. 345; Rem. 3.1, p. 346) gives

$$1/\widehat{\nu_\varepsilon}(s) \;=\; e^{g(s)}\,(1+\varepsilon s)^2 \;=\; e^{\delta_\varepsilon s}\prod_i \bigl(1+\lambda_i^{(\varepsilon)} s\bigr), \qquad \lambda_i^{(\varepsilon)} \geq 0, \quad \textstyle\sum_i \lambda_i^{(\varepsilon)} < \infty,$$

the constant being 1 at $s = 0$. Both sides are exponentials of Bernstein functions; taking logarithms on $s > 0$ and matching the representations — drifts from the $s \to \infty$ limit, then Lévy tails through their transforms and Proposition 3.10 — gives, for a.e. $t > 0$,

$$h(t) + 2\,e^{-t/\varepsilon} \;=\; \sum_i e^{-t/\lambda_i^{(\varepsilon)}} \;=\; \int_{(0,\infty)} e^{-\tau t}\, M^{(\varepsilon)}(d\tau),$$

where $M^{(\varepsilon)} := \sum_i \delta_{1/\lambda_i^{(\varepsilon)}}$ is a positive integer-atomic measure, locally finite on $(0,\infty)$ because $\sum_i \lambda_i^{(\varepsilon)} < \infty$. Comparing two values $\varepsilon \neq \varepsilon'$ through Proposition 3.10 once more shows that $M := M^{(\varepsilon)} - 2\,\delta_{1/\varepsilon}$ does not depend on ε; evaluating any single point with a third value of ε whose exceptional atom lies elsewhere shows $M \geq 0$ and integer-atomic. Hence

$$k(t/y) - k(t/x) \;=\; \int_{(0,\infty)} e^{-\tau t}\, M_{x,y}(d\tau) \qquad \text{for a.e. } t > 0,$$

with $M_{x,y}$ positive, locally finite, and integer-atomic.

Step 2: scaling and additivity. The tails satisfy $h_{\sigma x,\,\sigma y}(t) = h_{x,y}(t/\sigma)$ and $h_{x,z} = h_{x,y} + h_{y,z}$ for $x < y < z$, so by Proposition 3.10,

$$M_{\sigma x,\,\sigma y} \;=\; \text{the pushforward of } M_{x,y} \text{ under } \tau \mapsto \tau/\sigma, \qquad M_{x,z} \;=\; M_{x,y} + M_{y,z}.$$

Step 3: an atom replicates into a continuum. Suppose some $M_{x,y} \neq 0$; by the scaling relation we may take $M_{1,y_0} \neq 0$, with an atom at τ_0 of mass $m_0 \geq 1$. For $\sigma \geq 1$, Step 2 gives $M_{\sigma,\,\sigma y_0}$ an atom at τ_0/σ of mass m_0; and for $Y \geq \sigma y_0$, additivity and positivity give $M_{1,Y} \geq M_{\sigma,\,\sigma y_0}$. Hence $M_{1,Y}$ carries an atom of mass at least 1 at *every* point τ_0/σ with $1 \leq \sigma \leq Y/y_0$ — uncountably many atoms of mass ≥ 1 in the interval $[\tau_0 y_0/Y,\ \tau_0]$ — while

$$e^{-\tau_0 t}\, M_{1,Y}\big([\tau_0 y_0/Y,\ \tau_0]\big) \;\leq\; h_{1,Y}(t) \;\leq\; k(t/Y) \;<\; \infty$$

for any fixed $t > 0$: a contradiction. So $M_{x,y} = 0$ for every pair, i.e. $k(t/y) = k(t/x)$ for all $0 < x < y$ and a.e. t; a nonincreasing function that is a.e. dilation-invariant is constant on $(0,\infty)$, and $\int_1^\infty k(t)\, t^{-1} dt < \infty$ forces the constant to be 0. Hence $F(s) = b_0 s$. Conversely the pure delay's increments are the shifts $\delta_{b_0(y-x)}$, which are variation-diminishing, so the degenerate case does occur and the statement is sharp. □

Remark 8.16 (ladder monotonicity survives discrete covariance). The impossibility is a continuous-covariance phenomenon, and it is sharp in the ratio: increments of a *single* ratio can all be variation-diminishing. For an integer-atom family (Proposition 8.13(2)) and a fixed $q > 1$, the increments $\Phi_{x,qx}$, $x > 0$, are variation-diminishing precisely when the atom multiset satisfies $\mathrm{mult}(q\tau) \geq \mathrm{mult}(\tau)$ for every τ: then each pole of $e^{g_{x,qx}}$ at $-\tau/x$ cancels against a zero, the leftover product is of Schoenberg form, and the representation theorem runs backwards to sufficiency (necessity is Step 1 of the proof above, applied at ratio q). The time-causal limit kernel of Remark 13.1 is the fundamental example: its Thorin measure is the geometric train $\sum_{i\geq 1} \delta_{q^i/c}$, the condition holds, and every q-step increment exponent is $F(qxs) - F(xs) = \log(1 + cxs)$ — each rung one exponential (Gamma shape-1) kernel, a Pólya frequency density. Along any geometric ladder of ratio q the sign-change count of the record is therefore monotone from rung to rung: scale-monotone non-creation, impossible under full covariance, is available in the discrete-covariance stratum, and the published limit-kernel construction has it — in retrospect a further virtue, and a sharper entry on each side of the containment comparison of Remark 13.1.

Two facts about the Gamma family belong here although they are proved later. It is the family §11 implements, exactly, by first-order recursive sections, one per unit of shape. And the memory-line development of §9 needs more of it than admissibility: the standing hypothesis stated there requires a negative moment $\mathbb{E}\, T_1^{-\zeta} < \infty$ of some order $\zeta > 1$, which for a Gamma law of shape γ holds exactly when $\gamma > 1$. The propositions above hold for every $\gamma > 0$; the memory line uses $\gamma > 1$, and $\gamma \leq 1$ is listed among the open problems in §13.

The Bessel family is cut by the third principle, locality of the memory-line evolution, and it is the family on which the second half of the paper is worked. It is introduced here by its law and transform; what it is the kernel *of* is the subject of §§9–10.

Proposition 8.17 (the Bessel-K family; preview of the memory line)**.** *For $a > 0$ let $T_1 \overset{d}{=} 1/(2\gamma_a)$ be the scaled inverse-gamma law, where γ_a has the Gamma density of shape a, and let F be its Laplace exponent. Then*

$$e^{-F(s)} = \frac{2^{1-a}}{\Gamma(a)} \big(\sqrt{2s}\big)^a K_a\big(\sqrt{2s}\big),$$

and T_1 is self-decomposable, so F is of the form (7.1) and the family is admissible. The moments satisfy $\mathbb{E}\, T_x^n < \infty$ if and only if $a > n$.

Proof. Write γ_a for a Gamma$(a, 1)$ variable and $T_1 = 1/(2\gamma_a)$. Substituting $u = 1/(2t)$ in the Gamma density gives the density of T_1,

$$\phi_1(t) = \frac{t^{-a-1}\, e^{-1/(2t)}}{2^a\, \Gamma(a)}, \qquad t > 0.$$

The classical integral representation $\int_0^\infty t^{\nu-1} \exp(-A/t - Bt)\, dt = 2\,(A/B)^{\nu/2} K_\nu\big(2\sqrt{AB}\big)$, taken at $\nu = -a$, $A = \frac{1}{2}$, $B = s$, together with $K_{-a} = K_a$, gives

$$\mathbb{E}\, e^{-sT_1} = \frac{1}{2^a \Gamma(a)} \int_0^\infty t^{-a-1} e^{-st - 1/(2t)}\, dt = \frac{2\,(2s)^{a/2}\, K_a\big(\sqrt{2s}\big)}{2^a \Gamma(a)} = \frac{2^{1-a}}{\Gamma(a)} \big(\sqrt{2s}\big)^a K_a\big(\sqrt{2s}\big).$$

Self-decomposability of T_1 is a theorem of Halgreen ([12], for the generalized inverse Gaussian laws, of which this is a limiting case); by Lemma 7.1 its exponent is of the form (7.1), and both integrability conditions of Proposition 8.1 hold for the resulting (b_0, k) by Lemma 8.2, since $F(1) = -\log \mathbb{E}\, e^{-T_1} < \infty$; so F is finite and Theorem 7.3 makes the family admissible. Finally, since $T_x \overset{d}{=} x\, T_1$ in the canonical gauge, for every $x > 0$ and $n \in \mathbb{N}$,

$$\mathbb{E}\, T_x^n = x^n\, \mathbb{E}\, T_1^n = \frac{x^n}{2^n}\, \mathbb{E}\big[\gamma_a^{-n}\big], \qquad \mathbb{E}\big[\gamma_a^{-n}\big] = \frac{1}{\Gamma(a)} \int_0^\infty u^{a-n-1} e^{-u}\, du = \frac{\Gamma(a-n)}{\Gamma(a)},$$

the last integral (hence the moment) finite exactly when $a - n > 0$; so $\mathbb{E}\, T_x^n < \infty$ iff $a > n$. This is a direct computation from the defining substitution $T_1 = 1/(2\gamma_a)$ and cites no Lévy-density asymptotics. □

In the parabolic gauge ξ of Remark 6.4, $\tilde{x} = \xi^2$, these kernels are the laws of the first hitting time of the boundary $\xi = 0$ by the Bessel-type diffusion with generator $\frac{1}{2}\partial_\xi^2 + \frac{\beta}{\xi}\partial_\xi$, $a = \frac{1}{2} - \beta$, started at ξ; that identification is proved in Example 9.27, where it appears as the statement that the family's signaling form is local, and §10 shows it is the only admissible family with that property. Figure 6 shows the family across a: the tails are power laws of exponent $-(a+1)$, so the shape parameter is a dial on the moments, and $a = \frac{1}{2}$ is the stable member.

Remark 8.18 (heavy tails are the semigroup's doing). The heavy tails of the 2005 kernels are an artifact of the semigroup axiom, not of causality together with covariance. Within Theorem 7.3 the semigroup sub-case (Corollary 7.4) is the subset with k a pure power $c\, t^{-\alpha}$, $0 < \alpha < 1$, or the pure delay, $k \equiv 0$ with $b_0 > 0$ — the only data compatible with homogeneity. A pure power is never integrable at infinity, so by Proposition 8.3 no stable member has a mean (Proposition 8.6). Any other nonincreasing k is admitted, and its moments are governed by Proposition 8.4 alone: $\mathbb{E}\, T_x^n < \infty$ exactly when $\int_1^\infty t^{n-1} k(t)\, dt < \infty$. For $k(t) = \gamma e^{-t}$ every moment is finite (Proposition 8.12); for the Bessel family with shape a exactly the moments of order below a are finite (Proposition 8.17). The comparison at matched delay is drawn in Figure 10.

Remark 8.19 (the tail dichotomy: how light can an aperture be?). The moment criterion (Proposition 8.4) reads the *polynomial* weight of a kernel's tail off the tail of k. The super-exponential

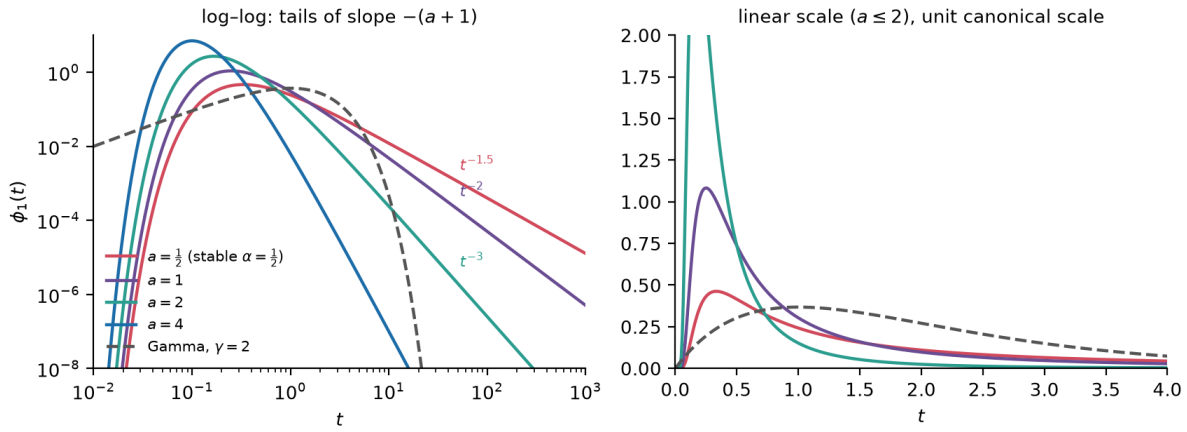


Figure 6: The moment threshold, seen in the tails. Left, log–log: the kernels ϕ_1 of the Bessel family of Proposition 8.17 for shapes $a = \frac{1}{2}, 1, 2, 4$ are straight lines of slope $-(a+1)$ at large t, so the moments of order $n < a$ are finite and no others; $a = \frac{1}{2}$ is the extremal stable member of the semigroup theory, the heaviest tail in the family, with no mean. The dashed curve is the Gamma kernel with $\gamma = 2$: an exponential tail, every moment finite (Proposition 8.12). Right: the same kernels on a linear scale at unit canonical scale. Raising a buys moments at the price of temporal reach — a heavier tail is a longer scale-invariant memory, not a defect. The comparison at matched delay, which is what an implementation sees, is in §11.

end is governed by a theorem that needs only infinite divisibility — axioms (A1)–(A7), no covariance — and so binds every stratum at once ([32], Thm. 26.1, p. 168). Off the drift ray, a kernel's tail satisfies $\mu_{0,x}\big((r,\infty)\big) \geq e^{-c\,r\log r}$ for some c and all large r. The mechanism is the cascade itself: the delay is assembled from independent increments, and independent increments can always conspire — n repetitions of one available delay cost only the Poisson probability $\lambda^n/n! \approx e^{-n\log n}$ — so no admissible tail can be suppressed faster than $e^{-c\,r\log r}$. Consequently no kernel of compact support and none with Gaussian-or-lighter tail belongs to any causal cascade family: apart from the pure delay, every admissible smoother has infinite impulse response, and truncation is a violation of the cascade, not an implementation detail.

The theorem is two-sided, and the dichotomy ranks the families. If the Lévy support is *bounded* — k vanishing beyond a cutoff τ — the floor is attained: $-\log \mu_{0,x}\big((r,\infty)\big) \sim (r/\tau x)\log r$. If it is *unbounded*, the tail is strictly heavier than every bounded-support rate. A completely monotone k is strictly positive on all of $(0,\infty)$, so the entire Thorin subclass sits on the heavy side of the dichotomy, the Gamma family's plain exponential tail being typical. The bounded-support members — the Dickman rays of Remark 7.9 and their superpositions — attain the lightest tails the axioms permit and, their k being compactly supported, have all moments finite as well: the extreme rays of the cone are its best-localized members. The trade-off is legible in the kernel k itself: a hard cutoff of long delays buys super-exponential tails and a full set of moments at the price of the echo signatures of Remark 7.9, unattainable within the Thorin subclass — the delay-equation dynamics behind them are developed in a companion note (in preparation); a completely monotone k buys echo-free kernels realizable by first-order sections (§11) at the price of a tail no lighter than exponential. Both dials — the polynomial one of Proposition 8.4 and the super-exponential one here — are turned by the same object, the tail of k.

Remark 8.20 (the classical catalogue). Theorem 7.3 is a machine: any single self-decomposable law on $[0,\infty)$, with Laplace exponent F, generates a complete time-causal scale space $\phi_x = \mathcal{L}^{-1}[e^{-F(xs)}]$. The three families above are therefore worked representatives, not an enumeration; the classical catalogue of class L supplies more, each with its own physiognomy as a family of

apertures. The *generalized inverse Gaussian* laws are self-decomposable in their entirety ([12] — the citation Proposition 8.17 already rests on, the Bessel family being the inverse-Gamma member); the inverse Gaussian member has the closed-form exponent $F(s) = \sqrt{1+2s} - 1$, the first-passage time of a drifting Brownian motion, so its kernels are hitting-time delays with all moments finite. The *lognormal* law is admissible: Thorin proved its infinite divisibility by exhibiting it as a generalized Gamma convolution ([37]) — the theorem for which the Thorin class was created. Its kernels have a delay that is a product of many independent small factors, multiplicative rather than additive accumulation of latency. The *Pareto* laws are admissible by the companion result ([38]); a Pareto kernel of index a has exactly the moments of order below a, the Bessel family's position on the moment dial with an exact power tail. The *Mittag-Leffler* laws are admissible: the law with transform $(1+s^\alpha)^{-\gamma}$, $0 < \alpha < 1$ — the delay $Y^{1/\alpha}X_\alpha$ of a Gamma variable's power times an independent stable factor — is a generalized Gamma convolution with explicit Thorin density ([2], Ex. 3.2.4, p. 38, where the object appears unnamed). It interpolates the Gamma and stable corners on one dial. The class is closed under a multiplicative calculus of delays: the pdf of a ratio of independent Gamma variables is hyperbolically completely monotone ([2], Ex. 4.3.1, p. 59), as is any power X^q, $|q| \geq 1$, of a Gamma variable (Ex. 4.3.4, p. 60), and HCM laws are GGC (Thm. 4.3.1, p. 57), with closure under independent products and ratios (Thm. 5.1.1, p. 69; Thm. 6.2.1, p. 92) — so compound latencies assembled multiplicatively from Gamma factors stay admissible. And the phenomenon behind the Bessel family is general, in a precisely weaker sense: for a nonsingular one-dimensional diffusion on an interval, every interior first-passage time τ_{ab}, $r_0 < a < b < r_1$, lies in the *Bondesson class* of infinitely divisible laws on $[0,\infty]$ — possibly defective, with completely monotone Lévy density ([16], p. 207; the class is Definition 2.1 there, p. 208). The Bondesson class strictly contains the Thorin subclass, and its members need not be self-decomposable — a completely monotone Lévy density ℓ does not make $t\,\ell(t)$ nonincreasing — so a general diffusion hitting time need not be admissible at all. The Bessel family is the corner where the hitting-time structure meets the axioms. Two boundaries frame the catalogue. Outward: nothing with tail lighter than $e^{-ct\log t}$ is admissible at all (Remark 8.19), so no windowed or truncated Gaussian appears however the data (b_0, k) are chosen. Inward: every admissible law just named lies in the Thorin subclass — the classical catalogue is classical largely because completely monotone structure is where the classical proofs live — while the admissible class is strictly larger. Its non-Thorin territory, with plateaus, kinks and cutoffs in k, is new ground as a source of scale spaces; the Dickman family of Remark 7.9 is its paradigm, and a companion note develops its theory — the delay-equation dynamics, the echo structure, and the delay-line implementation that replaces the first-order cascades of §11 outside the Thorin subclass.

9 The signaling form

Everything so far describes the observer from outside: the record at scale x is $u(t,x) = (\mu_{0,x} * f)(t)$, a convolution reaching back into the raw past. The embodiment principle of §2 demands more — the past is gone, and whatever of it influences the present must be carried in the observer's own state. This section shows that the demand is met, in the strongest sense: the field can be evolved out of its present state alone, with the signal entering only where it physically enters, at the fine-scale end. In the language of classical PDE this is a *signaling* problem rather than a Cauchy problem: the data live on the spatial boundary for all time, not on an initial time-slice. The model is a rod driven at its end:

$$\partial_t u \;=\; A\,u, \qquad u(t,0) = f(t), \qquad u(0,x) = 0,$$

with A an operator in the scale variable alone (Figure 7). Scale becomes space, time the evolution variable, and the memory line a medium whose interior transport law is A. Theorem 4

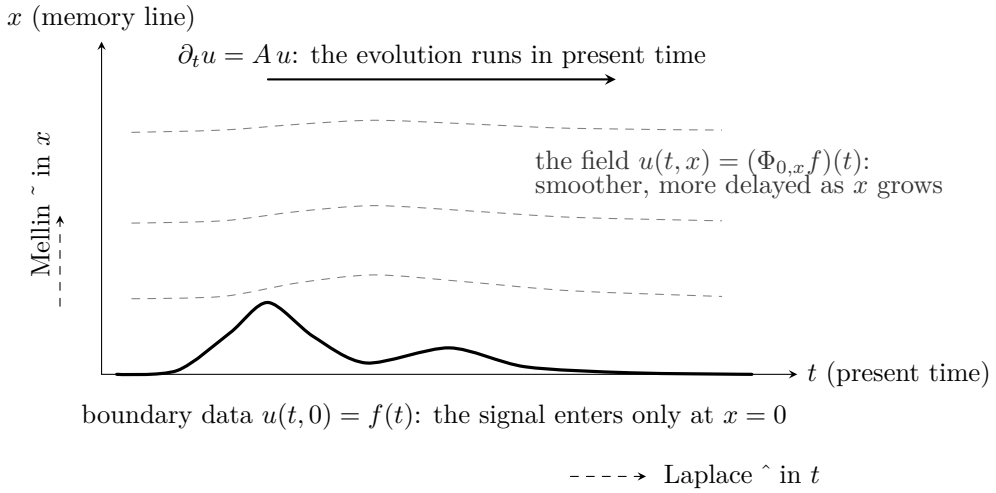


Figure 7: The signaling geometry. The signal is never stored: it enters the device only as boundary data at zero scale, and what the observer carries is the field $u(t,x)$ over the memory half-line — at each fixed x a record of the signal at that scale, smoother and more delayed as x grows. Theorem 9.17 inverts the scale evolution into an evolution *in present time*: $\partial_t u = Au$, with A acting along the memory line only. The section works with two transforms in two different variables: the Laplace transform ˆ in t, in which the field at fixed s is a dilate $\hat{f}(s)H(sx)$ of one profile, and the Mellin transform ˜ in x, in which A is diagonal with symbol B.

of [8] solved this in the homogeneous case, with $A = D^{1/\alpha}_{x,-}$ the right-sided Liouville derivative ([31]) — a *translation-invariant* operator in x, available because the semigroup's scale axis carries an additive structure. The hemigroup removes that structure, and with it the operator; this section constructs the general inversion from what survives.

Throughout we work in the canonical gauge, so that $\mu_{0,x}$ is the law of $x\,T_1$ with $\mathbb{E}[e^{-sT_1}] = H(s)$, $H := e^{-F}$; we write $\hat{g}(s) := \int_0^\infty e^{-st} g(t)\,dt$ for the Laplace transform in the time variable, $\tilde{g}(z) := \int_0^\infty x^{z-1} g(x)\,dx$ for the Mellin transform in the scale variable, $\theta := x\,\partial_x$ for the Euler operator, and $(I^z f)(t) := \frac{1}{\Gamma(z)} \int_0^t (t-r)^{z-1} f(r)\,dr$ for the Riemann–Liouville integral of complex order z, $\operatorname{Re} z > 0$ ([31]).

What survives is dilation structure, and it is enough. By Theorem 7.3 the Laplace-domain profiles of the field,

$$\hat{u}(s,x) = \hat{f}(s)\,H(sx),$$

are *dilates of a single profile* H, with the time-eigenvalue s appearing as the dilation parameter; and ∂_t acts on the causal mode at rate s by multiplication by s. So the sought operator must read off where a profile sits on its dilation orbit and return that position as a number: $A[H(s\,\cdot)] = s\,H(s\,\cdot)$ for every s. Operators diagonal on dilation orbits are the subject of multiplicative harmonic analysis, so A lives in the **Mellin calculus** — the multiplicative-group counterpart of the Wendel step of §4, now run in reverse: there an operator was given and its kernel extracted; here the spectral data are prescribed and the operator must be synthesized. Covariance fixes its shape outright: under $(t,x) \mapsto (\sigma t, \sigma x)$ an inversion must scale like ∂_t, and in the canonical gauge x carries the dimension of time while the Euler operator $\theta = x\partial_x$ is dimensionless, so $A = x^{-1} B(\theta)$. The eigenvalue demand then determines the symbol with no freedom left: on transforms the weight x shifts the Mellin argument by one, forcing $B(-z) = \tilde{H}(z+1)/\tilde{H}(z)$, the recursion of the profile's Mellin data. The theorem of this section (Theorem 9.17) makes all of this precise: *the signaling problem has a covariant inversion; the scale-space field solves it; and no other covariant Mellin operator does.* What stands between the ansatz and the theorem is analytic honesty — the symbol is a ratio and may have poles, so the strip, the transforms,

and a careful functional calculus must be built first; that is the business of the lemmas below.

The transforms of the profile and of the field are the section's working objects, and they need a strip of the complex plane to live on; the standing hypothesis is what opens one. The strip must moreover have width greater than one, because the equation shifts the Mellin argument by exactly one — differentiation in time and the weight x are the same shift seen from two sides — which is where the condition $z_* > 1$ comes from.

Definition 9.1 (the standing hypothesis (H))**.** An admissible exponent F satisfies the *standing hypothesis* (H) if $F(\infty) = \infty$ and

$$z_* \;:=\; \sup\big\{\, \zeta > 0 :\; \mathbb{E}\big[T_1^{-\zeta}\big] < \infty \,\big\} \;>\; 1.$$

Both clauses are conditions at the zero-delay end of the kernels, and Lemmas 9.2 and 9.3 below make the readings precise: the first clause says exactly that the kernels put no atom at zero delay, is equivalent to $b_0 > 0$ or $\int_0^\infty k(t)\, t^{-1} dt = \infty$, and is automatic for every admissible $F \not\equiv 0$, so all of (H)'s bite is in the second clause. The second implies $\int_0^\infty e^{-F(s)}\, ds = \mathbb{E}[T_1^{-1}] < \infty$, the converse failing only at the endpoint $z_* = 1$, an endpoint genuinely reached: the Gamma family at $\gamma = 1$ has $z_* = 1$ with $\mathbb{E}[T_1^{-1}] = \infty$. The second clause holds with $z_* = \infty$ whenever $b_0 > 0$, for the stable family, and for the Bessel-K family of Proposition 8.17; for the Gamma family $z_* = \gamma$, so (H) requires $\gamma > 1$ there, and $\gamma \le 1$ is among the open problems of §13. The rate is what Lemma 9.4 computes in general: $z_* = \lim F(s)/\log s$, which is $k(0^+)$ when $b_0 = 0$ and ∞ otherwise, so (H) reads: drift, or catalogue height above one at the origin. The construction extends below $z_* = 1$ by meromorphic continuation, which we do not pursue.

Lemma 9.2 (kernel readings of the standing hypothesis)**.** *In the canonical gauge, with T_1 as in Definition 9.1:*

1. *$F(\infty) = \infty$ if and only if the kernels have no atom at zero delay: $\mu_{0,x}(\{0\}) = \Pr[T_1 = 0] = 0$ for every $x > 0$.*
2. *Suppose $F(\infty) = \infty$. Then $\int_0^\infty e^{-F(s)}\, ds = \mathbb{E}[T_1^{-1}]$ in $[0, \infty]$; consequently $z_* > 1$ implies $\int_0^\infty e^{-F(s)}\, ds < \infty$, and $\int_0^\infty e^{-F(s)}\, ds < \infty$ implies $z_* \ge 1$.*

Proof. (1) $s \mapsto H(s) := \mathbb{E}[e^{-sT_1}]$ decreases to $\Pr[T_1 = 0]$ as $s \to \infty$: the integrand e^{-sT_1} is dominated by the constant 1 and converges pointwise to the indicator of $\{T_1 = 0\}$, so dominated convergence gives $H(s) \to \Pr[T_1 = 0]$. Since $H(s) = e^{-F(s)}$, $F(\infty) = \infty$ iff $H(s) \to 0$ iff $\Pr[T_1 = 0] = 0$. For general $x > 0$, $\mu_{0,x}$ is the law of xT_1 (the canonical gauge), and multiplication by $x \neq 0$ carries $\{0\}$ to itself and nowhere else, so $\mu_{0,x}(\{0\}) = \Pr[xT_1 = 0] = \Pr[T_1 = 0]$ for every $x > 0$; the case $x = 1$ already gives the stated equivalence, and every other x agrees with it.

(2) By Tonelli, $\int_0^\infty e^{-F(s)}\, ds = \int_0^\infty \mathbb{E}[e^{-sT_1}]\, ds = \mathbb{E}\big[\int_0^\infty e^{-sT_1}\, ds\big] = \mathbb{E}[T_1^{-1}]$ in $[0, \infty]$, the inner integral $\int_0^\infty e^{-sT_1} ds = T_1^{-1}$ being licensed pointwise because $T_1 > 0$ almost surely by (1). For the two consequences, the finiteness set $\{\zeta > 0 : \mathbb{E}[T_1^{-\zeta}] < \infty\}$ is downward closed — $t^{-\zeta'} \le t^{-\zeta} + 1$ for $t > 0$ and $\zeta' \le \zeta$, so a finite ζ-th moment bounds every smaller one up to an additive unit of mass — so $z_* > 1$ puts a real $\zeta > 1$ in the set, and hence 1 in it too, giving $\mathbb{E}[T_1^{-1}] < \infty$; conversely if $\mathbb{E}[T_1^{-1}] < \infty$ then 1 is itself a member of the set whose supremum is z_*, so $z_* \ge 1$. □

Lemma 9.3 (Lévy-data reading; the first clause is automatic)**.** *Let F be admissible with representation (7.1). Then:*

1. *$F(\infty) = \infty$ if and only if $b_0 > 0$ or $\int_0^\infty k(t)\, t^{-1} dt = \infty$ — i.e. the exponent carries drift or infinite Lévy mass.*
2. *If $F \not\equiv 0$ then $F(\infty) = \infty$; within the admissible class the standing hypothesis therefore reduces to its second clause, $z_* > 1$.*

Proof. (1) ($\Leftarrow$) If $b_0 > 0$ then $F(s) \geq b_0 s \to \infty$. If $\int_0^\infty k(t)t^{-1}dt = \infty$, monotone convergence in (7.1) — $1 - e^{-st}$ increases pointwise to 1 on $(0,\infty)$ as $s \to \infty$ — gives $F(s) \to b_0 \cdot \infty + \int_0^\infty k(t)t^{-1}dt = \infty$. ($\Rightarrow$) Contrapositive: if $b_0 = 0$ and $\int_0^\infty k(t)t^{-1}dt =: M < \infty$, the same monotone limit gives $F(s)$ increasing to $M < \infty$, so $F(\infty) \neq \infty$.

(2) If $k \not\equiv 0$, nonincreasing k that is positive somewhere is bounded below by that value on the interval below it: pick t_0 with $k(t_0) > 0$, then $k(t) \geq k(t_0)$ for $t \in (0, t_0]$, so $\int_0^{t_0} k(t)t^{-1}dt \geq k(t_0)\int_0^{t_0} t^{-1}dt = \infty$, and (1) gives $F(\infty) = \infty$. If $k \equiv 0$ and $F \not\equiv 0$, (7.1) reduces to $F(s) = b_0 s$ with $b_0 > 0$, and (1) again gives $F(\infty) = \infty$. □

Lemma 9.4 (the log-growth reading of z_*)**.** *Let $F \not\equiv 0$ be admissible, with (b_0, k) as in (7.1). Then:*

1. *The limit $\lim_{s\to\infty} F(s)/\log s$ exists in $[0,\infty]$ and equals z_*.*
2. *The limit is ∞ if $b_0 > 0$, and equals $k(0^+) := \sup_{t>0} k(t)$ if $b_0 = 0$ (finite or infinite).*
3. *Consequently, (H) holds if and only if $b_0 > 0$ or $k(0^+) > 1$.*
4. *z_* is homogeneous of degree one on the admissible cone: $z_*(cF) = c\, z_*(F)$ for every $c > 0$.*

Proof. Since $F \not\equiv 0$ is admissible, Lemma 9.3(2) gives $F(\infty) = \infty$, and Lemma 9.2(1) turns that into $T_1 > 0$ almost surely, so $\mathbb{E}[T_1^{-\zeta}]$ is an honest integral over $(0,\infty)$ for every $\zeta > 0$.

The exponent's slope. By Lemma 7.10, $F'(s) = b_0 + \int_0^\infty e^{-st}k(t)\,dt$, so $B(s) := sF'(s) = b_0 s + \int_0^\infty e^{-u}\,k(u/s)\,du$ after the substitution $u = st$. For $0 < s_1 \leq s_2$: $b_0 s_1 \leq b_0 s_2$, and k nonincreasing gives $k(u/s_1) \leq k(u/s_2)$ pointwise, since $u/s_1 \geq u/s_2$; so B is nondecreasing — no appeal to Lemma 7.1 or to B being a Bernstein function as a class, only to the explicit formula and the definition of nonincreasing. And since $k(u/s)$ increases pointwise to $k(0^+)$ as $s \to \infty$, monotone convergence gives $B(\infty) = b_0 \cdot \infty + k(0^+)\int_0^\infty e^{-u}\,du$: ∞ if $b_0 > 0$, and $k(0^+)$ if $b_0 = 0$. That is clause (2).

(1) $G(u) := F(e^u)$ has $G'(u) = e^u F'(e^u) = B(e^u)$ nondecreasing by the above, so G is convex in $u = \log s$. A convex function's difference quotient $G(u)/u$ increases, past any zero of u, to its terminal slope, so $L := \lim_{s\to\infty} F(s)/\log s = \lim_{u\to\infty} G(u)/u = \lim_{s\to\infty} B(s)$ exists in $[0,\infty]$, with value given by clause (2). To identify L with z_*: by Tonelli, $\Gamma(\zeta)\,\mathbb{E}[T_1^{-\zeta}] = \int_0^\infty s^{\zeta-1}e^{-F(s)}\,ds$ for every $\zeta > 0$, the exchange licensed exactly as in Lemma 9.2(2), now at general ζ rather than at $\zeta = 1$. For $0 < \zeta < L$, eventually $F(s) \geq (\zeta + \varepsilon)\log s$ for some $\varepsilon > 0$, so $e^{-F(s)} \leq s^{-\zeta-\varepsilon}$ eventually and the integral converges; for $\zeta > L$, eventually $F(s) \leq (\zeta - \varepsilon)\log s$, so the integrand is eventually $\geq s^{\varepsilon-1}$ and the integral diverges. Hence $\{\zeta > 0 : \mathbb{E}[T_1^{-\zeta}] < \infty\}$ contains every $\zeta < L$ and excludes every $\zeta > L$, so its supremum, z_*, is L.

(3) is (1)–(2) read against (H)'s two clauses. The first is automatic here by Lemma 9.3(2), so (H) reduces to $z_* > 1$, which by (2) is exactly $b_0 > 0$ or $k(0^+) > 1$ — an equality throughout, so with no boundary subtlety.

(4) cF is admissible by Lemma 7.6, with data (cb_0, ck), so $k_{cF}(0^+) = c\,k(0^+)$ and (cF)'s drift is cb_0; applying (1)–(2) to cF in place of F gives $z_*(cF) = c\,z_*(F)$, whether $b_0 > 0$ or $b_0 = 0$. □

The verified proof never differentiates F: it obtains (2) from two direct bounds on the integral in (7.1), and takes the existence of the limit in (1) from (2) rather than from the convexity of G.

The threshold 1 is not an artifact of one estimate. It arises three separate times, for three different reasons: in the convergence of the Mellin integral of the delayed field (Lemma 9.11), in the application of that lemma at $z - 1$ in Theorem 9.17(2), and in the convergence of $\int_\rho^t (t-r)^{z-2}dr$ inside the same proof — three obligations, one endpoint. And it is where the cone's extreme rays sit: every extreme ray with $b_0 = 0$ has $F(s) \sim \log s$, hence $z_* = 1$

(Proposition 7.8; also Lemma 9.4(2), since $k(0^+) = 1$ on every Dickman ray), so those rays lie exactly *on* the boundary of (H). The correspondence is not exact in either direction — the pure delay is an extreme ray and satisfies (H) outright, having $b_0 > 0$, and the Gamma family at $\gamma \le 1$ fails (H) without being extreme — but the coincidence of the three obligations at the extreme rays' abscissa marks the boundary of (H) as characteristic of the theory rather than of any one argument.

Figure 5 locates the cut in the cone of §7, and the location says what the hypothesis costs. Nothing from the preceding sections is given up: an admissible family failing (H) is still a causal scale space — the representation, the cascade, and the cone geometry all stand. What is withheld is only this section's reading of it: the covariant inversion needs a Mellin strip of width greater than one, and the excluded members — the slowest media, those whose exponents grow at most logarithmically (Lemma 9.4(1)) — do not open one. The shape of the excluded set follows from homogeneity (Lemma 9.4(4)). The Thorin condition selects rays: complete monotonicity of k survives positive multiples, so the subclass is itself a cone. (H) instead cuts across rays: a ray of finite z_* enters (H) beyond the threshold multiple $1/z_*$, so the excluded set is a collar around the apex $F = 0$, pinching to nothing wherever $z_* = \infty$ — every exponent with drift, the stable and Bessel-K families — and reaching exactly the unit point of each Dickman ray. The two cuts are independent: multiples of a single Dickman ray fall on both sides of (H) while never being Thorin — a step is not completely monotone — and the Gamma family falls on both sides while always being Thorin.

Lemma 9.5 (Mellin data of the profile)**.** *Under (H), for* $0 < \operatorname{Re} z < z_*$,

$$\tilde{H}(z) \;=\; \Gamma(z)\,\mathbb{E}\big[T_1^{-z}\big],$$

with $|\tilde{H}(c+i\tau)| \le \mathbb{E}[T_1^{-c}]\,|\Gamma(c+i\tau)|$ *for* $0 < c < z_*$.

Proof. By Tonelli,

$$\tilde{H}(z) = \int_0^\infty s^{z-1}\,\mathbb{E}[e^{-sT_1}]\,ds = \mathbb{E}\Big[\int_0^\infty s^{z-1}e^{-sT_1}ds\Big] = \mathbb{E}[T_1^{-z}]\,\Gamma(z),$$

the inner integral being the Gamma integral at rate T_1. The exchange is legitimate because the same computation run on absolute values gives $\Gamma(c)\,\mathbb{E}[T_1^{-c}]$ with $c = \operatorname{Re} z$, finite exactly when $c < z_*$ — so the strip condition is characteristic of the exchange, not merely sufficient for it. The first clause of (H) is used here, and only here: it gives $T_1 > 0$ almost surely, without which the identity is false rather than merely unproved — at an atom $t = 0$ the inner integral $\int_0^\infty s^{c-1}ds$ diverges, while $\mathbb{E}[T_1^{-c}]$, an integral over $(0,\infty)$, does not see the atom. Since $|T_1^{-z}| = T_1^{-\operatorname{Re} z}$, the identity also gives the stated bound. □

Lemma 9.6 (the profile's transform is integrable on vertical lines)**.** *Under (H), for every* $c \in (0, z_*)$ *the function* $\tau \mapsto \tilde{H}(c+i\tau)$ *is absolutely integrable on* $\mathbb{R}$.

Proof. $|\tilde{H}(c+i\tau)| \le \mathbb{E}[T_1^{-c}]\,|\Gamma(c+i\tau)|$ by Lemma 9.5, the first factor being finite because $c < z_*$. For the second, quadratic decay suffices, and quadratic decay is the functional equation twice. Euler's integral gives $|\Gamma(\sigma+i\tau)| \le \Gamma(\sigma)$ for $\sigma > 0$, the imaginary part only rotating the integrand; and $\Gamma(z+2) = (z+1)z\Gamma(z)$ with $|z|, |z+1| \ge |\tau|$ — both have imaginary part τ — gives $|\Gamma(c+i\tau)|\,\tau^2 \le \Gamma(c+2)$. Adding them,

$$|\Gamma(c+i\tau)|\,(1+\tau^2) \;\le\; \Gamma(c) + \Gamma(c+2),$$

and $(1+\tau^2)^{-1}$ is integrable on the line. (The classical asymptotic $|\Gamma(c+i\tau)| \sim \sqrt{2\pi}\,|\tau|^{c-1/2}e^{-\pi|\tau|/2}$ is stronger, but it needs Stirling in the complex plane and nothing below uses it.) □

The stage set, the ansatz of the lead is made honest in three steps: the symbol exists as a meromorphic function, the functional calculus $x^{-1}B(\theta)$ is given a definition that survives the symbol's poles, and the profiles are verified to be its eigenfunctions.

Lemma 9.7 (the inversion symbol)**.** *Under (H), $\tilde{H}$ is analytic on the strip $0 < \operatorname{Re} z < z_*$ and does not vanish identically there. Define the* inversion symbol

$$B(-z) \;:=\; \frac{\tilde{H}(z+1)}{\tilde{H}(z)} \;=\; z\,\frac{\mathbb{E}\big[T_1^{-z-1}\big]}{\mathbb{E}\big[T_1^{-z}\big]}, \qquad 0 < \operatorname{Re} z < z_* - 1.$$

Then B is meromorphic on that strip, with poles only at the isolated zeros of $\tilde{H}$.

Proof. Analyticity is the identity of Lemma 9.5 read backwards rather than a separate estimate: $m(z) := \mathbb{E}[T_1^{-z}] = \mathbb{E}[e^{-z\log T_1}]$ is the moment-generating function of $-\log T_1$, analytic on the interior of the set of exponents where the corresponding real exponential is integrable — which is $(0, z_*)$, the strip condition verbatim — and Γ contributes the rest. $\tilde{H}$ is not identically zero: both factors are strictly positive at real $z \in (0, z_*)$. Hence its zeros are isolated, and the quotient $\tilde{H}(z+1)/\tilde{H}(z)$ — which is $z\,\mathbb{E}[T_1^{-z-1}]/\mathbb{E}[T_1^{-z}]$ by $\Gamma(z+1) = z\Gamma(z)$ — is meromorphic with poles only there. □

Definition 9.8 (the inversion operator)**.** Fix $c \in (0, z_* - 1)$. For g such that $\tilde{g}$ is defined on the line $\operatorname{Re} z = c$ and $B(-z)\tilde{g}(z)$ is absolutely integrable there, set

$$\big(Ag\big)(x) \;:=\; \frac{1}{x}\cdot\frac{1}{2\pi i}\int_{(c)} x^{-z}\,B(-z)\,\tilde{g}(z)\,dz \;=\; \Big(\frac{1}{x}\,B(\theta)\,g\Big)(x),$$

the second expression being the functional-calculus reading, the Euler operator having Mellin symbol $-z$.

The second equality is an assertion and not a rewriting, and it is worth being explicit about what it asserts: $B(\theta)$ is a functional calculus for a symbol with poles, so what the display says is that there *exists* a function h on $(0,\infty)$ whose Mellin transform is $B(-z)\tilde{g}(z)$ on the line, and that the contour integral computes $h(x)/x$. Absolute integrability on the line is exactly the hypothesis under which the inversion integral recovers such an h. Given h, the rest is elementary and is Lemma 9.9; in particular the transform-level form is $\widetilde{Ag}(z) = B(1-z)\,\tilde{g}(z-1)$, wherever the identity defining h holds pointwise. Every application below *exhibits* its h rather than inferring it — for the profiles $g = H(s\,\cdot)$ it is $h(x) = s\,x\,H(sx)$, Lemma 9.10 — so the cited inversion theorem is available but never spent.

Lemma 9.9 (the operator computes the realising function)**.** *Fix $c > 0$ and g, and let h be a function on $(0,\infty)$ such that*

$$\tilde{h}(z) = B(-z)\,\tilde{g}(z)$$

for almost every z on the line $\operatorname{Re} z = c$, with both $\tilde{h}$ and the inversion integral absolutely convergent there. Then

$$(Ag)(x) = \frac{h(x)}{x} \quad \textit{at every } x > 0 \textit{ where } h \textit{ is continuous,}$$

and, if h is continuous on $(0,\infty)$,

$$\widetilde{Ag}(z) = \tilde{h}(z-1) \quad \textit{for every } z.$$

In particular $\widetilde{Ag}(z) = B(1-z)\,\tilde{g}(z-1)$ wherever the first display holds pointwise.

Proof. The integrand of the contour integral agrees almost everywhere on the line with $\tilde{h}$, so the integral is the inverse Mellin transform of $\tilde{h}$ at height c; under the two convergence hypotheses that recovers $h(x)$ at every point of continuity, which is the first display. The second is then the Mellin transform of $x^{-1}h(x)$, i.e. $\tilde{h}$ with its argument shifted by one. (That the identity on the line can only be asked for almost everywhere is forced: where $\tilde{H}$ vanishes the product $B(-z)\tilde{g}(z)$ has no value while $\tilde{h}(z)$ does; the zeros are isolated by Lemma 9.7, hence meet the line in a null set, and the inversion integral does not see them.) □

Lemma 9.10 (the profiles are eigenfunctions)**.** *Assume (H), let $c \in (0, z_* - 1)$ and $s > 0$. Then the dilate $H(s\,\cdot)$ lies in the domain of Definition 9.8 — its realising function is $h(x) = s\,x\,H(sx)$, exhibited — and*

$$A\big[H(s\,\cdot)\big](x) \;=\; s\,H(sx), \qquad x > 0.$$

Proof. $\widetilde{H(s\cdot)}(z) = s^{-z}\tilde{H}(z)$ and $\widetilde{s\,x\,H(sx)}(z) = s^{-z}\tilde{H}(z+1)$, the second because the weight x shifts the Mellin argument by one. Their ratio is $B(-z)$ wherever $\tilde{H}(z) \neq 0$, which is almost every point of the line. The transform of h converges there by Lemma 9.5 and its inversion integral by Lemma 9.6, both at height $c+1 \in (1, z_*)$, so Lemma 9.9 applies and gives $A[H(s\cdot)](x) = h(x)/x = s\,H(sx)$. □

The image of a profile is the same profile, scaled; so the profiles lie in the domain of every power of A, and by induction $A^n[H(s\,\cdot)] = s^n\,H(s\,\cdot)$, each iterate realised by the exhibited $h(x) = s^n x\,H(sx)$. This is what the jet construction of §12 iterates.

The first verification concerned the operator on single modes; the second concerns the field itself, and it is the section's central identity: what the memory line actually stores.

Lemma 9.11 (the memory line stores fractional integrals)**.** *Under (H), let $f \in X_0$ be causal and $u(\cdot, x) := \Phi_{0,x} f$. Then for every $t > 0$ and $1 < \operatorname{Re} z < z_*$,*

$$\widetilde{u(t,\cdot)}(z) \;=\; \int_0^\infty x^{z-1}\,u(t,x)\,dx \;=\; \tilde{H}(z)\,\big(I^z f\big)(t),$$

absolutely convergent. If moreover $f \in \mathcal{D}$, the same holds for $\partial_t u$ — the X_0-derivative of Lemma A.16, i.e. $u(\cdot, x)$ is the primitive of $\mathbb{E}[f'(\cdot - xT_1)]$ — with $I^z f$ replaced by $I^z f' = I^{z-1} f$; and in that case the identity extends to the wider strip $0 < \operatorname{Re} z < z_*$.

Proof. In the canonical gauge $u(t,x) = \mathbb{E}\big[f(t - xT_1)\big]$. Tonelli applies, since for $c = \operatorname{Re} z > 1$, $\mathbb{E}[T_1^{-c}] \int_0^t y^{c-1}|f(t-y)|\,dy \le \mathbb{E}[T_1^{-c}]\,t^{c-1}\|f\|_1 < \infty$, and gives

$$\int_0^\infty x^{z-1}\,\mathbb{E}\big[f(t - xT_1)\big]\,dx \;=\; \mathbb{E}\big[T_1^{-z}\big] \int_0^t y^{z-1} f(t-y)\,dy \;=\; \mathbb{E}\big[T_1^{-z}\big]\,\Gamma(z)\,(I^z f)(t),$$

which is the claim by Lemma 9.5. (For bounded f the inner integral converges for every $c > 0$, which is the wider strip.) For $f \in \mathcal{D}$, f is bounded — $f(0) = 0$ and $f' \in X_0$ give $\|f\|_\infty \le \|f'\|_1$ — so the weight x^{z-1} is then integrable at the origin on its own, which is where that extension is used. Also for $f \in \mathcal{D}$: $u(\cdot, x) = \mu_{0,x} * f$ and $f = \mathbf{1}_{[0,\infty)} * f'$, so $u(\cdot, x) = \mathbf{1}_{[0,\infty)} * (\mu_{0,x} * f')$ — that is, $u(\cdot, x)$ is the primitive of $\mathbb{E}[f'(\cdot - xT_1)]$, which is what $\partial_t u = \mathbb{E}[f'(\cdot - xT_1)]$ asserts in X_0. (It is *not* an assertion about pointwise derivatives, and would be false as one: f' exists only a.e., and $\mu_{0,x} * f'$ is an L^1 function with no pointwise values to be a derivative at.) Finally $I^z f' = I^{z-1} f$: substituting $f(r) = \int_0^r f'$ and exchanging the order of integration over $0 < \rho \le r \le t$ leaves $\int_\rho^t (t-r)^{z-2} dr = (t-\rho)^{z-1}/(z-1)$, and $\Gamma(z) = (z-1)\Gamma(z-1)$; the semigroup identity $I^{z-1} I^1 = I^z$ is not needed. □

Remark 9.12 (what the core supplies). Of the three conditions defining $\mathcal{D}$ the proofs use exactly three consequences and no others: boundedness, which is what widens the strip; that f is the

primitive of f', which is what the derivative clause asserts; and $f \in X_0$ itself, needed for the Laplace form of Theorem 9.17(2) and *not* implied by the other two, a primitive of an L^1 function lying in L^1 only when $\int_0^\infty f' = 0$. Density and invariance under τ_r and $\Phi_{x,y}$ — the substance of Lemma A.16 — are not used here.

Remark 9.13 (reading of Lemma 9.11; embodiment). Lemma 9.11 gives the motivation of [8], §5, an exact form: at each instant t, the Mellin analysis of the memory line holds precisely the analytic family of Riemann–Liouville integrals $\{(I^z f)(t)\}$ of the *past* signal, weighted by the negative delay-moments $\mathbb{E}[T_1^{-z}]$. The observer's embodiment of its past is fractional integration, with the admissible weightings classified by Theorem 7.3.

It remains to show that nothing else could have been built. The uniqueness argument needs an arena — what counts as a competitor — and one rigidity lemma.

Lemma 9.14 (rigidity of the symbol)**.** *Assume (H), and let B_1, B_2 satisfy*

$$B_1(-z)\,\tilde{H}(z) \;=\; B_2(-z)\,\tilde{H}(z), \qquad 0 < \operatorname{Re} z < z_* - 1.$$

Then $B_1 = B_2$ as meromorphic functions on that strip: they agree on a punctured neighbourhood of every point of it. Where in addition both are continuous — in particular at any point that is a pole of neither — they agree there too.

Proof. Where $\tilde{H}(z) \neq 0$, cancel it. By Lemma 9.7 the zeros of $\tilde{H}$ are isolated, so that cancellation is available on a punctured neighbourhood of *every* point of the strip, the zeros included; hence B_1 and B_2 agree near every point, the point itself excepted. If both are continuous at such a point, the punctured neighbourhood determines the value there and the agreement extends to it. □

Definition 9.15 (the covariant Mellin class)**.** Fix $c \in (0, z_* - 1)$. An operator A' belongs to the *covariant Mellin class* at height c if there is a function $B'(-z)$ on the line $\operatorname{Re} z = c$ such that A' is of the form $x^{-1}B'(\theta)$ in the sense of Definition 9.8: every g in its domain has a realising function h with $\tilde{h}(z) = B'(-z)\tilde{g}(z)$ at every point of the line where $\tilde{H}(z) \neq 0$, and $A'g = h/x$. The weight x^{-1} makes every member scale like ∂_t under $(t, x) \mapsto (\sigma t, \sigma x)$, which is the covariance (A8) requires of an inversion — the class is the covariant one.

Lemma 9.16 (uniqueness of the symbol)**.** *If two operators of the form $x^{-1}B_1(\theta)$ and $x^{-1}B_2(\theta)$ both satisfy $A[H(s\,\cdot)] = s\,H(s\,\cdot)$ for some $s > 0$, at every height $c \in (0, z_* - 1)$ — that is, at every such c both are of that form on $H(s\,\cdot)$ at height c and both send it to $s\,H(s\,\cdot)$ — then $B_1 = B_2$ on the strip. Hence A is the unique inversion within the covariant Mellin class (Definition 9.15).*

Proof. "Of the form $x^{-1}B_i(\theta)$" is the hypothesis that each operator has a realising function in the sense of Definition 9.8. Fix a height $c \in (0, z_* - 1)$. Both operators send $H(s\,\cdot)$ to $s\,H(s\,\cdot)$, so on $(0, \infty)$ their realising functions at height c agree, since $x^{-1}h_1 = x^{-1}h_2$ on $(0, \infty)$ is $h_1 = h_2$; call the common value h. Then $B_1(-z)\,\tilde{g}(z) = \tilde{h}(z) = B_2(-z)\,\tilde{g}(z)$ at every point of the line $\operatorname{Re} z = c$ where the identity holds, and $\tilde{g}(z) = s^{-z}\tilde{H}(z)$, so cancelling the nonvanishing factor s^{-z} gives $B_1(-z)\tilde{H}(z) = B_2(-z)\tilde{H}(z)$ at every point of the line where $\tilde{H}(z) \neq 0$, and trivially at its zeros, where both sides vanish. As c ranges over $(0, z_* - 1)$ the lines fill the strip, which is the hypothesis of Lemma 9.14, and that lemma concludes. (This is where the hypothesis at every height is spent: with the identity on a single line one would need B_1, B_2 meromorphic on the strip and the identity theorem to reach the rest of it, and neither is assumed.) Two economies are worth recording, since both tools are natural to reach for and neither is needed: no injectivity of the inverse Mellin transform, the shared realising function carrying the information that injectivity would otherwise have to recover; and a single dilation rather than all $s > 0$, the dilate contributing only the factor s^{-z}, which never vanishes. □

The pieces assemble. Recall the problem: evolve the field in present time over the memory line, the signal entering as boundary data,

$$\partial_t u = A\,u, \qquad u(t,0) = f(t), \qquad u(0,x) = 0.$$

The theorem delivers, over the covariant Mellin class: the inversion exists and has the field's Laplace profiles as eigenfunctions (1); the scale-space field solves the problem, the equation holding mode-by-mode in the Laplace variable and as an identity of Mellin transforms in the scale variable (2); and the inversion is unique (3) — the *operator*, that is, not yet the solution. What "solves" means, and the sense in which the problem is genuinely *solved*, is settled after the theorem: Definition 9.19 names the notion of solution that clause (2) informally reads off, Corollary 9.21 adds *solution*-uniqueness, and Corollary 9.24 completes Hadamard's demands.

Theorem 9.17 (Theorem 4′: the signaling form)**.** *Assume (H) and work in the canonical gauge. Then:*

1. (Eigenfunctions.) *For every $s > 0$ the profile $H(s\,\cdot)$ lies in the domain of Definition 9.8 for every $c \in (0, z_* - 1)$, and*

$$A\big[H(s\,\cdot)\big](x) \;=\; s\,H(sx), \qquad x > 0.$$

2. (The field solves the signaling problem.) *Let $f \in \mathcal{D}$ and $u(\cdot,x) := \Phi_{0,x} f$. Then u is causal in t with $u(\cdot,x) \to f$ in X_0 as $x \downarrow 0$, and*

   - *Laplace form: for every $s > 0$, $A\big[\hat{u}(s,\cdot)\big] = s\,\hat{u}(s,\cdot)$, with $\hat{u}(s,0+) = \hat{f}(s)$;*
   - *time-domain (Mellin) form: for every $t > 0$ and every z with $1 < \operatorname{Re} z < z_*$ and $\tilde{H}(z-1) \neq 0$ — all of the strip off an isolated set, Remark 9.18 —*

$$\widetilde{\partial_t u(t,\cdot)}(z) \;=\; B(1-z)\,\widetilde{u(t,\cdot)}(z-1),$$

   both transforms absolutely convergent; that is, $\partial_t u = A\,u$ read through the Mellin transform in x.

3. (Uniqueness.) *A is the unique operator in the covariant Mellin class (Definition 9.15) with property (1), at every height c; in particular, it is the operator of the signaling problem (2).*

Proof. (1) The realising function is exhibited: put $h(x) := s\,x\,H(sx)$, so that $\tilde{h}(z) = s^{-z}\tilde{H}(z+1)$, the weight x shifting the Mellin argument by one and the two powers of s combining as $s \cdot s^{-(z+1)}$. On the other side $\tilde{g}(z) := \widetilde{H(s\,\cdot)}(z) = s^{-z}\tilde{H}(z)$, so

$$B(-z)\,\tilde{g}(z) = \frac{\tilde{H}(z+1)}{\tilde{H}(z)} \cdot s^{-z}\tilde{H}(z) = s^{-z}\tilde{H}(z+1) = \tilde{h}(z)$$

at every point of the line where $\tilde{H}(z) \neq 0$, which is all but an isolated — hence null — set of it. The cancellation is the whole content, being Lemma 9.7's recursion with the denominator cleared; it should *not* be described as a product containing no division, since the division is there and clearing it is what costs the exceptional set. Both convergence hypotheses of Definition 9.8 hold for h at height c, being Lemma 9.5 and Lemma 9.6 at height $c+1 \in (1, z_*)$, and h is continuous because H is. So h realises $B(\theta)H(s\,\cdot)$ and Lemma 9.9 gives $A[H(s\cdot)](x) = h(x)/x = s\,H(sx)$: the weight x^{-1} of Definition 9.8 is exactly what cancels the weight x that shifted the transform, so the exhibition of h and the eigenfunction relation are one statement rather than two.

(2) Causality and the boundary attainment are (A3) and (A7). The Laplace form is (1) applied to $\hat{u}(s,\cdot) = \hat{f}(s)H(s\,\cdot)$. The Mellin form combines Lemma 9.11 with Lemma 9.7 — note that it applies the former at $z-1$, so it is the wider strip that is used, available because $f \in \mathcal{D}$ is bounded; and that the cancellation below is of $\tilde{H}(z-1)$, hence valid wherever that does not vanish, which by Lemma 9.7 is all of the strip off an isolated set:

$$B(1-z)\,\widetilde{u(t,\cdot)}(z-1) = \frac{\tilde{H}(z)}{\tilde{H}(z-1)} \cdot \tilde{H}(z-1)\,(I^{z-1}f)(t) = \tilde{H}(z)\,(I^{z-1}f)(t) = \widetilde{\partial_t u(t,\cdot)}(z).$$

(3) is Lemma 9.16. □

Remark 9.18 (classical interpretation; poles). The time-domain equation in Theorem 9.17(2) is stated at the level of Mellin transforms, and that is the honest general formulation: B may have poles, at zeros of $\tilde{H}$, and polynomial growth along vertical lines, so $Au(t,\cdot)$ need not be given by an absolutely convergent integral for rough f. On the field the identity is nevertheless regular, the factor $\tilde{H}(z-1)$ cancelling every denominator. That last statement is about the product as a *meromorphic function*; read pointwise it needs $\tilde{H}(z-1) \neq 0$, which holds off an isolated set by Lemma 9.7, and Theorem 9.17(2) is stated accordingly. The distinction recurs through this section — it is the same one that makes Lemma 9.14's conclusion an equality of meromorphic functions rather than of functions — and is worth keeping in view: a cancellation performed on symbols has to be justified at each point where the identity is used.

When the symbol B is a *polynomial*, A is a local differential operator and the equation is a classical PDE. That is precisely the situation of Theorem 5′ in §10, and of Example 9.27 below.

Theorem 9.17(2) closes with an informal gloss — "that is, $\partial_t u = A\,u$ read through the Mellin transform in x". Clause (3) proves uniqueness of the *operator* A within the covariant Mellin class, not uniqueness of the *solution* u: two different claims the single word "unique" can hide. The gap is not merely presentational. The eigenvalue recursion $s\,\tilde{g}(z) = B(1-z)\,\tilde{g}(z-1)$ that pins $\hat{u}(s,\cdot)$'s Mellin data has the classical period-one ambiguity: its general solution is $\tilde{g}(z) = s^{-z}\tilde{H}(z)\,p(z)$ for an *arbitrary* 1-periodic p, the same obstruction Bohr–Mollerup closes for Γ, one level down. What follows closes it here. Definition 9.19 names what a solution of the signaling problem *is*, mirroring Theorem 9.17(2)'s own clauses rather than restating them. Lemma 9.20 kills the periodic ambiguity, under a normalization the field satisfies for free: its own modes are dominated by the profile, uniformly over one period, the theory's counterpart of the log-convexity hypothesis Bohr–Mollerup imposes. And Corollary 9.21 assembles existence with *solution*-uniqueness under that normalization — existence and uniqueness in the profile-dominated Mellin class, which Theorem 9.17 alone does not deliver. Of Hadamard's three demands this supplies the first two, and Corollary 9.24 completes the third from the representation, with stability under perturbation of the exponent besides. The problem is therefore well posed in the stated class; Remark 9.25 marks the boundary of the claim. Corollary 9.22 adds a classical-solution clause, under an extra decay hypothesis on the negative-moment function m that the Gamma, stable and Bessel-K families satisfy and the pure delay fails — correctly, that family belonging instead to Theorem 10.6's territory.

Definition 9.19 (a Mellin solution of the signaling problem)**.** Assume (H) and work in the canonical gauge, and let $f \in \mathcal{D}$. A *Mellin solution of the signaling problem for* f is a field $u : \mathbb{R} \times (0,\infty) \to \mathbb{R}$ such that $u(\cdot,x) \in X_0$ for every $x > 0$ — causal in t at every scale, the honest form of the boundary condition $u(0,\cdot) = 0$, since X_0 is by definition the causal elements of X — with $u(\cdot,x) \to f$ in X_0 as $x \downarrow 0$, and, writing $\hat{u}(s,x) := \int_0^\infty e^{-st}u(t,x)\,dt$ for $s > 0$:

1. *(Laplace form.)* For every $s > 0$: $x \mapsto \hat{u}(s,x)$ is continuous on $(0,\infty)$; its Mellin transform $z \mapsto \widetilde{\hat{u}(s,\cdot)}(z)$ is analytic on the whole strip $0 < \operatorname{Re} z < z_*$; at every height $c \in (0, z_*-1)$, $\hat{u}(s,\cdot)$ lies in the domain of A (Definition 9.8); and

$$A\big[\hat{u}(s,\cdot)\big] \;=\; s\,\hat{u}(s,\cdot), \qquad \hat{u}(s,0+) \;=\; \hat{f}(s);$$

2. *(Mellin form.)* For every $t > 0$ and every z with $1 < \operatorname{Re} z < z_*$ and $\tilde{H}(z-1) \neq 0$,

$$\widetilde{\partial_t u(t,\cdot)}(z) \;=\; B(1-z)\,\widetilde{u(t,\cdot)}(z-1),$$

both transforms absolutely convergent.

Lemma 9.20 (rigidity of a profile-dominated mode)**.** *Assume (H) and fix $s > 0$. Let $\tilde{g}$ be analytic on the strip $0 < \operatorname{Re} z < z_*$ and satisfy the pole-free eigenvalue recursion*

$$s\,\tilde{g}(z)\,\tilde{H}(z-1) \;=\; \tilde{H}(z)\,\tilde{g}(z-1), \qquad 1 < \operatorname{Re} z < z_*,$$

an identity of holomorphic functions on that range, with no exceptional set. Suppose the periodic factor

$$p(z) \;:=\; \frac{s^z\,\tilde{g}(z)}{\tilde{H}(z)},$$

a priori meromorphic on $0 < \operatorname{Re} z < z_*$ with possible poles only at the isolated zeros of $\tilde{H}$ (Lemma 9.7), is bounded on some period substrip $\{c \le \operatorname{Re} z \le c+1\} \subset (0, z_*)$, off those zeros. Then p is constant: there is $\kappa \in \mathbb{C}$ with

$$\tilde{g}(z) \;=\; \kappa\, s^{-z}\,\tilde{H}(z), \qquad 0 < \operatorname{Re} z < z_*.$$

Proof. Where $\tilde{H}(z) \neq 0$ and $\tilde{H}(z-1) \neq 0$, dividing the recursion by $s\,\tilde{H}(z)\tilde{H}(z-1)$ and multiplying by s^z gives $p(z) = p(z-1)$, so the periodic factor satisfies the shift relation off the (isolated) zeros of $\tilde{H}$ on both sides.

Step 1: p extends to a meromorphic, 1-periodic function on $\mathbb{C}$. $\tilde{H}$'s zeros in $(0, z_*)$ are isolated (Lemma 9.7), hence locally finite there: every compact subset of the open strip contains finitely many of them — accumulation at the strip's *boundary* is not excluded, and is not needed; so, writing $S := (0, z_*)$ as a set of vertical strips $S_n := S+n = \{z : n < \operatorname{Re} z < n+z_*\}$ for $n \in \mathbb{Z}$, and $p_n(z) := p(z-n)$, meromorphic on S_n with possible poles only at the (locally finite) translated zeros of $\tilde{H}$: for $n \le k \le m$ and $z \in S_n \cap S_m$, z lies in every intermediate S_k (since $\operatorname{Re} z > m \ge k$ and $\operatorname{Re} z < n+z_* \le k+z_*$), so the shift relation applied at each consecutive pair $(k, k+1)$, $n \le k < m$, gives $p_n(z) = p_{n+1}(z) = \cdots = p_m(z)$: every two translates agree wherever they overlap. Since $z_* > 1$ (H), consecutive strips overlap and $\{S_n\}_{n\in\mathbb{Z}}$ covers $\mathbb{C}$, so there is a well-defined meromorphic $P : \mathbb{C} \to \mathbb{C}$ with $P|_{S_n} = p_n$ for every n; it is 1-periodic, since for $z \in S_n$, $z+1 \in S_{n+1}$ and $P(z+1) = p_{n+1}(z+1) = p(z-n) = p_n(z) = P(z)$.

Step 2: P is bounded off its singular set, on all of $\mathbb{C}$. $P|_{S_0} = p$ is bounded on the given substrip $\{c \le \operatorname{Re} z \le c+1\}$, off the zeros of $\tilde{H}$ there, by hypothesis. By periodicity, P is bounded by the *same* constant on every unit translate $\{c+n \le \operatorname{Re} z \le c+1+n\}$, $n \in \mathbb{Z}$, off P's singular set there; and these translates cover $\mathbb{C}$, consecutive ones sharing an edge. So P is bounded, by one constant, off its singular set, everywhere.

Step 3: removability and Liouville. At each point w_0 of P's singular set — a priori a pole, though this step shows it is not — P is bounded on a punctured neighbourhood of w_0 (Step 2), so by Riemann's removable singularity theorem P extends to be analytic at w_0, with value the limit from nearby — hence bounded by the same constant, since nearby values are. Doing this at every such point is licensed because the singular set is locally finite in $\mathbb{C}$: every point of $\mathbb{C}$ has a compact neighbourhood inside some open S_n — again $z_* > 1$, the open intervals $(n, n+z_*)$ covering $\mathbb{R}$ — and there the possible poles are the finitely many translated zeros of Step 1. The result is an *entire* function $\hat{P}$, equal to P off its singular set and bounded on all of $\mathbb{C}$ by the constant of Step 2. Liouville's theorem gives $\hat{P} \equiv \kappa$ for some $\kappa \in \mathbb{C}$.

Closing. On $S_0 = (0, z_*)$, $\hat{P} = P = p$ off the zeros of $\tilde{H}$, so $s^z\tilde{g}(z) = \kappa\,\tilde{H}(z)$ for every $z \in (0, z_*)$ off that isolated set. Both sides are analytic on the whole (connected, open) strip $(0, z_*)$ — the left because $\tilde{g}$ is, by hypothesis; the right because $\tilde{H}$ is (Lemma 9.7) — and agree off a set with no accumulation point in the strip, so by the identity theorem they agree everywhere on it. Dividing by $s^z \neq 0$ gives $\tilde{g}(z) = \kappa\, s^{-z}\tilde{H}(z)$ for every $z \in (0, z_*)$. □

The verified proof exchanges the order of the steps: it propagates the bound over the strip $(0, z_*)$ and removes the singularities there first, so that only analytic pieces are glued into P and nothing meromorphic appears.

Corollary 9.21 (the signaling problem: existence and uniqueness in the profile-dominated Mellin class)**.** *Assume (H), let $f \in \mathcal{D}$, and let A be the inversion operator of Definition 9.8. Then:*

1. (Existence.) *The scale-space field $u(t,x) := (\Phi_{0,x} f)(t)$ is a Mellin solution of the signaling problem for f (Definition 9.19); its periodic factor $p_s(z) := s^z \widetilde{\hat{u}(s,\cdot)}(z)/\tilde{H}(z)$ (Lemma 9.20) is the* constant *$\hat{f}(s)$, for every $s > 0$ — in particular profile-dominated.*

2. (Uniqueness.) *It is the* unique *Mellin solution of the signaling problem for f whose modes are profile-dominated — i.e., such that for every $s > 0$ the periodic factor of Lemma 9.20 is bounded on some period substrip: if v is another such solution, then $v(\cdot,x) = u(\cdot,x) = (\Phi_{0,x} f)(\cdot)$ in X_0, for every $x > 0$.*

Proof. (1) is Theorem 9.17(1) for the domain and eigenfunction clauses and Theorem 9.17(2) for causality, the boundary limit, the Laplace and Mellin forms; continuity of $x \mapsto \hat{u}(s,x) = \hat{f}(s)H(sx)$ and analyticity of its transform on the whole strip are as in the statement. Since $\widetilde{\hat{u}(s,\cdot)}(z) = \hat{f}(s)\, s^{-z}\tilde{H}(z)$, the periodic factor is $p_s(z) = s^z \cdot \hat{f}(s)\, s^{-z}\tilde{H}(z)/\tilde{H}(z) = \hat{f}(s)$ identically.

(2) Let v be a Mellin solution for f with profile-dominated modes, and fix $s > 0$. Write $\tilde{g}(z) := \widetilde{\hat{v}(s,\cdot)}(z)$, analytic on the whole strip by Definition 9.19(1). At every height $c \in (0, z_* - 1)$, $\hat{v}(s,\cdot)$ lies in the domain of A with $A[\hat{v}(s,\cdot)] = s\,\hat{v}(s,\cdot)$; taking Mellin transforms (Definition 9.8's transform identity, as in the proof of Theorem 9.17(1)) gives $s\,\tilde{g}(z) = B(1-z)\tilde{g}(z-1)$ off the zeros of $\tilde{H}(z-1)$ on the line $\operatorname{Re} z = c+1$; multiplying out $B(1-z) = \tilde{H}(z)/\tilde{H}(z-1)$ clears the denominator to $s\,\tilde{g}(z)\tilde{H}(z-1) = \tilde{H}(z)\tilde{g}(z-1)$ off an isolated set of the line, and both sides are analytic on the strip $1 < \operatorname{Re} z < z_*$ (the left by hypothesis, the right by Lemma 9.7), so by the identity theorem the identity holds *everywhere* on that range — the pole-free recursion Lemma 9.20 asks for. By hypothesis its periodic factor is bounded on some period substrip, so Lemma 9.20 gives $\kappa_s \in \mathbb{C}$ with $\tilde{g}(z) = \kappa_s\, s^{-z}\tilde{H}(z)$ for every $z \in (0, z_*)$ — that is, $\widetilde{\hat{v}(s,\cdot)}$ and $\kappa_s \widetilde{H(s\cdot)}$ agree on the strip.

Both $x \mapsto \hat{v}(s,x)$ (continuous, Definition 9.19(1)) and $x \mapsto \kappa_s H(sx)$ (continuous, H a Laplace transform) have this same Mellin transform, absolutely convergent on the line $\operatorname{Re} z = c+1$ for any $c \in (0, z_* - 1)$ (Lemma 9.6, up to the constant κ_s). Mellin inversion on a line of absolute convergence recovers a continuous function from its transform, so

$$\hat{v}(s,x) \;=\; \kappa_s\, H(sx), \qquad x > 0.$$

Letting $x \downarrow 0$ on both (continuous) sides and using $\hat{v}(s,0+) = \hat{f}(s)$ (Definition 9.19(1)) and $H(0) = 1$ (H the Laplace transform of the law of T_1) gives $\kappa_s = \hat{f}(s)$. So $\hat{v}(s,x) = \hat{f}(s)H(sx) = \hat{u}(s,x)$ for every $x > 0$ and every $s > 0$: $t \mapsto v(t,x)$ and $t \mapsto u(t,x)$, both in X_0 (Definition 9.19(1) for v, Lemma A.16 for $u = \Phi_{0,x} f \in \mathcal{D} \subset X_0$), have the same Laplace transform for every $s > 0$, so by Corollary 3.11 applied to their difference they agree: $v(\cdot,x) = u(\cdot,x)$ in X_0, for every $x > 0$. □

Corollary 9.22 (classical solutions of the signaling problem)**.** *Assume (H), let $f \in \mathcal{D}$ with f' additionally bounded and continuous, and let $u(t,x) := (\Phi_{0,x} f)(t)$ be the scale-space field. Suppose that for some $c \in (0, z_* - 1)$,*

$$\int_{\mathbb{R}} (1 + |\tau|)\, \big|m(c + 1 + i\tau)\big|\, d\tau \;<\; \infty, \qquad m(z) := \mathbb{E}\big[T_1^{-z}\big]$$

— decay of the negative-moment function along one vertical line. Then for every $t > 0$, the function $x \mapsto u(t,x)$ lies in the domain of A at height c, with realising function $h(x) = x\,\partial_t u(t,x)$, and

$$\partial_t u(t,x) \;=\; \big(A\,u(t,\cdot)\big)(x) \qquad \text{at every } x > 0 :$$

the field solves the signaling equation classically*, pointwise in x and at every t, not merely at the level of Mellin transforms.*

Proof. By Lemma 9.11, $u(\cdot,x)$ is the primitive of the field of f': $u(t,x) = \int_0^t \mathbb{E}[f'(\rho - xT_1)]\,d\rho$ for every t. The integrand $\rho \mapsto \mathbb{E}[f'(\rho - xT_1)]$ is continuous, by dominated convergence (f' bounded and continuous, $|f'| \le \|f'\|_\infty$ an integrable dominating constant on any compact ρ-range), so the fundamental theorem of calculus gives $t \mapsto u(t,x)$ a genuine pointwise derivative at *every* t, equal to $g(x) := \mathbb{E}[f'(t - xT_1)]$ — continuous in x by the same dominated convergence argument, run in x instead of ρ. Set $h(x) := x\,g(x) = x\,\partial_t u(t,x)$, continuous.

$\tilde{H}(z)(I^{z-1}f)(t) = \widetilde{\partial_t u(t,\cdot)}(z)$ for $0 < \operatorname{Re} z < z_*$ (Lemma 9.11, the derivative clause, extended strip since $f \in \mathcal{D}$ is bounded), and the weight x shifts the Mellin argument by one, so $\tilde{h}(z) = \widetilde{\partial_t u(t,\cdot)}(z+1)$. By Lemma 9.7's recursion $B(-z) = \tilde{H}(z+1)/\tilde{H}(z)$ and Theorem 9.17(2)'s transform identity shifted by one, $\tilde{h}(z) = B(-z)\,\widetilde{u(t,\cdot)}(z)$ for $0 < \operatorname{Re} z < z_* - 1$, off the isolated zeros of $\tilde{H}(z)$: the realising-function condition of Definition 9.8, with $u(t,\cdot)$ in the role of its g, at symbol B, height c.

For the vertical-integrability bound, write $z = c + 1 + i\tau$. By definition of I, $\Gamma(z-1)(I^{z-1}f)(t) = \int_0^t (t-r)^{z-2} f(r)\,dr$, and $f \in \mathcal{D}$ is bounded ($\|f\|_\infty \le \|f'\|_1$, Lemma A.16), so

$$\big|\Gamma(z-1)(I^{z-1}f)(t)\big| \;\le\; \|f\|_\infty \int_0^t (t-r)^{\operatorname{Re} z - 2}\,dr \;=\; \|f\|_\infty \frac{t^{\operatorname{Re} z - 1}}{\operatorname{Re} z - 1} \;=\; \|f\|_\infty \frac{t^c}{c}$$

(using $\operatorname{Re} z = c + 1 > 1$). Since $\Gamma(z) = (z-1)\Gamma(z-1)$, $|\Gamma(z)(I^{z-1}f)(t)| \le |z-1|\,\|f\|_\infty\, t^c/c$, and $\widetilde{\partial_t u(t,\cdot)}(z) = m(z)\,\Gamma(z)(I^{z-1}f)(t)$ (Lemma 9.5), so

$$\big|\tilde{h}(c+i\tau)\big| \;=\; \big|\widetilde{\partial_t u(t,\cdot)}(c+1+i\tau)\big| \;\le\; \frac{\|f\|_\infty\, t^c}{c}\,|c+i\tau|\,\big|m(c+1+i\tau)\big|.$$

Since $|c+i\tau| \le (c+1)(1+|\tau|)$, the displayed hypothesis makes the right side integrable in τ over $\mathbb{R}$: $\tilde{h}$ is vertically integrable at height c.

So $u(t,\cdot)$ lies in the domain of A at height c with realising function h, which is continuous; Lemma 9.9 gives $A[u(t,\cdot)](x) = h(x)/x = g(x) = \partial_t u(t,x)$, at every $x > 0$. □

Remark 9.23 (which families are classical). Corollary 9.22's hypothesis is a statement about $m(z) = \mathbb{E}[T_1^{-z}]$'s decay along one vertical line, checked against every family the article names.

Gamma ($F(s) = \gamma\log(1+s)$, $\gamma > 1$, Example 9.26): $m(z) = \Gamma(\gamma - z)/\Gamma(\gamma)$, analytic for $\operatorname{Re} z < \gamma$. For $c \in (0, \gamma - 1)$, $\gamma - (c+1) \in (0, \gamma - 1)$ avoids Γ's poles, and the classical vertical-line decay $|\Gamma(\sigma + i\tau)| \sim \sqrt{2\pi}\,|\tau|^{\sigma - 1/2} e^{-\pi|\tau|/2}$ (already used at Lemma 9.6) gives $|m(c+1+i\tau)|$ exponential decay in $|\tau|$: the hypothesis holds.

Stable, $\alpha < 1$ ($F(s) = s^\alpha$, Example 9.28): $m(z) = \Gamma(z/\alpha)/(\alpha\Gamma(z))$. The same asymptotic applied to numerator and denominator gives $|m(c+1+i\tau)|$ decaying like $e^{-(\pi/2)(1/\alpha - 1)|\tau|}$ up to a polynomial factor, exponential since $\alpha < 1$: the hypothesis holds.

Bessel-K ($T_1 = 1/(2\gamma_a)$, Proposition 8.17; $m(z) = 2^z\Gamma(a+z)/\Gamma(a)$, Example 9.27): $|2^{c+1+i\tau}| = 2^{c+1}$ is constant in τ, so the decay of $|m(c+1+i\tau)|$ is exactly that of $|\Gamma(a+c+1+i\tau)|$ — exponential, by the same asymptotic as the Gamma family. **Checked directly, not assumed**: the closed form $2^z\Gamma(a+z)$ looks, read along the real axis, like it grows without bound, but $2^{i\tau}$ has modulus 1 and contributes nothing to the decay along a *vertical* line; the family is classical.

Pure delay ($F(s) = b_0 s$, $b_0 > 0$): $T_1 \equiv b_0$ a.s., so $m(z) = b_0^{-z}$, and $|m(c+1+i\tau)| = b_0^{-(c+1)}$ is constant in τ: not integrable against $(1+|\tau|)\,d\tau$. The hypothesis fails, correctly. The transport

operator's eigenfunctions are the constant-modulus exponentials, whose Mellin data does not decay, and this family's inversion is instead Theorem 10.6's territory — a polynomial (here, first-order) symbol, local by construction, not reached by Corollary 9.22's route at all.

Hadamard's third demand, continuous dependence, is all that remains, and the representation supplies it — together with a stability under perturbation of the exponent itself that the equation alone would not suggest.

Corollary 9.24 (the signaling problem is well posed)**.** *Assume (H), work in the canonical gauge, and let $f \in \mathcal{D}$, with $u(t,x) := (\Phi_{0,x} f)(t)$ the scale-space field. Then the signaling problem for f satisfies all three of Hadamard's demands — existence, uniqueness, continuous dependence on the data — and is stable under perturbation of the exponent besides:*

1. (Existence.) *u is a Mellin solution of the signaling problem for f (Definition 9.19), with profile-dominated modes (Corollary 9.21(1)).*

2. (Uniqueness.) *It is the unique Mellin solution for f with profile-dominated modes (Corollary 9.21(2)).*

3. (Continuous dependence on the data.) *For $g \in \mathcal{D}$ with field $v(t,x) := (\Phi_{0,x} g)(t)$,*

$$\sup_{x>0} \|u(\cdot,x) - v(\cdot,x)\|_1 \;\le\; \|f - g\|_1,$$

and along the evolution,

$$\|u(\cdot,y) - v(\cdot,y)\|_1 \;\le\; \|u(\cdot,x) - v(\cdot,x)\|_1 \qquad (0 < x < y).$$

4. (Stability under perturbation of the exponent.) *Let F_n be admissible exponents with $F_n(s) \to F(s)$ for every $s > 0$, and let $u_n(t,x) := (\mu_{0,x}^{(n)} * f)(t)$ be the corresponding fields, $\mu_{0,x}^{(n)}$ the canonical-gauge kernel of F_n — the law of $x\,T_1^{(n)}$ with $\mathbb{E}[e^{-sT_1^{(n)}}] = e^{-F_n(s)}$. Then for every $x > 0$,*

$$u_n(\cdot,x) \;\to\; u(\cdot,x) \quad \text{in } X_0.$$

When every F_n satisfies (H), the left side is, by (1)–(2) applied to F_n, the unique profile-dominated Mellin solution of the perturbed problem: the solutions of the perturbed problems converge, at every scale, to the solution of the limit problem.

Proof. (1) and (2) are Corollary 9.21, restated clause for clause.

(3) By Theorem 7.3 the solution operator at scale x is convolution with the probability measure $\mu_{0,x}$, so $u(\cdot,x) - v(\cdot,x) = \mu_{0,x} * (f - g)$; and convolution with a probability measure μ carried by $[0,\infty)$ is nonexpansive on X: by Tonelli, $\int |(\mu * h)(t)|\,dt \le \iint |h(t-y)|\,d\mu(y)\,dt = \mu([0,\infty))\,\|h\|_1 = \|h\|_1$. That gives the first display, uniformly in x. For $0 < x < y$ the hemigroup factorizes $\Phi_{0,y} = \Phi_{x,y}\Phi_{0,x}$, so $u(\cdot,y) - v(\cdot,y) = \mu_{x,y} * \big(u(\cdot,x) - v(\cdot,x)\big)$ with $\mu_{x,y}$ again a probability measure, and the same bound applies.

(4) Two steps: the kernels converge weakly, and weak convergence convolves into X_0-convergence.

The kernels. Fix $x > 0$. The Laplace transform of $\mu_{0,x}^{(n)}$ at s is $e^{-F_n(xs)}$, which converges to $e^{-F(xs)}$, the transform of $\mu_{0,x}$, for every $s > 0$; so $\mu_{0,x}^{(n)} \to \mu_{0,x}$ weakly by Proposition 3.8. (The same argument runs on any increment: the transform of $\mu_{x,y}^{(n)}$ is $e^{-(F_n(ys) - F_n(xs))}$, so every $\mu_{x,y}^{(n)} \to \mu_{x,y}$ weakly as well.)

The fields. We show that $\mu_n \to \mu$ weakly, for probability measures on $[0,\infty)$, forces $\mu_n * h \to \mu * h$ in X for every $h \in X$, and apply this at $\mu_n = \mu_{0,x}^{(n)}$, $h = f$. First let g be continuous with compact support and $g \ge 0$. Then $(\mu_n * g)(t) = \int g(t-y)\,d\mu_n(y) \to (\mu * g)(t)$ at every t,

the integrand being bounded and continuous in y; and the integrals are all equal: by Tonelli, $\int(\mu_n * g) = \mu_n([0,\infty)) \int g = \int g$, and likewise for μ. Pointwise convergence of nonnegative integrable functions whose integrals converge to the limit's integral upgrades to L^1 convergence — Scheffé's lemma, and here it is one line: writing $a_+ := \max(a, 0)$,

$$\int \left|\mu_n * g - \mu * g\right| \;=\; \int (\mu_n * g - \mu * g) \;+\; 2\int \big(\mu * g - \mu_n * g\big)_+,$$

the first term zero as just computed, and the second tending to zero by dominated convergence, since $0 \le (\mu * g - \mu_n * g)_+ \le \mu * g \in L^1$ and the integrand tends to zero pointwise. A general continuous compactly supported g splits as $g_+ - g_-$, both again continuous with compact support. Finally, for $h \in X$ and $\varepsilon > 0$ pick such a g with $\|h - g\|_1 < \varepsilon$ — continuous functions of compact support are dense in L^1 — and the contraction of (3) closes the estimate:

$$\|\mu_n * h - \mu * h\|_1 \;\le\; \|\mu_n * (h - g)\|_1 + \|\mu_n * g - \mu * g\|_1 + \|\mu * (g - h)\|_1 \;\le\; 2\varepsilon + \|\mu_n * g - \mu * g\|_1,$$

whose limit superior over n is at most 2ε, for every $\varepsilon > 0$. Applied at $\mu_n = \mu^{(n)}_{0,x}$ and $h = f$ this gives $u_n(\cdot, x) \to u(\cdot, x)$ in X; every field here is causal, f and the kernels being carried by $[0,\infty)$, so the convergence is in X_0. The closing sentence of (4) is (1)–(2) applied to F_n, whose hypotheses — (H) for the exponent, $f \in \mathcal{D}$ for the data — are exactly what is assumed. □

Remark 9.25 (what the stability rests on; what is not claimed). The stability clauses of Corollary 9.24 come from the representation rather than from the equation: the solution operator at each scale is convolution with a probability measure (Theorem 7.3), and the exponent stability is the continuity theorem for Laplace transforms (Proposition 3.8) read on the kernels. Neither uses (H), which enters only through the solution concept of clauses (1)–(2); both therefore hold for every admissible family and every signal in X, not only on the class the signaling problem is posed on. In the scale-Cauchy reading of Appendix A, the second contraction of clause (3) says that a perturbation of the record at one scale does not grow at any coarser one. What is *not* claimed is a stability estimate in a topology adapted to A itself — continuity of the solution in the Mellin-class data, or convergence of the operators behind clause (4). The conclusion there is convergence of the solution fields, never of the symbols, and the distinction is real: $F(s) + s/n \to F(s)$ pointwise with every perturbed exponent carrying drift, so $z_* = \infty$ along the whole sequence (Lemma 9.4(2)) while the limit's z_* may be finite, and the strips the symbols live on do not converge to the limit's. Whether the signaling problem is well posed in a topology adapted to A is a question this article leaves open.

Three worked cases calibrate the construction, ordered by the growth of the symbol — bounded, quadratic, fractional — which is the order of the media's regularity: an integral operator, a classical PDE, and a fractional operator whose medium, for $\alpha < \frac{1}{2}$, does not exist at all.

Example 9.26 (the Gamma family; scale-recursive form)**.** For $F(s) = \gamma \log(1 + s)$ with $\gamma > 1$ we have $\tilde{H}(z) = \Gamma(z)\Gamma(\gamma - z)/\Gamma(\gamma)$ and $B(-z) = z/(\gamma - 1 - z)$, i.e.

$$B(\theta) = -\,\theta\,\big(\theta + \gamma - 1\big)^{-1}, \qquad A = -\,\frac{1}{x}\,\theta\,\big(\theta + \gamma - 1\big)^{-1},$$

a *rational* Mellin symbol — the memory-line mirror of the time-side resolvent $\gamma\,\partial_t(1 + x\partial_t)^{-1}$ of Example A.22 in Appendix A. Explicitly, $w := (\theta + \gamma - 1)^{-1}u$ solves $x\,w_x + (\gamma - 1)w = u$, with the regular-at-0 solution

$$w(t, x) = x^{-(\gamma-1)} \int_0^x y^{\gamma-2}\,u(t, y)\,dy,$$

so the time evolution at scale x is computed from the field at *finer* scales by a single weighted average — scale-recursive, one-sided from the boundary, matching the causal architecture of the time side.

Example 9.27 (parabolic gauge; the Bessel family is the quadratic case)**.** In the α-gauge (similarity variable $x^{1/\alpha}s$; Remark 6.4) the profiles are $H(x^{1/\alpha}s)$, the Euler operator acts on the profile through $\alpha^{-1}\theta$, and the inversion becomes $A = x^{-1/\alpha}\,B(\alpha\,\theta)$. Take the parabolic gauge $\alpha = \frac{1}{2}$, i.e. the coordinate ξ of Remark 6.4, and the Bessel-K family of Proposition 8.17, $T_1 = 1/(2\gamma_a)$: then $\mathbb{E}[T_1^{-z}] = 2^z\,\Gamma(a+z)/\Gamma(a)$, so

$$B(-z) = 2z(z+a), \qquad A = \xi^{-2}\Big(\tfrac{\theta^2}{2} - a\theta\Big) = \tfrac{1}{2}\,\partial_\xi^2 + \frac{\beta}{\xi}\,\partial_\xi, \quad \beta = \tfrac{1}{2} - a.$$

The symbol is a *quadratic polynomial*, so A is local: the signaling form of the Bessel-K scale space is exactly the Bessel diffusion-with-drift medium, and Theorem 9.17(2) becomes the classical signaling problem $\partial_t u = \frac{1}{2}u_{\xi\xi} + \frac{\beta}{\xi}u_\xi$ with $u(t,0) = f(t)$ and $u(0,\xi) = 0$.

Example 9.28 (extremal stable; recovering Theorem 4 of 2005)**.** For $F(s) = s^\alpha$ we have $\tilde{H}(z) = \Gamma(z/\alpha)/\alpha$ and

$$B(-z) \;=\; \frac{\Gamma\big(\frac{z+1}{\alpha}\big)}{\Gamma\big(\frac{z}{\alpha}\big)} \;\sim\; \Big(\tfrac{z}{\alpha}\Big)^{1/\alpha} \quad (z \to \infty),$$

a symbol of order $1/\alpha$. On the profiles $e^{-(sx)^\alpha}$ the right-sided Liouville derivative acts by $D^{1/\alpha}_{x,-}e^{-cx} = c^{1/\alpha}e^{-cx}$ with $c = s^\alpha$, i.e. it also produces the eigenvalue s; so the Mellin representative and the operator of [8], Thm. 4, agree on the span of the field profiles. The translation-invariant representative is a coincidence of homogeneity; the Mellin representative is the one that survives in general.

Remark 9.29 (Markov media). Within the class of Theorem 9.17, the operators A that additionally satisfy the positive maximum principle (Definition 10.1) — i.e. generate a sub-Markov process on the memory half-line, so that the medium is a physical transport medium — are, after the logarithmic substitution $x = e^\xi$, exactly those whose symbol B is the Laplace exponent of a Lévy process. We record this without proof here. By the Lamperti correspondence ([21]) these media are positive self-similar Markov processes absorbed at the boundary, and the kernels are the boundary-arrival times $T_x \overset{d}{=}$ absorption time from depth x.

The spectral orientation of the jumps is *not* constrained. For spectrally positive media the process creeps downward through every level and the hemigroup (A6) holds *pathwise*, the kernels being genuine level-passage times; but media with downward jumps are also admissible, the factorization then holding in law though not pathwise, being inherited from the exponent F rather than from the paths.

The Gamma family is the standard example: unfolding Example 9.26, its medium is the multiplicative jump process that waits at scale x an Exp(mean x) time and jumps to σx with $\sigma \sim \mathrm{Beta}(\gamma - 1, 1)$, reaching the boundary by accumulation of jumps; a first-jump computation confirms $\mathbb{E}_x[e^{-s\tau_0}] = (1+sx)^{-\gamma}$.

A necessary condition in all cases — again recorded without proof — is quadratic symbol growth, $B(\theta) = O(\theta^2)$. By Example 9.28 the homogeneous members with $\alpha < \frac{1}{2}$, of symbol order $1/\alpha > 2$, admit *no* Markov medium, although Theorem 9.17 always supplies the analytic signaling form: the operator exists, the process does not. The locality theorem settles the local sub-case — a polynomial symbol of degree at most two — but the general classification of which admissible F are Markov-realizable is the memory-line inverse problem, and it is left open; it is listed with the open problems in §13.

The remark is context, not part of this paper's proved record: the two claims it records without proof — the Lévy–Lamperti form of the PMP media after the logarithmic substitution, and the necessity of quadratic symbol growth — belong to the separate article on the memory-line inverse problem announced in §13, and nothing proved in this paper depends on them. What *is* proved here about media is Theorem 10.6's classification of the local case, and, for the Gamma medium, the first-jump computation quoted in the remark.

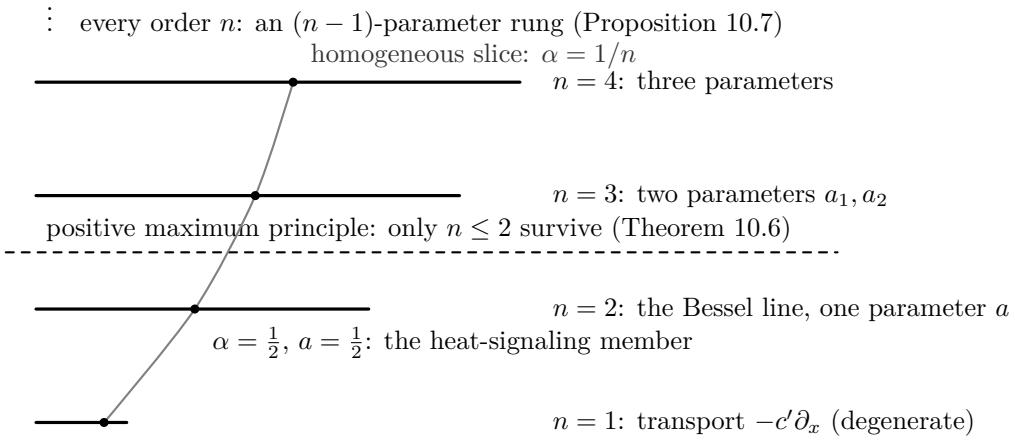


Figure 8: The ladder of local media. Locality alone admits media of every order: rung n is the $(n-1)$-parameter family of Proposition 10.7, with $n = 1$ the degenerate transport member. The positive maximum principle cuts the ladder between orders two and three (Theorem 10.6), leaving the transport member and the Bessel line. The homogeneous (semigroup) members are one point per rung, the stable cases $\alpha = 1/n$ of [8] (Remark 10.9); the maximum principle acts on them the same way, and what survives of the slice is the single point $\alpha = \frac{1}{2}$ — the heat-signaling equation — which under the hemigroup axioms unfolds into the full Bessel rung.

10 The locality theorem

Theorem 5 of [8] singles out $\alpha = \frac{1}{2}$ — the heat signaling equation — as the unique member of the stable family whose memory-line evolution is a local, positivity-respecting differential equation. This section proves the hemigroup analogue.

The structure is genuinely two-layered: locality *alone* opens a ladder of admissible local media of every order, each rung an $(n-1)$-parameter family. It is the positive maximum principle — the requirement that the medium be a physical transport medium — that closes the ladder at order two, where it leaves the Bessel family together with the degenerate transport member of case (1) (Figure 8). Throughout we assume (H), work in the canonical gauge, write $m(z) := \mathbb{E}[T_1^{-z}]$, and take A and B as in §9.

Definition 10.1 (locality; positive maximum principle)**.** The inversion operator A is *local of order n* if it agrees with a differential expression $\sum_{j=0}^{n} c_j(x)\,\partial_x^j$, with $c_j \in C((0,\infty))$ and $c_n \not\equiv 0$, on $C_c^\infty((0,\infty))$ *and on the profiles* $H(s\,\cdot)$, $s > 0$. It satisfies the *positive maximum principle* (PMP) if $(Ag)(x_0) \le 0$ whenever $g \in C_c^\infty((0,\infty))$ attains a nonnegative maximum at x_0.

The profiles are part of the test class because the ($\Rightarrow$) direction of Lemma 10.2 must reach the symbol, and $C_c^\infty((0,\infty))$ is disjoint from the only class on which B's behavior is controlled; the maximum principle stays on $C_c^\infty((0,\infty))$, where a profile, having no interior maximum, would say nothing.

Why the maximum principle, and why on $C_c^\infty((0,\infty))$? The cascade is positive at the level of the axioms — kernels are probability measures, so a nonnegative signal has a nonnegative field — but that constrains A only along the fields the cascade produces. The positive maximum principle is the standard infinitesimal form of the same demand: the generator-level certificate of a positivity-preserving evolution on *all* data, not merely on the solution manifold. Courrège's theorem — the citation carrying the order bound in Theorem 10.6's proof — classifies the operators that satisfy it. It is also this theory's counterpart of a classical scale-space axiom. The requirement descends from Koenderink's causality hypothesis — every feature at a coarse resolution must have a cause at a finer one, enforced at the extrema of the image ([18], p. 364).

It is formalized as *non-enhancement of local extrema*: the value at a local maximum must not grow as the evolution runs ([22], Def. 3; the relation to the maximum principle — close kin, not identical — is worked out in its Appendix B). In the spatial theory that axiom singles out the Gaussian; here, on the memory line, its operator-side form will single out the Bessel rung. Requiring it of a candidate medium is thus no extra aesthetic: it asks that the differential expression transport nonnegative quantities nonnegatively at every point — the property the kernels already have globally.

The first step is a dictionary. Locality speaks about the operator, while everything the theory controls — the recursion, the poles, the profiles' transforms — speaks about the symbol. The Euler calculus $x^j\partial_x^j = \theta(\theta-1)\cdots(\theta-j+1)$ is the translation between them, and covariance leaves a local operator no freedom beyond a polynomial in θ.

Lemma 10.2 (covariant local = polynomial symbol)**.** *A is local of order n if and only if B is a polynomial of degree n; in that case the covariance built into the Mellin class (Definition 9.15) forces the coefficients*

$$c_j(x) \;=\; \gamma_j\, x^{\,j-1}, \qquad A \;=\; \frac{1}{x}\sum_{j=0}^{n}\gamma_j\,\theta(\theta-1)\cdots(\theta-j+1) \;=\; \frac{1}{x}\,B(\theta).$$

Proof. ($\Leftarrow$) $x^j\partial_x^j = \theta(\theta-1)\cdots(\theta-j+1)$, so a polynomial B, expanded in falling factorials, gives $x^{-1}B(\theta) = \sum \gamma_j x^{j-1}\partial_x^j$, a differential expression of order $\deg B$. This has to be checked on both classes of Definition 10.1. On $C_c^\infty((0,\infty))$ it is the Euler-factor identity and the linearity of the inverse Mellin transform along the line. On a profile $H(s\,\cdot)$ the same argument applies to $h(x) = \sum_j \gamma_j x^j \partial_x^j H(sx)$, whose transform is $P(z)s^{-z}\tilde H(z)$ by the identity used in ($\Rightarrow$) below; that transform is absolutely integrable on the line because $|\Gamma(c+i\tau)|$ decays faster than every power of τ, so inverting recovers h and $x^{-1}h$ is $A[H(s\,\cdot)]$.

($\Rightarrow$) Every operator of the covariant Mellin class (Definition 9.15) satisfies the covariance $A\Delta_\sigma = \sigma^{-1}\Delta_\sigma A$, where $(\Delta_\sigma g)(x) = g(x/\sigma)$: dilation multiplies the Mellin transform by σ^z, and multiplying the symbol integrand by σ^z translates the inverse transform's argument, both reducing to $x^{-(c+i\tau)}\sigma^{c+i\tau} = (x/\sigma)^{-(c+i\tau)}$. Imposing this on $\sum c_j(x)\partial_x^j$ and comparing coefficients of $g^{(j)}(x/\sigma)$ gives $c_j(x)\sigma^{-j} = \sigma^{-1}c_j(x/\sigma)$, i.e. $c_j(x) = c_j(1)x^{j-1}$. Rewriting via falling factorials, A agrees with $x^{-1}P(\theta)$ for the polynomial $P(w) = \sum_j \gamma_j E_j(w)$, where $E_j(w) = \prod_{i<j}(-w-i)$.

It remains to identify P with B, and the comparison is made on the profiles rather than on $C_c^\infty((0,\infty))$ — the class Definition 10.1 was widened to include, and the only one on which B's behavior is known. Fix $s>0$ and put $g = H(s\,\cdot)$. Locality gives $(Ag)(x) = \sum_j \gamma_j x^{j-1}\partial_x^j H(sx)$, while Theorem 9.17(1)'s eigenfunction relation gives $(Ag)(x) = sH(sx)$; equating them and clearing the weight,

$$s\,x\,H(sx) \;=\; \sum_{j=0}^{n}\gamma_j\,x^j\,\partial_x^j H(sx), \qquad x>0.$$

The left side is the realising function h of Definition 9.8 for g, with $\tilde h(w) = s^{-w}\tilde H(w+1)$. On the right, $\partial_x^j H(sx) = (-s)^j\int t^j e^{-sxt}\mu(dt)$, so $x\mapsto x^j\partial_x^j H(sx)$ has, by the Mellin shift x^j, the dilation factor, and Lemma 9.5 applied to the measure $t^j\mu(dt)$, the transform $E_j(w)\tilde g(w)$ throughout $0<\operatorname{Re} w<z_*$. Each term converges there, so the sum may be transformed term by term, giving $\tilde h(w) = P(w)\tilde g(w)$ — which is exactly the hypothesis of Lemma 9.16 with P in the place of a candidate symbol. Hence B and P agree as meromorphic functions on $0<\operatorname{Re} z<z_*-1$; equivalently, since $\tilde g(w) = s^{-w}\tilde H(w)$ never loses its nonvanishing factor,

$$\tilde H(z+1) \;=\; P(z)\,\tilde H(z)$$

at every point of the strip, and $B(z) = P(z)$ at every point where $\tilde H(z)\neq 0$. No injectivity of the inverse Mellin transform is used, and none is available. □

With the dictionary in hand the word *operator* can be retired: local means B polynomial, and the remainder of the section is arithmetic of the moment function $m(z) = \mathbb{E}[T_1^{-z}]$. Three consequences of a polynomial symbol come at once.

Lemma 10.3 (the moment recursion)**.** *Suppose A is local of order n, i.e. B is a polynomial of degree n. Then:*

1. *$B(0) = 0$; write $B(-z) = z\,Q(z)$ with Q a polynomial of degree $n-1$;*
2. *$z_* = \infty$: all negative moments of T_1 are finite;*
3. *$m(z+1) = Q(z)\,m(z)$ for all $z > 0$, and $Q > 0$ on $(0,\infty)$.*

Proof. (1) By Lemma 9.7, $B(-z) = z\,m(z+1)/m(z)$ on the strip. As $z \downarrow 0$, $m(z) \to 1$ by dominated convergence — the integrand t^{-z} tends to 1 and is dominated by $1 + t^{-1}$, integrable because (H) places 1 strictly below z_* — and $m(z+1)$ stays bounded, splitting the integral at $t = 1$ against a fixed $\zeta \in (1, z_*)$. So $B(-z) \to 0$, and B being a polynomial, $B(0) = 0$; $Q(z) := B(-z)/z$ is then a polynomial of degree $n-1$.

(2) On $0 < z < z_* - 1$ we have the identity $\tilde{H}(z+1) = B(-z)\tilde{H}(z)$ between functions analytic on the strip $0 < \operatorname{Re} z < z_*$. If $z_* < \infty$, the right side is analytic at $z = z_* - 1 + \varepsilon$ for small ε, which extends $\tilde{H}$ analytically through the real point z_*. But $\tilde{H}(z) = \int_0^\infty x^{z-1} H(x)\,dx$ has nonnegative integrand — $H(x) = \mathbb{E}\,e^{-xT_1} \geq 0$ — which is exactly the hypothesis of the cited theorem: for a Laplace transform with nonnegative integrand, the real endpoint of the convergence strip is a singularity when finite ([42], Ch. II, §5, Thm. 5b, pp. 58–59 — usually called the Pringsheim–Landau theorem, though Widder himself credits Hamburger). Hence $z_* = \infty$.

(3) The recursion is now Lemma 9.7 on all of $(0,\infty)$, and positivity of Q follows from $m > 0$. □

Clause (2) is worth a geometric pause: locality does not merely respect the standing hypothesis, it forces $z_* = \infty$, so a local medium sits at infinite distance from the excluded collar of Figure 5 — (H), assumed throughout this section, is the one hypothesis that costs nothing here. Clause (3) is the engine, and it arrives with the section's central difficulty attached: the recursion $m(z+1) = Q(z)\,m(z)$ determines m only up to a 1-periodic factor. This is the period-one ambiguity of §9 again, met there in the mode recursion and closed analytically (Lemma 9.20: boundedness over one period, gluing, Liouville); here it is closed by the other classical mechanism, Bohr–Mollerup — log-convexity plus normalization pins the solution. The hypothesis is free, which is the next lemma; the uniqueness it buys, at the linear Q the theorem needs, is the one after.

Lemma 10.4 (log-convexity)**.** *$z \mapsto m(z)$, valued in $[0,\infty]$, is log-convex on $(0,\infty)$: for $z_1, z_2 > 0$ and $\theta \in [0,1]$,*

$$m\big(\theta z_1 + (1-\theta) z_2\big) \;\leq\; m(z_1)^{\theta}\, m(z_2)^{1-\theta}.$$

If moreover T_1 has no atom at the origin (the first clause of (H)), then $\log m$ is a real-valued convex function on $(0, z_*)$, hence on $(0,\infty)$ when $z_* = \infty$.

Proof. Hölder at the exponent pair $\big(\frac{1}{\theta}, \frac{1}{1-\theta}\big)$, applied to $T_1^{-z_1}$ and $T_1^{-z_2}$, gives

$$m\big(\theta z_1 + (1-\theta) z_2\big) \;=\; \mathbb{E}\big[(T_1^{-z_1})^{\theta}\,(T_1^{-z_2})^{1-\theta}\big] \;\leq\; m(z_1)^{\theta}\, m(z_2)^{1-\theta}$$

at every $\theta \in [0,1]$ at once — Cauchy–Schwarz being the case $\theta = \frac{1}{2}$. Since that inequality is the definition of log-convexity, nothing remains to upgrade. □

Lemma 10.5 (the shifted Gamma recursion; Krull–Webster at a linear Q)**.** *Let $c, a > 0$ and let $m : (0,\infty) \to (0,\infty)$ be log-convex and satisfy*

$$m(z+1) \;=\; c\,(z+a)\,m(z) \quad (z>0), \qquad m(z) \to 1 \quad (z \downarrow 0).$$

Then

$$m(z) \;=\; c^{\,z}\,\frac{\Gamma(a+z)}{\Gamma(a)} \qquad (z>0).$$

Proof. Choose an integer $n \geq a$ and put $d := n - a \geq 0$. Define, for $y > 0$,

$$f(y) \;:=\; \frac{m(y+d)\;c^{-(y+d)}}{P(y)}, \qquad P(y) := \prod_{j=0}^{n-1}(y+j),$$

which uses m only on $(d,\infty) \subseteq (0,\infty)$. Two shifts cancel: the recursion gives $m(y+1+d) = c\,(y+n)\,m(y+d)$, since $d + a = n$, while $P(y+1)\,y = P(y)\,(y+n)$; together with $c^{-(y+1+d)} = c^{-(y+d)}c^{-1}$ this leaves

$$f(y+1) \;=\; y\,f(y) \qquad (y>0).$$

f is positive, and $\log f(y) = \log m(y+d) - (y+d)\log c - \log P(y)$ is convex on $(0,\infty)$: the first term is Lemma 10.4 precomposed with a translation, the second is affine, and the third is convex because $-\log(y+j)$ is, log being concave. Bohr–Mollerup applied to $f/f(1)$ therefore gives $f = f(1)\,\Gamma$ on $(0,\infty)$.

Write $C := f(1)$ and $G(z) := c^z\Gamma(z+a)$. Unwinding the definition of f at $y = z - d$ and using $P(y)\,\Gamma(y) = \Gamma(y+n)$ with $y + n = z + a$ gives $m(z) = C\,G(z)$ for every $z > d$. Both m and G satisfy the same recursion on $(0,\infty)$, so $m(z+k)\,G(z) = m(z)\,G(z+k)$ for every $k \in \mathbb{N}$; choosing k with $z + k > d$ and cancelling $G(z+k) > 0$ extends $m = C\,G$ to all of $(0,\infty)$. Finally $z \downarrow 0$ gives $C\,\Gamma(a) = 1$, since $c^z \to 1$ and Γ is continuous at $a > 0$. □

The pieces assemble, in the order the proof runs: Courrège's theorem bounds the order at two; the moment recursion is solved rung by rung — a constant Q forces a deterministic delay, a linear Q a Gamma law; and Fourier uniqueness converts the moment data into the law of the kernel. What the maximum principle buys is the first step alone — everything after it is the arithmetic prepared above.

Theorem 10.6 (Theorem 5′: the locality theorem)**.** *Assume (H). The inversion operator A of Theorem 9.17 is local* and *satisfies the positive maximum principle* ***iff*** *one of the following holds.*

1. (Order 1; pure transport, degenerate.) *$B(\theta) = -c'\theta$ with $c' > 0$: then $A = -c'\,\partial_x$, the kernels are the deterministic delays $\mu_{0,x} = \delta_{x/c'}$, and the signaling problem is the transport equation. This member has no smoothing and is excluded by requiring absolutely continuous kernels (Remark 7.5).*

2. (Order 2; the Bessel family.) *After a normalization of the unit of time: $B(\theta) = 2\,\theta(\theta - a)$ for some $a > 0$; the kernels are the inverse-gamma family $T_1 \overset{d}{=} 1/(2\gamma_a)$ of Proposition 8.17; and in the parabolic gauge ξ (Remark 6.4) the signaling problem is the classical degenerate-parabolic PDE*

$$\partial_t u \;=\; \tfrac{1}{2}\,\partial_\xi^2 u + \frac{\beta}{\xi}\,\partial_\xi u, \qquad \beta = \tfrac{1}{2} - a, \qquad u(t,0) = f(t), \quad u(0,\xi) = 0.$$

Within case (2), the depth-homogeneous (semigroup) members are exactly $a = \frac{1}{2}$, i.e. $\beta = 0$ — the heat signaling equation — recovering [8], Thm. 5, as the zero-drift point of the family. The kernel moments in case (2) satisfy $\mathbb{E}\,T_x^n < \infty$ iff $n < a$.

Proof. ($\Rightarrow$) Definition 10.1's pair of hypotheses — locality (agreement with a differential expression on $C_c^\infty((0,\infty))$) together with the positive maximum principle there — is exactly Courrège's *local* positive maximum principle, whose Corollaire 2 ([5], p. 2-10) bounds the order of such an operator by 2: it is a pure second-order diffusion operator with no jump part. By Lemma 10.2, $\deg B \le 2$, and $B(0) = 0$ by Lemma 10.3(1) — equivalently, there is no killing, which is (A5).

Order 1. Here $Q \equiv c'$ is constant, so $m(z+1) = c'm(z)$ with $m(0+) = 1$. By Lemma 10.4, $\varphi := \log m$ is convex on $(0,\infty)$ and satisfies $\varphi(z+1) - \varphi(z) = \log c'$ for every $z > 0$. The difference quotients of a convex function are nondecreasing, and on each unit interval they average to $\log c'$; a nondecreasing function whose average over every unit interval is the same constant equals that constant identically, so φ is affine. Hence $m(z) = {c'}^z$, the constant fixed by $m(0+) = 1$. Thus $T_1 = 1/c'$ a.s., $B(-z) = c'z$, i.e. $B(\theta) = -c'\theta$ and $A = -c'x^{-1}\theta = -c'\partial_x$; positivity of Q gives $c' > 0$.

Order 2. Here $Q(z) = q_1 z + q_0$ with $q_1 > 0$, by positivity of Q as $z \to \infty$ in Lemma 10.3(3), and $q_0 = Q(0) = m(1) > 0$; write $Q(z) = q_1(z+a)$ with $a := q_0/q_1 > 0$. The recursion $m(z+1) = q_1(z+a)\,m(z)$ with $m(0+) = 1$ — the normalization established in the proof of Lemma 10.3(1) — has a *unique* log-convex solution by Lemma 10.5, namely

$$m(z) \;=\; q_1^z\,\frac{\Gamma(a+z)}{\Gamma(a)} \;=\; \mathbb{E}\Big[\big(q_1\,\gamma_a\big)^z\Big],$$

and m is log-convex by Lemma 10.4, hence equals it. So $m(z) = \mathbb{E}[(q_1\gamma_a)^z]$ for every real $z > 0$. Both sides are, as functions of z, the moment functions of $\log(1/T_1)$ and of $\log(q_1\gamma_a)$: they are analytic for $\operatorname{Re} z > 0$ ($m = \tilde{H}/\Gamma$ by Lemma 9.7 and Γ's own analyticity, the right side being entire), and, since $|T_1^{-z}|$ and $|(q_1\gamma_a)^z|$ are bounded by 1 on the line $\operatorname{Re} z = 0$, each is continuous up to it as well. The identity therefore extends, first by the identity theorem across the open half-plane, then by continuity to the line $\operatorname{Re} z = 0$, where it says that $\log(1/T_1)$ and $\log(q_1\gamma_a)$ have the same Fourier transform; Fourier uniqueness then identifies the two laws, giving $T_1 \overset{d}{=} 1/(q_1\gamma_a)$. (This is the one place in this proof that spends transform uniqueness on the scale variable itself, rather than on moment data alone.) The time normalization $q_1 = 2$ gives Proposition 8.17, the profile is the Bessel-K function, and the operator is computed in Example 9.27: $B(\theta) = 2\theta(\theta - a)$, with the parabolic-gauge form as displayed.

($\Leftarrow$) *Case (1).* Admissibility is the pure-drift member $F(s) = s/c'$ of (7.1), and a first-order operator satisfies the PMP trivially, since $g'(x_0) = 0$ at an interior maximum. *Case (2).* Admissibility of the inverse-gamma family is Proposition 8.17; locality is Example 9.27; and, in the parabolic gauge ξ, a second-order operator $\frac{1}{2}\partial_\xi^2 + \frac{\beta}{\xi}\partial_\xi$ with nonnegative leading coefficient satisfies the PMP, because at an interior nonnegative maximum $g'(\xi_0) = 0$ and $g''(\xi_0) \le 0$, so $(Ag)(\xi_0) = \frac{1}{2}g''(\xi_0) \le 0$.

The homogeneity claim. By Corollary 7.4, depth-homogeneous members have $F(s) = c\,s^\alpha$; within the inverse-gamma family the exponent is a pure power only for $a = \frac{1}{2}$. Indeed, by Proposition 8.17, $e^{-F(s)} = \frac{2^{1-a}}{\Gamma(a)}(\sqrt{2s})^a K_a(\sqrt{2s})$, and the large-argument asymptotic $K_a(w) \sim \sqrt{\pi/(2w)}\,e^{-w}$ as $w \to \infty$ gives

$$F(s) \;=\; \sqrt{2s} \;-\; \Big(\tfrac{a}{2} - \tfrac{1}{4}\Big)\log s \;+\; O(1), \qquad s \to \infty,$$

which is a pure power of s only when the logarithmic coefficient vanishes, i.e. $a = \frac{1}{2}$ — where $K_{1/2}$ is elementary and $H(y) = e^{-\sqrt{2y}}$ exactly, the $\frac{1}{2}$-stable, i.e. heat, case. The moment statement is Proposition 8.17. □

What if the maximum principle is dropped? Locality alone is far from empty, and where its families sit in the classification of §7 is specific: strictly inside it, twice over. Every rung of the ladder below has $z_* = \infty$ (Lemma 10.3(2)); and every rung — the real-spectrum families of

Proposition 10.7 — is hyperbolically completely monotone, hence a generalized gamma convolution: it lies in the *Thorin* subclass of Figure 5. §7 delivered a continuum of admissible media, and §8 cut the Thorin subcone as a tractable representation class; locality now cuts, out of the same cone, a countable union of finite-parameter slices, and lands inside Thorin of its own accord — the demand for a local embodiment is an independent road into the same subclass, so long as the spectrum is real (Remark 10.8). The homogeneous member of each rung is the stable point $\alpha = 1/n$ (Remark 10.9): what the semigroup could see of the ladder was one point per rung.

Proposition 10.7 (the ladder: locality without the maximum principle)**.** *Fix $n \geq 2$, $c' > 0$ and $a_1, \dots, a_{n-1} > 0$, and let*

$$T_1 \stackrel{d}{=} \frac{1}{c' \gamma_{a_1} \gamma_{a_2} \cdots \gamma_{a_{n-1}}} \quad (\textit{independent factors}), \qquad m(z) = c'^{\,z} \prod_{i=1}^{n-1} \frac{\Gamma(a_i + z)}{\Gamma(a_i)}.$$

Then the family is admissible in the sense of Theorem 7.3 and satisfies (H), and its inversion operator is local of order *n, with polynomial symbol*

$$B(-z) \;=\; c'\, z \prod_{i=1}^{n-1} (z + a_i);$$

the kernels are Meijer G-function densities. Conversely, every local admissible family of order n has m equal to the unique log-convex solution of $m(z+1) = Q(z)m(z)$ for its polynomial Q; when the roots of Q are real — necessarily ≤ 0, giving $Q(z) = c' \prod (z + a_i)$ with $a_i > 0$ — this is the family above, and the laws are identified from m as in the proof of Theorem 10.6.

Proof. Admissibility. Gamma densities are hyperbolically completely monotone (HCM), and the HCM class is closed under products of independent variables and under reciprocals, while HCM laws are generalized gamma convolutions and hence self-decomposable ([2], Thm. 5.1.1, p. 69; Thm. 4.3.1, p. 57, whose regularity hypotheses are removed on p. 68; and the inclusion GGC $\subset$ SD, a remark on p. 30). So $-\log \mathbb{E}e^{-sT_1}$ is of the form (7.1), and (H) holds because all negative moments are finite and $\mathbb{P}(T_1 = 0) = 0$.

The symbol. $B(-z) = z\, m(z+1)/m(z) = c' z \prod (z + a_i)$, a polynomial, so A is local of order n by Lemma 10.2.

The converse. This is Lemma 10.3 and Lemma 10.4 together with the Krull–Webster uniqueness theorem ([40], Thm. 3.1, p. 609), as in the proof of Theorem 10.6 — the general theorem now, the recursion carrying $n - 1$ Gamma factors, for which Lemma 10.5's single-factor case is not enough. Two hypotheses of Webster's theorem must be discharged, not assumed. The asymptotic condition is automatic: Q is a polynomial, so $Q(z+w)/Q(z) \to 1$ as $z \to \infty$ for each fixed $w > 0$. The normalization is not automatic — Webster pins the solution with $f(1) = 1$, where this proposition's normalization is $m(0+) = 1$ — so apply Webster to $\hat{m} := m/m(1)$, which satisfies $\hat{m}(1) = 1$ and the same recursion; this gives uniqueness of $\hat{m}$ and hence of m, the constant $m(1) = Q(0)\, m(0+) = Q(0)$ being read off the recursion at $z = 0$. A real root $-a_i \geq 0$ of Q would contradict $Q > 0$ on $(0, \infty)$ together with $Q(0) = m(1) > 0$, so real roots give $a_i > 0$, and the Krull–Webster solution is the displayed gamma product.

The Meijer G identification. The product $\gamma_{a_1} \cdots \gamma_{a_{n-1}}$ has a $G^{\,n-1,0}_{0,n-1}$ density, for arbitrary distinct shapes, by [35], Thm. 1, p. 722; the reciprocal and the scale factor c' are then absorbed by the G-function transformation rules ([31], eqs. (1.96) and (1.97), p. 22). □

Remark 10.8 (complex roots). Whether complex-conjugate root pairs can support an admissible family is left open; we conjecture not. The question is taken up in a companion note (in preparation), where spectrally leading pairs are excluded and order three is settled.

Remark 10.9 (the stable slice; Gauss multiplication). Depth-homogeneity intersects the ladder at the equally spaced parameters $a_j = j/n$ and $c' = n^n$: by the Gauss multiplication formula,

$$m(z) \;=\; n^{nz} \prod_{j=1}^{n-1} \frac{\Gamma(j/n+z)}{\Gamma(j/n)} \;=\; n\,\frac{\Gamma(nz)}{\Gamma(z)}, \qquad \tilde{H}(z) = n\,\Gamma(nz), \qquad H(y) = e^{-y^{1/n}},$$

the $\frac{1}{n}$-stable laws — recovering the classical identity that represents the $\frac{1}{n}$-stable subordinator as a scaled reciprocal product of $n-1$ gamma variables. Thus the local cases $\alpha = 1/n$ of [8] are precisely the homogeneous slice of the ladder, and each unfolds under the hemigroup axioms into an $(n-1)$-parameter inhomogeneous family of local media. The maximum principle eliminates $n \geq 3$ in both settings; what remains at $n = 2$ is the single point $\alpha = \frac{1}{2}$ there, and the Bessel line here.

Remark 10.10 (what the theorem settles). Theorem 10.6 completes the resolution of the practicality problem of Remark 8.18 *within the local class*: the drift weight $\beta = \frac{1}{2} - a$ is a design parameter, and any prescribed number of finite kernel moments is achievable by a local, positivity-respecting, continuously scale-covariant, time-causal medium, namely $n < a$. The residual trade-off is the tail: inverse-gamma kernels have polynomial tails t^{-1-a} for every finite a, so *all* moments — exponential localization — still require leaving the local class for the Gamma-type corner of Proposition 8.8.

Homogeneity, locality and moment control thus interact as follows: homogeneity together with covariance forces stable kernels, which have no mean; locality together with covariance and the PMP allows any finite number of moments but polynomial tails; full moment control requires nonlocal delay accrual.

11 Implementation

This section records how the scale spaces of Theorem 7.3 are computed, and compares the three practically distinguished corners — Gamma, Bessel, stable — from the implementer's point of view. The story is a three-step reduction: the scale direction can be sampled *exactly*, for any family (Proposition 11.1); for the Gamma family the time direction then needs only γ first-order states per increment (Proposition 11.2); and the whole Thorin subclass — every admissible family with completely monotone memory kernel, the Bessel and stable corners included — is approximated from inside the axiom class by finite Thorin truncations, realized as banks of such sections exactly when the weights are integers (Proposition 11.6). Only the genuinely non-Thorin members are left to direct convolution. The structural fact that organizes everything is the first step — the *scale direction never needs to be discretized approximately*:

Proposition 11.1 (exactness at the knots)**.** *Let $0 = x_0 < x_1 < \cdots < x_M$ be any finite set of scale knots and define the discrete cascade*

$$u_0 := f, \qquad u_{k+1} := \mu_{x_k,\, x_{k+1}} * u_k.$$

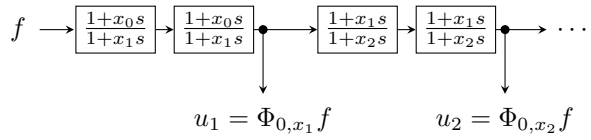
Then $u_k = \Phi_{0,x_k} f$ exactly*, for every knot set; and refining the knot set changes nothing at the old knots.*

Proof. Immediate induction from the hemigroup (A6): $\mu_{0,x_{k+1}} = \mu_{x_k,x_{k+1}} * \mu_{0,x_k}$. □

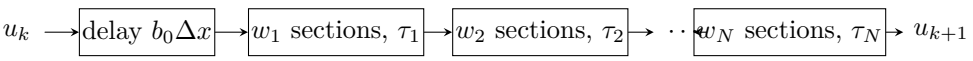
So the only discretization error in a digital implementation is the time discretization of the increment filters, and intermediate scales can be added at any time without recomputation error. This is a genuine advantage over discretizing the memory-line evolution of §9, where the scale direction carries truncation error; the cascade is not an approximation scheme for the family but the family itself, sampled.

(a) the Gamma cascade over the knot ladder $x_0 < x_1 < x_2 < \cdots$ (here $\gamma = 2$)

γ identical first-order sections per increment; each tap is exact (Prop. 11.1)



(b) the Thorin chain for one increment: one Gamma cascade per node (τ_i, w_i), in series



node i's sections at time constants x/τ_i; the transfers multiply — the kernels convolve — so the chain is serial, in any order; positivity and unit mass exact at every truncation level (Prop. 11.6)

Figure 9: The implementation substrate. (a) For the Gamma family the increment between adjacent knots is a cascade of γ identical first-order pole–zero sections, and the record at each knot is read off a tap; the taps are exact for every knot set (Proposition 11.1), so the scale direction is sampled, not approximated. (b) For a general Thorin-subclass member the increment is approximated from inside the axiom class by a serial chain of Gamma cascades, one per atom (τ_i, w_i) of the Thorin measure, led by the drift's pure time shift: the Thorin sum lives in the exponent, so the transfer functions multiply and the kernels convolve — a series connection, in any order — and every truncation level is itself an admissible family (Proposition 11.6).

For an implementer the property to want next is *rationality* of the increment transfer: a rational transfer function is realized by finitely many first-order states, time-recursively, with no stored history. The Gamma family has it (Figure 9).

Proposition 11.2 (the Gamma cascade)**.** *For the Gamma family with integer γ, the increment filter is rational:*

$$\widehat{\mu}_{x_k,x_{k+1}}(s) \;=\; \Big(\frac{1 + x_k\, s}{1 + x_{k+1}\, s}\Big)^{\gamma},$$

a cascade of γ identical *first-order pole–zero sections (pole at $-1/x_{k+1}$, zero at $-1/x_k$). Each section admits the one-state realization, free of input differentiation,*

$$\dot{w} = \frac{y - w}{x_{k+1}}, \qquad v \;=\; \frac{x_k}{x_{k+1}}\, y \;+\; \Big(1 - \frac{x_k}{x_{k+1}}\Big)\, w,$$

(y input, v output), whose transfer is $(1+x_k s)/(1+x_{k+1}s)$. The output is a convex combination *of input and state.*

Proof. The increment transfer is the ratio of the endpoint transforms: by the hemigroup (A6), $\widehat{\mu}_{0,x_{k+1}} = \widehat{\mu}_{x_k,x_{k+1}}\, \widehat{\mu}_{0,x_k}$, and $\widehat{\mu}_{0,x}(s) = (1+xs)^{-\gamma}$ is nonvanishing for $\operatorname{Re} s \geq 0$, so dividing gives the display. The transfer identity of the section: $v = \frac{x_k}{x_{k+1}} y + \big(1 - \frac{x_k}{x_{k+1}}\big) \frac{y}{1+x_{k+1}s} = y\, \frac{1+x_k s}{1+x_{k+1}s}$; the convex-combination clause is $0 < x_k/x_{k+1} < 1$. □

The convex combination is what makes positivity and unit mass manifest at the realization level. In discrete time, write Δ for the sample interval and $y_n := y(n\Delta)$; with sampled-and-held input each section admits the exact exponential-integrator update

$$w_{n+1} = e^{-\Delta/x_{k+1}}\, w_n + \big(1 - e^{-\Delta/x_{k+1}}\big)\, y_n,$$

unconditionally stable (real pole in $(0, 1)$) and free of overshoot and ringing. Each *section* is exact for zero-order-held input; the cascade is not exact end-to-end, the input to the next section

being the previous section's response rather than a held signal, so time discretization is a genuine approximation — what it preserves exactly is positivity, unit mass and stability, the update being a convex combination; what it does not preserve is the end-to-end kernel and covariance under dilations not commensurate with Δ. A full scale space over knots $x_1 < \cdots < x_M$ costs γM first-order states and $O(\gamma M)$ multiply-adds per sample.

Remark 11.3 (covariance in practice). With geometric knots $x_k = q^k x_0$ the *continuum* cascade is carried to itself by every rescaling $\sigma \in q^{\mathbb{Z}}$, as a shift of the ladder; in the sampled system the time constants are tied to the sample interval Δ, so the shift is exact on the knot set but not on the sampled filters. Because the underlying family exists at *every* scale, an arbitrary σ is handled by instantiating the same family at the shifted knots — a new filter bank with new time constants and re-initialized state, not a relabelling of the running one; the operation is principled exactly because the kernels between knots belong to the family. This is the operational content of continuous covariance, and the point of the pole–zero structure: a pure-pole cascade (truncated exponentials with distinct time constants), the finite form of the time-causal limit-kernel construction of [23], also marches a hemigroup, but its kernels are hypoexponential laws whose family is *not* dilation-closed, so only the discrete covariance of the ladder itself survives. The zeros are what the hemigroup increments of the Sato–Gamma process add, and they are exactly what buys the continuum.

Remark 11.4 (implementing the local corner). The Bessel/inverse-gamma increments have transcendental transforms (ratios of Bessel-K profiles), hence *no* finite-dimensional time-recursive realization; the same holds for the stable family ($e^{-(x_{k+1}^{\alpha}-x_k^{\alpha})\,s^{\alpha}}$ in the homogeneous gauge). The implementation options are: (i) direct convolution against tabulated increment kernels per knot — burdened by the polynomial tails t^{-1-a}, so truncation error decays only polynomially in the window length; (ii) marching the memory-line PDE, in the parabolic gauge ξ of Remark 6.4, $\partial_t u = \frac{1}{2}u_{\xi\xi} + \frac{\beta}{\xi}u_\xi$ in time (implicit in ξ, with care at the degenerate boundary), which reintroduces scale-direction discretization error; or (iii) rational approximation of the irrational symbol, in the manner standard for fractional-order elements in filter and control design ([28]); Proposition 11.6 below keeps that route inside the axiom class, and its state-count economy is discussed after the proposition, as a heuristic. All three are workable; none matches the cascade of Proposition 11.2 in simplicity or exactness. This is the practical face of Theorem 10.6: locality of the medium is bought at the price of heavy-tailed, non-rational kernels.

Remark 11.5 (finite differences on the memory line). The explicit finite-difference scheme of [8], §§6–7, for the heat signaling equation extends verbatim to the Bessel family. Working in the parabolic gauge ξ of Remark 6.4, on the logarithmic grid $\eta = \log\xi$ the operator is constant-coefficient up to a rate factor,

$$\tfrac{1}{2}\,\partial_\xi^2 + \frac{\beta}{\xi}\,\partial_\xi \;=\; e^{-2\eta}\Big[\tfrac{1}{2}\,\partial_\eta^2 + \big(\beta - \tfrac{1}{2}\big)\,\partial_\eta\Big],$$

so the stencil is uniform along the scale ladder and only the update rate varies with position — the discrete form of the Lamperti time change ([21]). With the first-order term upwinded, the explicit update matrix is sub-stochastic, so positivity and non-amplification are automatic, and the discrete scheme is itself a hemigroup of sub-probability kernels: a random-walk medium whose absorption-time laws are discrete analogues of the inverse-gamma kernels. We record this for continuity with the 2005 paper but do not recommend it: the explicit step is constrained by the finest cell, $\Delta t \lesssim \xi_{\min}^2\,\Delta\eta^2$, while the coarse-scale dynamics live on horizons $\sim \xi_{\max}^2$, so the cost of covering the ladder grows like $(\xi_{\max}/\xi_{\min})^2$ (an implicit march trades this for per-sample band solves plus additional discretization error). The finite-moment regime $a > 1$, i.e. $\beta < -\frac{1}{2}$ — the very reason to prefer the Bessel family — is advection-dominated near the singular boundary and numerically the least pleasant. And, decisively, any marching of the signaling equation approximates the scale direction, which Proposition 11.1 shows can be sampled exactly. The principled numerics for the local corner is instead the following.

Proposition 11.6 (the Thorin subclass; gamma cascades approximate the local corner)**.** *For an admissible F (Theorem 7.3) the following are equivalent:*

1. *the memory kernel k is completely monotone — the classical Sonine case of Proposition A.9(2);*
2. *F has a* Thorin representation
$$F(s) \;=\; b_0\, s + \int_{(0,\infty)} \log\Big(1+\frac{s}{\tau}\Big)\, U(d\tau), \qquad U \ge 0,\ \int_{(0,1]} \log\tfrac{1}{\tau}\, U(d\tau) + \int_{(1,\infty)} \frac{U(d\tau)}{\tau} < \infty;$$
3. *the kernels are generalized gamma convolutions (GGC).*

In that case, for any atomic $U_N = \sum_{i=1}^{N} w_i\, \delta_{\tau_i}$ such that $F_N(s) := b_0 s + \sum_i w_i \log(1 + s/\tau_i)$ converges to F pointwise — as it does for the standard quadratures of U, the weights carrying the two integrability conditions — the exponents F_N are themselves admissible*, finite multi-Gamma members of Theorem 7.3, and the kernel families converge weakly at every scale; positivity and unit mass hold* exactly at every truncation level*, the approximation staying inside the axiom class rather than merely near it. For integer weights the increments over any knot ladder are rational of order $\sum_i w_i$ up to the pure delay $e^{-b_0(x_{k+1}-x_k)s}$ contributed by the drift — itself a pure time shift — realized by the cascade of Proposition 11.2 applied per Thorin node, with time constants x/τ_i.*

Proof. (1) $\Leftrightarrow$ (2): by Bernstein's theorem ([33], Thm. 1.4, p. 3), k completely monotone means $k(t) = \int e^{-\tau t}\, U(d\tau)$ for a positive measure U on $[0,\infty)$, finite on compacts since the integral converges; an atom of U at $\tau = 0$ would put a constant component into k, making $\int_1^\infty k(t)\, t^{-1} dt$ — hence F — infinite, so U lives on $(0,\infty)$. By Lemma 7.10, $F'(s) = b_0 + \int U(d\tau)/(s+\tau)$, and integrating in s from 0 gives the Thorin form, the integrability condition being finiteness of $F(1)$ — three regimes: $\log(1+1/\tau) \asymp \log(1/\tau)$ as $\tau \downarrow 0$, $\asymp 1/\tau$ as $\tau \to \infty$, and bounded on the compact middle, where U is finite; conversely differentiate. (2) $\Leftrightarrow$ (3) is Thorin's characterization of the GGC class ([38], [37]). In the Lévy-data form — GGC iff the Lévy density ℓ has $y\,\ell(y)$ completely monotone, which is clause (1) verbatim, $y\,\ell(y)$ being this paper's k — it is [2], Thm. 3.1.1, p. 30; the exponent-side form is the Thorin–Bernstein class of [33], Thm. 8.2, p. 109. Admissibility of F_N: $sF_N'(s) = b_0 s + \sum_i w_i\, s/(s+\tau_i)$ is a Bernstein function, so Lemma 7.1 applies and F_N is of the form (7.1); $F_N(0+) = 0$, the integrability conditions are trivial for a finite sum, and Theorem 7.3 makes the family admissible. Pointwise convergence of F_N gives convergence of the transforms $e^{-F_N(xs)}$, hence weak convergence of the kernels by the continuity theorem (Proposition 3.8). □

The four working families of this paper — Gamma, Bessel, stable, and the gamma-product ladder — all lie in the Thorin subclass. The Gamma family is its *single-atom* member, $U = \gamma\, \delta_1$ — the cascade of Proposition 11.2 is literally the atomic case of Proposition 11.6. The stable family has the explicit Thorin density $U(d\tau) = \frac{\alpha \sin(\pi\alpha)}{\pi}\, \tau^{\alpha-1}\, d\tau$. The inverse-gamma/Bessel family is GGC ([12]; [2]), with Thorin measure supported on the spectral cut of the Bessel operator and density expressible through the Bessel functions of the first and second kind, $u(\tau) \propto \big[\tau\,\big(J_a^2 + Y_a^2\big)\big(\sqrt{2\tau}\big)\big]^{-1}$ ([14]), decaying like $\tau^{-1/2}$ at infinity in accordance with $F(s) \sim \sqrt{2s}$. And the gamma-product ladder of Proposition 10.7 is GGC since HCM $\subset$ GGC ([2]). The conclusion, stated with its three statuses separated: **the Gamma family is the implementation substrate for the entire Thorin subclass** — all four working families — in the following graded sense. *Proved, any positive weights:* atomic truncations of the Thorin measure are themselves admissible families converging weakly at every scale, so the subclass is approximated from *inside* the axiom class (Proposition 11.6). *Proved, integer weights:* the truncated member is realized *exactly*, finitely many first-order sections per increment, the cascade

Table 3: The three corners from the implementer's point of view.

	Gamma (Prop. 8.9)	Bessel (Prop. 8.17)	stable (Prop. 8.5)
axiomatic role	nonlocal, rational symbol (Ex. 9.26)	local + PMP (Thm 10.6)	homogeneous (Cor. 7.4)
kernels	$\Gamma(\gamma,\, x)$	inverse-gamma, shape a	one-sided α-stable
moments	all finite; $\mathbb{E}\,T_x = \gamma x$	n-th iff $n < a$	none
tails	exponential	t^{-1-a}	$t^{-1-\alpha}$
non-creation from the signal (Prop. 8.13)	yes (integer γ; VD)	no	no
increment transfer	rational, order γ	transcendental (Bessel-K)	transcendental
recursive states per knot†	γ	— (approx. only)	— (approx. only)
scale covariance	continuous	continuous	continuous
medium* (Rmk 9.29)	multiplicative jumps	diffusion + drift	$\alpha \geq \frac{1}{2}$: Lévy; $\alpha < \frac{1}{2}$: none

† proved exact for the Gamma column (Prop. 11.2); "approx. only" is the in-class approximation of Prop. 11.6, exactly rational at integer Thorin weights and a fitting heuristic otherwise (the graded statuses stated after that proposition).

* the medium row is context, stated without proof (Remark 9.29); every other row is proved at the reference it cites.

of Proposition 11.2 run per Thorin node. *Heuristic, real weights:* realizing a non-integer-weight truncation by first-order sections is the rational-fitting practice of Remark 11.4(iii) ([28]); no error theorem is claimed for it here, and a constructive integer-weight approximation with error bounds is left open. Even when Bessel semantics are wanted, then, the machine one builds is a bank of first-order Gamma sections — exactly so when the weights are integers, by fitting otherwise. And by Proposition 8.13, the substrate is essentially the regularity class of §8: both select drift plus integer Thorin atoms of finite mean, the variation-diminishing property admitting countably many atoms where rationality keeps finitely many. Outside the subclass — the non-completely-monotone kernels, the Dickman rays with their echo singularities foremost — no Gamma bank converges from inside the class, the GGC class being closed under weak limits ([2], p. 29), and an implementer is left with direct convolution.

As a design heuristic: the Bessel Thorin density is asymptotically a power law, hence scale-invariant over the covered band, so logarithmically spaced nodes — a fixed number per decade of bandwidth — are the natural quadrature. We state this as a heuristic, with no error bound claimed — the form in which the state-count claim of Remark 11.4(iii) should be read. The lognormal — the delay law of the scale-time aperture, a Thorin member by [37]; Remark 13.1 — is the idealized instance: its $(\log s)^2$ exponent growth corresponds to an asymptotically log-uniform Thorin measure, a constant number of sections per decade. In practice the node data (τ_i, w_i) come either from quadrature of U or from direct rational fitting of the increment transfer functions; the Thorin structure guarantees that a fit with *positive* weights exists. One caveat on the drift: the pure delay $e^{-b_0 \Delta x\, s}$ is a time shift by $b_0 \Delta x$, which a sampled implementation realizes exactly only when that shift is an integer number of samples — otherwise it is a fractional delay, approximated by the standard interpolation and allpass designs, again with no error claim made here.

Comparison of the three corners. Table 3 collects the comparison.

The recommendation implicit in Table 3: the Gamma family as the default measurement front-end; the Bessel family when a local PDE medium or a diffusion interpretation is required; the stable family when strict depth-homogeneity is required and heavy tails are acceptable.

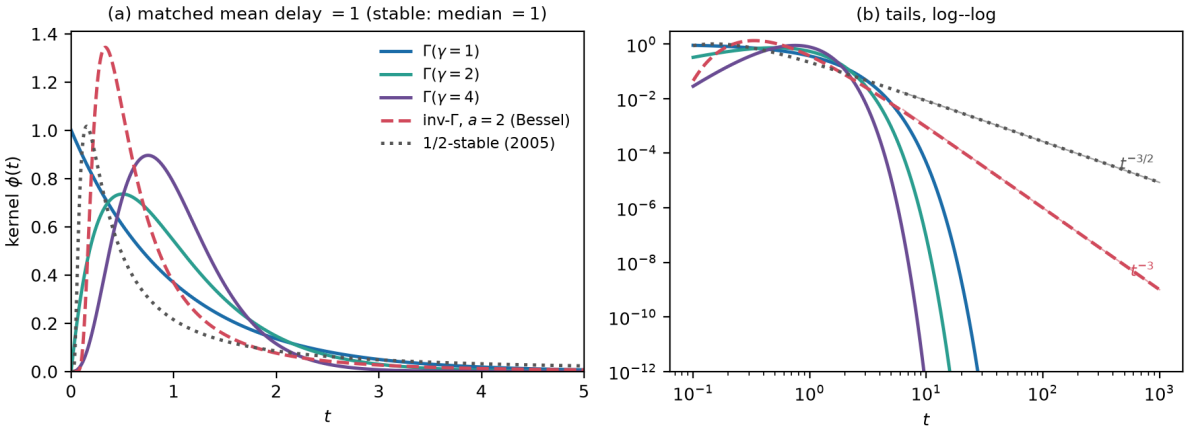


Figure 10: Time-causal scale-space kernels at matched mean delay 1 (stable: median 1, its mean being infinite). (a) linear scale; (b) log–log, with the power-law guides t^{-3} (inverse-gamma, $a = 2$) and $t^{-3/2}$ ($\frac{1}{2}$-stable).

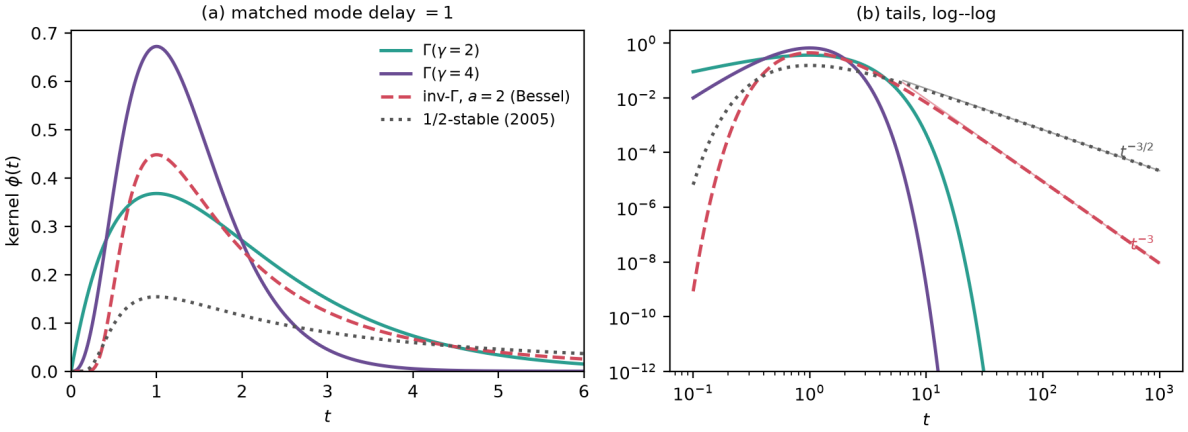


Figure 11: The same comparison at matched mode delay = 1, the normalization of the 2005 influence-curve discussion. (a) linear scale; (b) log–log, with the exact power-law asymptotes.

Example 11.7 (numerical illustration)**.** Figure 10 compares the kernels of the three corners at matched delay: Gamma kernels with $\gamma \in \{1, 2, 4\}$ and scale $x = 1/\gamma$ (mean delay 1); the Bessel/inverse-gamma kernel with $a = 2$ and inverse-gamma scale parameter $\vartheta = a - 1$ (mean delay 1); and the $\frac{1}{2}$-stable kernel of [8] normalized to median delay 1, its mean being infinite.

Panel (a) shows the bodies: at matched mean the Gamma kernels are the most concentrated, sharpening with γ (mode $(\gamma - 1)/\gamma \to$ mean), while the inverse-gamma kernel buys its early sharp peak by exporting mass into the tail, and the stable kernel does so to the point of losing the mean altogether; beyond $t \approx 2$–3 the ordering reverses and the heavy-tailed members dominate. Panel (b) shows the mechanism: the Gamma tails fall exponentially, the inverse-gamma and stable tails follow the powers t^{-1-a} and $t^{-3/2}$ of Remark 10.10. The practical consequence is quantified by the window length T_p capturing a fraction p of the kernel mass ($\int_0^{T_p} \phi = p$), the quantity governing the truncation error of route (i) in Remark 11.4 (Table 4):

For the Gamma family each extra decade of captured mass costs a fixed additive window increment (exponential tails); for the inverse-gamma with $a = 2$ it costs a factor $10^{1/a} \approx 3.2$; for the $\frac{1}{2}$-stable, of index $\alpha = \frac{1}{2}$, a factor $10^{1/\alpha} = 100$ — capturing 99.9% of its mass requires a window five orders of magnitude longer than its median delay. This is the quantitative content of the statement that the 2005 kernels are too heavy-tailed for practical measurement, and of

Table 4: Window lengths T_p capturing mass p, at matched delay 1 (stable: median 1).

kernel (delay 1)	$T_{0.99}$	$T_{0.999}$
$\Gamma(\gamma{=}1)$	4.61	6.91
$\Gamma(\gamma{=}2)$	3.32	4.62
$\Gamma(\gamma{=}4)$	2.51	3.27
inverse-gamma, $a = 2$	6.73	22.0
$\frac{1}{2}$-stable	2.90×10^3	2.90×10^5

Table 5: The same comparison at matched mode delay 1, the 2005 normalization.

kernel (mode 1)	mean	$T_{0.99}$	$T_{0.999}$
$\Gamma(\gamma{=}2)$	2	6.64	9.23
$\Gamma(\gamma{=}4)$	4/3	3.35	4.35
inverse-gamma, $a = 2$	3	20.2	66.1
$\frac{1}{2}$-stable	∞	1.91×10^4	1.91×10^6

its resolution within the hemigroup class.

Since the stable mean is infinite, the delay proxy used in the influence-curve discussion of [8], §8, is the *mode*, and Figure 11 repeats the experiment in that normalization: matched mode delay 1, i.e. $x = 1/(\gamma - 1)$ for the Gamma kernels ($\gamma = 1$ is omitted, its mode sitting at the onset), $\vartheta = a + 1 = 3$ for the inverse-gamma, and $\vartheta = 3/2$ for the $\frac{1}{2}$-stable.

Matching at the peak is the normalization most favorable to the heavy-tailed kernels in the body — panel (a) shows the $\Gamma(\gamma{=}2)$ and inverse-gamma bodies nearly coinciding through the peak, so the families are barely distinguishable by their peak response — and it makes the separation in the delay statistics all the starker (Table 5):

At matched mode, capturing 99.9% of the $\frac{1}{2}$-stable kernel requires a window six orders of magnitude beyond the peak. Both figures and both tables are generated by the accompanying script `make-figures.py`; the power-law guides in the tail panels are the exact asymptotes $\vartheta^a\, t^{-1-a}/\Gamma(a)$ and $\sqrt{\vartheta/\pi}\, t^{-3/2}$, with ϑ the scale parameter of the inverse-gamma law ($\vartheta = \tilde{x}/2$ at canonical scale $\tilde{x}$) in the respective normalization.

The kernels are compared here in isolation. The machine itself — the recursive cascade of Proposition 11.2 run over a fine knot ladder on a structured signal, with its jets — is exercised end to end in Example 12.7, once the jet architecture of §12 is in place.

12 The temporal N-jet: the memory embodies its derivatives

The output of a scale space is not the smoothed field alone but its jet — the temporal derivatives $\partial_t^n u(t, x)$ across scales, the causal analogue of the derivative-of-Gaussian receptive fields. Computed naively, $\partial_t^n u$ requires differencing the field in time, i.e. a *second* memory recording the recent history of the scale space on top of the memory line itself — a history whose length is the window T_p of Tables 4 and 5, since the stencil must see the kernel's mass. This section shows that the axioms exclude the need for it: the inversion structure of §9 turns temporal differentiation into an instantaneous spatial readout of the present memory state — the memory line embodies its own jet.

The hypothesis space iterates the core of §3: what the theorem needs of f is that each derivative up to order n is again a causal L^1 function that is the primitive of the next — the vanishing boundary values are exactly what will turn $I^z f^{(n)}$ into $I^{z-n} f$ in the proof. So let

$$\mathcal{D}^n \;:=\; \big\{\, f \in X_0 : \text{there are } f^{(1)}, \dots, f^{(n)} \in X_0 \text{ with } f^{(j)} = \textstyle\int_0^{\cdot} f^{(j+1)},\ 0 \le j < n \,\big\},$$

so that $\mathcal{D}^1 = \mathcal{D}$.

Definition 12.1. The *temporal N-jet* of the scale space at (t,x) is $\{\partial_t^n u(t,x)\}_{0\le n\le N}$, and the *scale-normalized jet* is $\{x^n\partial_t^n u(t,x)\}_{n\le N}$, which in the canonical gauge transforms under the joint dilation $(t,x)\mapsto(\sigma t,\sigma x)$ like the field itself.

Theorem 12.2 (the embodied jet)**.** *Assume (H), canonical gauge, $f\in\mathcal{D}^n$, and let $u(\cdot,x)=\Phi_{0,x}f$. Then:*

1. *For every $t>0$ and $1<\operatorname{Re} z<z_*$,*

$$\widetilde{\partial_t^n u(t,\cdot)}(z) \;=\; \tilde H(z)\,\big(I^z f^{(n)}\big)(t) \;=\; \tilde H(z)\,\big(I^{z-n}f\big)(t),$$

the second equality holding as a Riemann–Liouville integral for $\operatorname{Re} z>n$ and by analytic continuation — a fractional derivative of f — below.

2. *Consequently, whenever $z_*>n+1$, the n-th temporal derivative is the n-th power of the inversion operator applied to the* present *memory state: in the Mellin sense on $n+1<\operatorname{Re} z<z_*$,*

$$\partial_t^n u(t,\cdot) \;=\; A^n\,u(t,\cdot),\qquad A^n=\frac{1}{x^n}\,B^{(n)}(\theta),\qquad B^{(n)}(-z):=\frac{\tilde H(z+n)}{\tilde H(z)}=\prod_{j=0}^{n-1}B(-z-j)$$

for $0<\operatorname{Re} z<z_*-n$ — the identity evaluating the symbol at the shifted argument, as at order one.

In particular the temporal jet at time t, up to every order $n<z_*-1$, is an instantaneous linear functional of the single profile $x\mapsto u(t,x)$; no record of past states is required. The identity is Mellin-level, with the pointwise reading subject to the same pole discipline as order one (Remark 9.18).

Proof. (1) By induction on the order: for $j<n$, $f^{(j)}$ is the primitive of $f^{(j+1)}\in X_0$, so Lemma 9.11's derivative clause applies at each step — $\partial_t\,\mathbb{E}[f^{(j)}(\cdot-xT_1)]=\mathbb{E}[f^{(j+1)}(\cdot-xT_1)]$ as X_0-derivatives, the boundary conditions of $\mathcal{D}^n$ licensing each step — and n steps give $\partial_t^n u(t,x)=\mathbb{E}\big[f^{(n)}(t-xT_1)\big]$. The lemma applied to the causal $f^{(n)}\in X_0$ then gives the first equality; the second is $I^z f^{(n)}=I^{z-n}I^n f^{(n)}=I^{z-n}f$, $I^n f^{(n)}=f$ being the primitive structure of $\mathcal{D}^n$ again. (2) Telescoping the shift relation of Definition 9.8 n times, $\mathcal{M}[A^n g](z)=\prod_{j=1}^n B(j-z)\,\tilde g(z-n)=\frac{\tilde H(z)}{\tilde H(z-n)}\,\tilde g(z-n)$; insert $\tilde g=\widetilde{u(t,\cdot)}$ from Lemma 9.11 and compare with (1). □

Remark 12.3 (differentiation is re-indexing). Combined with Remark 9.13, Theorem 12.2(1) says that the memory line stores, once and simultaneously, the analytic family $\{(I^z f)(t)\}$ of fractional integrals of the past — and the jet of every order is a *re-reading* of that same stored family at shifted index, order n living at Mellin depth $\operatorname{Re} z>n$ of the strip. Nothing new is stored when one differentiates; the embodiment of the signal already contains its jet, at every order below z_*-1.

Proposition 12.4 (the Gamma family: jets by subtraction along the shape column)**.** *For the Gamma family write $u_m(\cdot,x)$ for the field of shape m ($u_0:=f$). From $s(1+xs)^{-m}=x^{-1}\big[(1+xs)^{-(m-1)}-(1+xs)^{-m}\big]$,*

$$x\,\partial_t\,u_m \;=\; u_{m-1}-u_m \;=\; -\,(\nabla u)_m,\qquad (\nabla u)_m:=u_m-u_{m-1},$$

and hence, iterating at fixed x,

$$x^n\,\partial_t^n\,u_\gamma \;=\; (-\nabla)^n u_\gamma \;=\; \sum_{j=0}^{n}(-1)^{n-j}\binom{n}{j}\,u_{\gamma-j}(t,x),\qquad 0\le n\le\gamma.$$

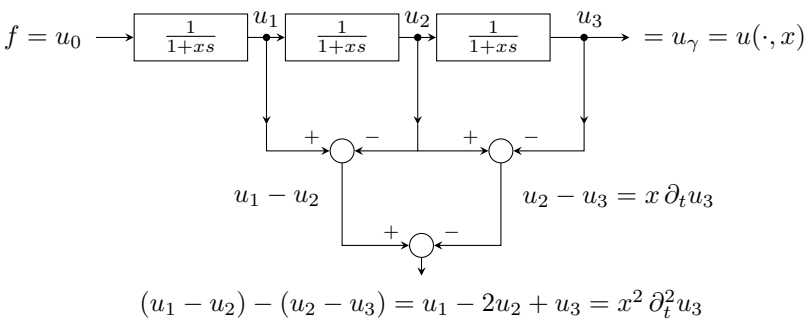


Figure 12: Jets by subtraction along the shape column ($\gamma = 3$). The column of γ identical leaky integrators at time constant x — the pure-pole cascade from scale zero, the gamma memory of [39] — carries the fields $u_0, \ldots, u_\gamma$ of increasing shape as coexisting states. The scale-normalized jet at scale x is read off the taps by binomial-weighted subtraction, $x^n \partial_t^n u_\gamma = \sum_j (-1)^{n-j} \binom{n}{j} u_{\gamma-j}$, realized as iterated differences of adjacent taps (Proposition 12.4): no temporal differencing, and no second memory, anywhere in the architecture.

The column $\{u_m(\cdot, x)\}_{m=0}^{\gamma}$ is realized by γ identical leaky integrators in series with time constant x, $u_m = (1 + x\,\partial_t)^{-1} u_{m-1}$ — the gamma memory of [39] — so the scale-normalized jet at scale x is a binomial-weighted subtraction of coexisting filter states*, with no temporal differencing anywhere in the architecture, γ states per scale, and jets available up to order γ.*

Proof. The transform identity is $(1 + xs)^{-m}\big[(1 + xs) - 1\big] = xs(1 + xs)^{-m}$; induction in n at fixed x; the realization is Proposition 11.2 at $x_k = 0$, where the section degenerates to the pure pole $(1 + x_{k+1}s)^{-1}$. □

The column is the pure-pole cascade from scale zero, fed by the signal (Figure 12) — an architecture parallel to the knot ladder of §11, not a reading of its states: the ladder's taps carry only the top shape u_γ at each knot, so jets at M scales cost one column of γ states per scale, the same total count as the ladder, fed from f rather than from the previous knot. And the identity is exact for all $n \le \gamma$, extending beyond the strip condition of Theorem 12.2 (the transforms are rational and continue meromorphically) — consistently with $z_* = \gamma$ marking the strip reading and with the relaxation of hypothesis (H) by meromorphic continuation left open in §13.

The subtraction rule is the single-atom case. For a Thorin bank (Proposition 11.6) the field is a convolution of per-node Gamma fields and no per-column subtraction reads its jet; there the jet is read through Theorem 12.2(2), whose symbol $B^{(n)}$ is again rational for rational B. We do not develop that readout here.

Remark 12.5 (receptive fields; Laguerre structure). By Rodrigues' formula the jet kernels are associated Laguerre: with $\phi_x^{(m)}$ the Gamma density of shape m at scale x and $\phi_x = \phi_x^{(\gamma)}$,

$$x^n\,\partial_t^n\,\phi_x \;=\; \frac{n!\,\Gamma(\gamma - n)}{\Gamma(\gamma)}\,\phi_x^{(\gamma-n)}\;L_n^{(\gamma-n-1)}\big(t/x\big), \qquad 0 \le n < \gamma,$$

a Gamma density of shape $\gamma - n$ times an associated Laguerre polynomial — the shape dropping as the order rises, each derivative spending one of the kernel's poles. And the shape column spans the order-γ rational function space with the single pole at $-1/x$: the classical Laguerre filter space at time constant x, exponentials times polynomials of degree $< \gamma$, whose orthogonal basis is the Laguerre functions. Both clauses are computations; the orthogonality belongs to the basis of the space, not to the jet kernels themselves, whose polynomial parts all have degree $\gamma - 1$. The comparison the clauses support is structural: the temporal receptive-field family stands to the causal scale space as the Hermite/derivative-of-Gaussian family stands to the

Gaussian one — a graded basis, read off the embodied state by subtraction. The difference is that the grading here is by the shape index rather than by differentiation order alone.

Remark 12.6 (the local corner: derivatives as curvatures of the memory trace). For the Bessel family, $z_* = \infty$ (Lemma 10.3(2)), so every order is available; and in the parabolic gauge ξ of Remark 6.4, Theorem 12.2(2) is the n-th power of the local operator of Example 9.27:

$$\partial_t^n u \;=\; \Big(\tfrac{1}{2}\,\partial_\xi^2 + \frac{\beta}{\xi}\,\partial_\xi\Big)^n u,$$

the jet read from spatial stencils of order $2n$ on the instantaneous memory profile. In the heat case $\beta = 0$ this is the statement that the observer reads temporal derivatives as curvatures, and iterated curvatures, of its memory trace — the embodiment reading already implicit in [8], Thm. 4, here extended from the local corner to the whole class by the Mellin form.

Example 12.7 (the machine run end to end)**.** Figures 13 and 14 run the whole architecture of §11 and this section on one signal: the recursive cascade of Proposition 11.2 at $\gamma = 2$, over a geometric knot ladder of 64 knots spanning $x \in [2, 400]$ (the time unit is one sample; 4096 samples), with the jets read by binomial subtraction along the shape column (Proposition 12.4) — no temporal differencing anywhere. The signal assembles one specimen per prediction: a sharp and a broad bump, a strong event with a small companion in its wake, a persistent step, and a broadband noise burst. The maps display the qualitative content of the preceding sections at a glance: every ridge shears rightward along its mean-delay curve $t_0 + \gamma x$ — the delay–smoothing coupling of Theorem 9.17 made visible — the events annihilate in order of their widths while the step persists, and the fingerprints close from below without re-opening.

Four checks tie the run to numbered results; both figures and every quoted number are produced by the accompanying script `make-fig-experiment.py`.

1. *Knot exactness* (Proposition 11.1). Refining the ladder from 64 to 128 knots moves the final tap by 4.3×10^{-5}, and the ladder agrees with the filter computing the full map $\Phi_{0,x_{\max}}$ in a single increment to 2.2×10^{-3} — both pure time discretization, the continuum transfers being identical. A forward-Euler march of the scale evolution, run for contrast, errs by 1.3×10^{-2}, halving to 6.0×10^{-3} at doubled scale resolution: the $O(\Delta x)$ that the exponential-integrator update of Proposition 11.2 avoids.

2. *Non-creation from the signal* (Proposition 8.13). The extremum count satisfies $N\big(u(\cdot, x)\big) \le N(f)$ at every one of the 64 scales (146 at the input, at most 94 across the ladder). The counting convention, identical for signal and field and implemented in `count_extrema`: signs of successive differences, with a deadband — differences of magnitude below 10^{-7} set to zero — and the zero signs then dropped, which compresses flat runs so that a plateau counts as one extremum, not two; the count is the number of sign alternations that remain, taken over the sampled window with no padding, so only interior extrema are counted. Scale-to-scale monotonicity is not merely unasserted but impossible in the class (Proposition 8.15) — the increments carry an atom and transform zeros and are not variation-diminishing (Remark 8.14) — and the run shows the count rising at 7 of the 64 steps. Inspected individually, the created pairs are micro-events, of prominence 6×10^{-3} and below, in the remnant of the noise burst and near the dissolving wake, where the sampled cascade retains more high frequency than the continuum kernels (the sections' feedthrough); recomputing the taps with transform-domain kernels moves the residual increases to the kernel-discretization floor. The run thus certifies the bound at every scale and shows step-monotonicity failing in the implemented system, without exhibiting a certified continuum creation event.

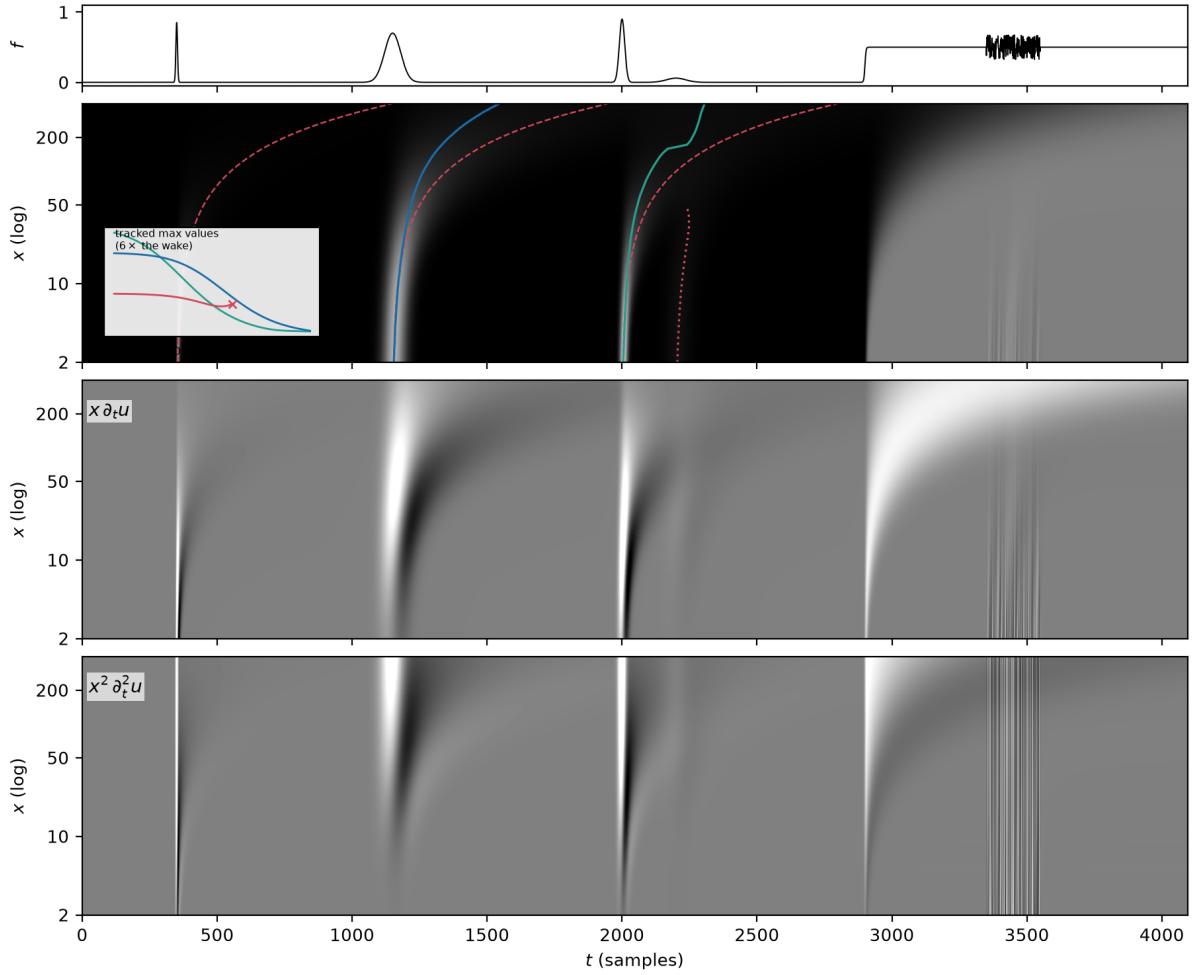


Figure 13: The machine run end to end (Example 12.7): the Gamma cascade at $\gamma = 2$ on a synthetic gray-level signal, in the orientation of Figure 7 — time horizontal, $\log x$ vertical, the signal entering along the bottom edge. Top: the input. Second panel: the field u, with the mean-delay curves $t_0 + \gamma x$ of the three isolated events (dashed), the tracked maxima of the strong (teal) and coarse (blue) events, and the wake companion's track (red, dotted); the inset plots the tracked values against scale — nonincreasing while past-dominating, the wake curve (magnified 6×) showing the echo uptick before its annihilation (×). The bend near the top of the strong event's track is its arch merging with a neighbour — structure, not a tracking failure. Lower panels: the scale-normalized jet maps $x\,\partial_t u$ and $x^2\,\partial_t^2 u$, computed by binomial subtraction along the shape column (Proposition 12.4); for the fine streaks under the noise burst in the second-order map, see the example's closing paragraph.

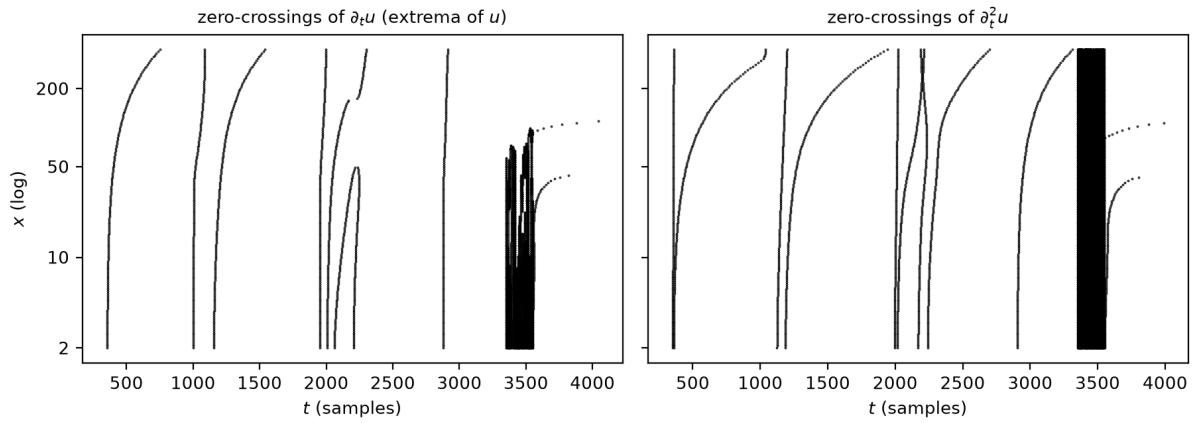


Figure 14: The fingerprints of the same run: zero-crossings of $\partial_t u$ (the extrema of u, left) and of $\partial_t^2 u$ (right) across scale. The classical picture — arches closing from below, never re-opening — holds in this run, sheared rightward along the delay direction: the causal counterpart of the Gaussian fingerprint diagrams. The noise burst's dense column annihilates pair by pair; the wake companion's arch closes at $x \approx 50$.

3. *Non-enhancement along the delay path* (Corollary A.21). The tracked maxima of the strong and the coarse event never increase at any scale where the past-dominating condition holds (0 violations; the condition holds at 15 and 63 of the 64 scales respectively). The wake companion realizes the corollary's sharp failure mode: its maximum decays to 0.0423 by $x = 35$, is then lifted 8% to 0.0456 by the strong event's arriving delayed mass — the echo — and annihilates near $x = 49$ (the inset of Figure 13).

4. *Jets by subtraction* (Proposition 12.4). The identity $x\,\partial_t u_2 = u_1 - u_2$ holds to 4.8×10^{-17} in transform semantics — the machine floor — and the recursive shape column satisfies it against temporal differencing to 6.7×10^{-3}, once more the time discretization.

One feature of the maps should be read off the symbol rather than dismissed as an artifact. In the fully normalized second-order map $x^2\partial_t^2 u$ the noise burst leaves fine streaks at *every* scale: the jet operator's transfer $x^n s^n (1 + xs)^{-\gamma}$ — here $n = \gamma = 2$ — behaves as $(xs)^{n-\gamma}$ at high frequency, so the *margin* $\gamma - n$ is the roll-off order of the composite filter. At $n = \gamma$ the margin is zero: the derivative order exhausts the kernel's pole order, the composite is all-pass at high frequency, and the normalization cancels the scale dependence exactly — which is why the streaks keep constant amplitude at every scale instead of fading upward. The top of the jet is noise-limited by construction; a clean n-th derivative wants poles to spare, $\gamma > n$, consistently with Proposition 12.4's count of jets available up to order γ. This is the Fourier face of the Mellin budget $z_* = \gamma$ of Theorem 12.2, both paid for by the kernel's onset behavior $t^{\gamma-1}$: onset smoothness, high-frequency decay, and negative-moment finiteness are one finiteness, and the single number γ rations all three.

13 Concluding remarks

One axiom of [8] was weakened — the one-parameter semigroup cascade to the two-parameter hemigroup, i.e. the assumption that measurement stages are interchangeable — and the entire derivation was rerun in the same order, with the same analytic apparatus. What the uncommitted observer of §2 can be is now settled: the admissible measurement families are exactly the self-decomposable Laplace exponents, one nonincreasing memory density k and one drift per family (Theorem 7.3). The class contains members with every moment finite, and the

heavy tails of the stable kernels are thereby exposed as the price of the semigroup axiom, not of causality or covariance. The observer's embodiment of its past is made exact: the memory line carries the analytic family of fractional integrals of the signal, the scale evolution inverts into an evolution in present time under the unique covariant inversion (Theorem 9.17), and the temporal jet is an instantaneous readout of that same state (§12). The local, positivity-respecting media among these are classified — a degenerate transport member and the Bessel family, the latter unfolding the isolated heat case of the semigroup theory (Theorem 10.6). Non-creation — the oldest axiom of the subject — is settled in both directions: the scale-monotone form is impossible for every nondegenerate member (Proposition 8.15), the endpoint form — no scale showing more sign changes than the signal — is classified and attained (Proposition 8.13), and along the delay path the field is non-enhancing at past-dominating extrema (Corollary A.21). And the hemigroup is itself the algorithm: the discrete cascade is exact at every set of scale knots (Proposition 11.1), with the memory-kernel and Sonine structure of Appendix A — the scale-Cauchy problem, and the exact right inverse of each scale's memory operator — reading the same function k as the physical memory kernel.

On scope: what this paper delivers is the foundations and an exact implementation route — the characterization, the solution theory over the memory line, and cascades that stay inside the axiom class — in the L^1 signal model, with the pointwise extension to bounded signals recorded in Remark 3.1. It makes no empirical claims about natural signals; evaluation on such signals belongs to work built on this foundation.

One methodological remark, since the formalization changed the paper rather than merely certifying it. Five of the proofs printed above are not the ones first written. In each case the original argument invoked a named theorem about a *class* of functions — Bernstein–Widder, closure of BF under pointwise limits, the semigroup property of the Riemann–Liouville family, the asymptotics of Γ on vertical lines, uniqueness of the Mellin transform — where an elementary argument about the *object* suffices, and is no longer. The formal development, which has no notion of complete monotonicity at all, could not take the classical route and so found the elementary one; having found it, there was no reason to keep the classical route on paper. The net effect is that the machine-checked results rest on one cited fact where the classical route would have spent six or seven. And the one irreducible appeal to the derivative-sign definition — Lemma 7.1, where $sF'(s)$ is the object of study — is now visible as the single place where that vocabulary is doing work rather than providing convenience. We record this because the direction of influence is the opposite of the usual expectation: verification is generally described as a check on a finished argument, and here it shortened the argument.

The limits of what is done here should be stated as plainly as the results. The theory is purely temporal; composition with a spatial scale space is untouched. The local corner is bought at a price the implementation section makes exact: the Bessel family admits no finite-dimensional time-recursive realization (Remark 11.4), and the homogeneous members with $\alpha < \frac{1}{2}$ admit no Markov medium at all (recorded without proof in Remark 9.29). The memory-line theory holds under the standing hypothesis (H), which the extreme rays with $b_0 = 0$ sit on the boundary of and the Gamma family with $\gamma \leq 1$ fails. And the paper's numerical component (Example 12.7) exercises the implementation on synthetic signals only: what is delivered is a characterization and an exact implementation route, with no validation on natural data.

Four problems remain. One is genuinely open here: the relaxation of hypothesis (H) by meromorphic continuation, which would admit the slowly growing exponents — for instance the Gamma family with $\gamma \leq 1$ — into §§9–10 and §12. The others are open in this paper and taken up elsewhere. The memory-line inverse problem asks which admissible exponents F are realized by genuine Markov media — a correspondence of Krein type ([20]; [6]) admitting jumps — together with the question of what surface data determine the medium. It is settled here only in its local, positivity-respecting corner (Theorem 10.6, discharging the promise of Remark 9.29) and is otherwise the subject of a separate article. The complex-root cases of Proposition 10.7

(Remark 10.8) are largely resolved in a companion note (in preparation), where spectrally leading pairs are excluded and order three is settled, with the subordinate-pair configuration remaining. And the discrete-covariance branch, whose place in the axiom system is identified in Remark 13.1 below, has its structure theorem, together with the canonical-selection question for the time-causal limit kernel, in the same companion note.

Remark 13.1 (the discrete-covariance stratum and the previous literature). Where do the earlier time-causal constructions sit relative to these axioms? The instrument that answers the question is a weakening of (A8) to the subgroup $\sigma \in q^{\mathbb{Z}}$, $q > 1$. Most of §6 survives verbatim: uniqueness of S_q, the group law along the subgroup, and the absence of interior fixed points, the orbit argument needing only the powers $q^{\pm n}$. The similarity solution generalizes: in the variables $(\log(xs),\, \log(x/s))$ the covariance relation $G(qx, s) = G(x, qs)$ says precisely that

$$G(x,s) \;=\; \mathfrak{F}\big(xs;\; \log(x/s)\big), \qquad \mathfrak{F}(\,\cdot\,;\; v + 2\log q) = \mathfrak{F}(\,\cdot\,;\; v),$$

with $\mathfrak{F}$ a profile, log-periodic in the second, *chirp* variable — a modulation of the continuous case's $F(xs)$. The probabilistic identification is exact: applying covariance once, the marginal exponents satisfy

$$\Psi_x(qs) - \Psi_x(s) \;=\; g_{x,\,qx}(s) \;\in\; \mathrm{BF}_0$$

— a Bernstein function vanishing at the origin, §3 — which is the definition of a *semi-selfdecomposable* law of span q ([26]), and the two-parameter families are the *semi-selfsimilar additive* subordinators of [27]. The axiom system thus carries a three-level stratification, and the earlier time-causal constructions occupy its strata rather than lying outside it:

1. *(A1)–(A7) alone: Bernstein hemigroups.* This stratum contains the Poisson scale space of [24], which belongs to no higher stratum: with exponent $\lambda(1 - e^{-\delta s})$ the q-step increment $\lambda(e^{-\delta s} - e^{-q\delta s})$ has a sign-changing derivative for every $q > 1$, so the fixed jump size δ is incompatible with covariance under any dilation subgroup.

2. *(A8) for $q^{\mathbb{Z}}$: semi-Sato families.* This stratum contains the time-causal limit kernel of [23]: its exponent $F(s) = \sum_{i\geq 1} \log(1 + c\, q^{-i} s)$ telescopes under one covariance step,

$$F(qs) - F(s) \;=\; \log(1 + c\, s) \;\in\; \mathrm{BF}_0,$$

   one q-step of scale per pole section. But more is true: the Lévy tail of F is $k(t) = \sum_{i\geq 1} e^{-tq^i/c}$, a sum of decreasing exponentials — nonincreasing, indeed completely monotone — so F is itself *self-decomposable*, and the similarity family $\{F(xs)\}_{x>0}$ generated by its exponent is a fully covariant member of Theorem 7.3: the Thorin member of Proposition 11.6 whose measure is the geometric atom train $\sum_i \delta_{q^i/c}$ in place of the Gamma family's single atom. The atoms carry unit weight with finite mean, $\sum_i c\, q^{-i} = c/(q-1)$, so the family lies in the variation-diminishing subclass of Proposition 8.13 as well: its kernels are Pólya frequency and create no extrema beyond the signal's — and by Remark 8.16 its q-step increments are variation-diminishing too, so along any geometric ladder of ratio q the counts are monotone rung to rung, the most that survives, Proposition 8.15 excluding the all-ratio form for every nondegenerate member. That similarity family is thus, in retrospect, the first-discovered member of the class characterized in this paper, and of its variation-diminishing subclass — the published construction itself is its $q^{\mathbb{Z}}$-sampled form. To it the knot-exactness of Proposition 11.1, the Sonine-pair structure of Appendix A, and the jets of §12 (under (H), which holds: $z_* = \infty$) transfer exactly; the finite *cascade* of §11 does not — the atom train is infinite, so the rational realizations of Proposition 11.6 reach it only approximately. Two different truncations must be kept apart here: the finite cascade of the *published construction*, whose time constants are fixed, is hypoexponential and not dilation-closed, carrying only the ladder's discrete covariance (Remark 11.3);

truncating the *exponent* instead — finitely many atoms of the Thorin train, their time constants moving with scale — gives a family that is itself admissible and fully covariant (Proposition 11.6), approximating the limit kernel from inside the class. What is intrinsically discrete about it is the cascade construction and the role of q as the log-spacing of its Thorin atoms. Strict members of the middle stratum, semi-selfdecomposable but not self-decomposable, exist and are natural — geometric jump ladders refining the Poisson scale space; the companion note constructs them and proves the stratum's structure theorem.

3. *(A8) in full: the Sato class of the present paper.*

The scale-time of [19] also finds its place, though at the kernel level rather than as a stratum. Pulled back from the logarithmically remapped past to the time axis, its aperture is exactly the *lognormal* density in the delay, the Jacobian $du = d\xi/\xi$ of the remap completing the Gaussian into the lognormal. And by Thorin's theorem ([37]) the lognormal is a generalized gamma convolution. The containment is deep rather than decorative, resting on the theorem the class of Proposition 11.6 was created to deliver, and no elementary property of the log-remap produces it. The lognormal delay law therefore generates an admissible similarity family in the Thorin subclass, and a well-behaved one: every moment, positive and negative, is finite, with $\mathbb{E}[T^{-z}] = e^{-z\mu+z^2\sigma^2/2}$ for all z, so $z_* = \infty$ and (H) holds with room to spare. The exponent grows like $(\log s)^2$ by the Laplace method, barely superlogarithmic, just above the Gamma rays in the admissible cone. What does *not* carry over is the construction's own scale parameter: varying the log-domain Gaussian width changes the lognormal's *shape*, and those blurrings compose in the log-delay domain — multiplicative convolution — not in time, which is the precise form of the non-recursivity conceded in §1. The admissible family runs the other axis, dilating the delay at fixed shape. In retrospect the 1988 kernel choice is vindicated by 1977 mathematics: the log-remapped Gaussian lands inside the admissible cone, in its Thorin corner, and the axioms decline only the composition structure it arrived with.

For the limit kernel the extension question has a clean answer, and the argument is short enough to record here: among all $q^{\mathbb{Z}}$-covariant families with its q-step increments $g_{x,qx}(s) = \log(1 + cxs)$, *exactly one* extends to full continuous covariance, namely the limit-kernel family itself. Full covariance forces the similarity form $\Psi_x = F(x\,\cdot)$; the q-difference equation $F(qs) - F(s) = \log(1 + cs)$ determines F up to a log-periodic summand; and a continuous log-periodic function with limit 0 at the origin vanishes. Whether the limit kernel is distinguished *within* its modulation orbit by an extremality principle that does not invoke the continuum extension is the selection question taken up in the companion note. The comparison with the previous literature is thereby one of containment rather than difference: the Poisson scale space lies strictly in the first stratum, the limit kernel in the third, scale-time's kernel in the third's Thorin corner with its composition structure left behind, and the strict middle stratum is populated by their common refinement (Figure 15).

A Memory kernels, Sonine conservation, and the scale-Cauchy problem

The main line reads the representation (7.1) through its transforms; this appendix reads it through the memory kernel. Lemma 7.10 identified F' as, up to the drift, the Laplace transform of the nonincreasing function k. Here that identity becomes the evolution of the scale space across scale — a general fractional relaxation equation whose kernel is k itself, dilated to the current scale — together with the Sonine conservation identity that gives each scale's memory operator an explicit right inverse, a Volterra equation from which the kernels can be

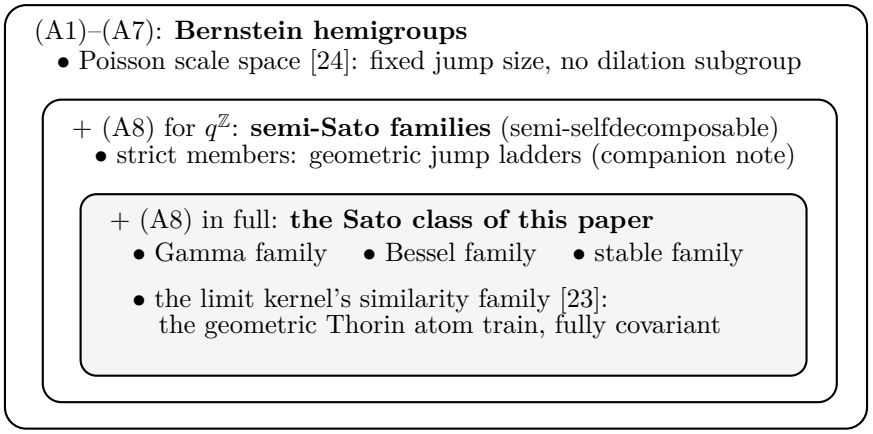


Figure 15: The three-level stratification of Remark 13.1. The axioms without covariance admit the Bernstein hemigroups, where the Poisson scale space lives and stays; covariance under a discrete subgroup $q^{\mathbb{Z}}$ cuts out the semi-Sato stratum, whose strict members are geometric refinements of it; full continuous covariance cuts out the class characterized in this paper, which contains the similarity family generated by the time-causal limit kernel's exponent. Containment, not competition: the earlier constructions occupy strata of one axiom system.

computed without Laplace inversion, and finally the classical scale-Cauchy problem (Theorem 3′). Throughout we work in the canonical gauge — so that $\mu_{0,x}$ is the law of $x\,T_1$ — with the causal vocabulary of §3. Throughout, F is an admissible exponent as in (7.1), with drift b_0 and kernel k, and $F \not\equiv 0$ (equivalently (ND); Corollary 5.3).

A.1 Memory kernels, Sonine conservation, and a Volterra derivation

Three readings of one function organize this subsection. The function k enters the paper in §7 as the family's catalogue of delays — the size-weighted density of the subordinator's jumps. The results below prove that the same k, dilated to the current scale, is also the *physical memory kernel* of the scale space. The evolution along the memory line is a conservation law in the (t, x)-plane whose flux is a k-weighted running average of the record — integrating over t kills the right-hand side, so mass is transported across scale, never created, which is axiom (A5) in differential form (Proposition A.3). The memory operation destroys no information, the algebraic reciprocal of its symbol being again the transform of a causal positive measure, so the memory operator at each scale has an explicit right inverse — what is inverted is the memory operator $D^{(x)}$, in the one composition order stated, not the cascade step $\Phi_{x,y}$ (Theorem A.7, Corollary A.8). And the temporal kernels themselves are recovered from k by forward integration, with no Laplace inversion (Propositions A.10–A.12). One function, three proven roles: delay catalogue, scale generator, and Volterra generator of the kernels. The subsection is the t-convolutional leg of the transform identity $\hat{u}(s, x) = e^{-F(xs)}\hat{f}(s)$, as §9 is its x-operational leg: differentiating in x and inverting the Laplace transform gives the evolution studied here; taking the Mellin structure in x at fixed t gives the signaling form there.

The first lemma restates, dilated to the current scale, the object already fixed at the tail of §7: the measure $\kappa^{(x)}$ there is the one defined here.

Lemma A.1 (the memory kernel's transform)**.** *For $x > 0$ define the locally finite measure*

$$\kappa^{(x)}(dt) \;:=\; b_0\,\delta_0(dt) \;+\; \tfrac{1}{x}\,k\big(\tfrac{t}{x}\big)\,dt.$$

Then $\partial_x F(xs) = s\,F'(xs) = s\,\widehat{\kappa^{(x)}}(s)$, so the symbol $\varphi_x(s) := sF'(xs)$ is s times the transform of the memory kernel.

Proof. The substitution $\tau = t/x$ in $F'(xs) = b_0 + \int_0^\infty e^{-xs\tau} k(\tau)\, d\tau$, which is Lemma 7.10 at xs, turns the integral into $\int_0^\infty e^{-st}\, \frac{1}{x} k(\frac{t}{x})\, dt$; the drift b_0 is the transform of $b_0\delta_0$, which is constant in s. □

Lemma A.2 (the transform equation)**.** *Let $(\Phi_{x,y})$ be as in Theorem 7.3, in the canonical gauge, let $f \in X$ be causal, and set $u(\cdot, x) := \Phi_{0,x} f$. Then for each $s > 0$ the function $\hat{u}(s, x) = e^{-F(xs)} \hat{f}(s)$ is continuously differentiable in x, with*

$$\partial_x \hat{u}(s, x) \;=\; -\, s\, F'(xs)\, \hat{u}(s, x).$$

Proof. F is smooth on $(0, \infty)$ by Proposition 3.4(3), so $x \mapsto e^{-F(xs)}$ is continuously differentiable there with derivative $-sF'(xs)e^{-F(xs)}$; multiply by the constant $\hat{f}(s)$. □

Proposition A.3 (scale evolution in memory-kernel form)**.** *Let $(\Phi_{x,y})$ be as in Theorem 7.3, in the canonical gauge, let $f \in X$ be causal, and set $u(\cdot, x) := \Phi_{0,x} f = \mu_{0,x} * f$. Then, with the transform equation $\partial_x \hat{u} = -sF'(xs)\hat{u}$ of Lemma A.2 in hand:*

For each $x > 0$ the causal function $w(\cdot, x) := \kappa^{(x)} * u(\cdot, x)$ lies in L^1_{loc}, and for all $0 \le a \le b$,

$$u(\cdot, b) - u(\cdot, a) \;=\; -\, \partial_t \int_a^b \big(\kappa^{(x)} * u(\cdot, x)\big)\, dx \qquad \text{in } \mathcal{D}'(\mathbb{R}).$$

Differentiating in x, u satisfies, in the distributional sense on $\mathbb{R}_t \times (0, \infty)_x$ and with boundary value $u(\cdot, 0) = f$,

$$\partial_x u \;=\; -\, \partial_t\big(\kappa^{(x)} * u\big) \;=\; -\, b_0\, \partial_t u \;-\; \partial_t \int_0^t \tfrac{1}{x}\, k\big(\tfrac{t-r}{x}\big)\, u(r, x)\, dr.$$

Since $u(\cdot, x)$ is causal, the outer ∂_t may equivalently be moved onto u inside the convolution: the Riemann–Liouville and Caputo forms of the operator coincide on this family.

Proof. Local integrability. For $T > 0$,

$$\int_0^T |w(t, x)|\, dt \le \kappa^{(x)}([0, T])\, \|u(\cdot, x)\|_1 \le \Big(b_0 + \int_0^{T/x} k\Big) \|f\|_1 < \infty,$$

locally uniformly in $x > 0$; joint measurability follows from (A7). By Tonelli, for $s > 0$, $\hat{w}(s, x) = \widehat{\kappa^{(x)}}(s)\, \hat{u}(s, x) = F'(xs)\, \hat{u}(s, x)$.

Integration in x. Integrate Lemma A.2 over $x \in [a, b]$. The exchange of $\int_a^b dx$ with the Laplace integral is justified by Tonelli, since $\int_a^b s\, F'(xs)\, \|u(\cdot, x)\|_1\, dx \le \|f\|_1 \int_a^b sF'(xs)\, dx = \|f\|_1 \big(F(bs) - F(as)\big) < \infty$, finite also for $a = 0$. This gives

$$\hat{u}(s, b) - \hat{u}(s, a) \;=\; -\, s \int_a^b \hat{w}(s, x)\, dx \;=\; -\, s\, \widehat{W}(s), \qquad W := \int_a^b w(\cdot, x)\, dx,$$

with $W \in L^1_{\mathrm{loc}}$ and causal.

Inversion. The causal, locally integrable function $V(t) := \int_0^t \big(u(r, b) - u(r, a)\big)\, dr + W(t)$ has Laplace transform $\big(\hat{u}(s, b) - \hat{u}(s, a)\big)/s + \widehat{W}(s) = 0$ for every $s > 0$, so $V = 0$ a.e. by Corollary 3.11, the transform of $|V|$ being finite by the two bounds above. That is the displayed identity. The final statement holds because $u(\cdot, x)$ vanishes on $(-\infty, 0)$, so no boundary term arises when moving ∂_t through the convolution. □

Remark A.4 (reading: a general fractional relaxation equation). Proposition A.3 is the memory-kernel form of the per-scale generator; Theorem A.19 below upgrades it to a classical scale-Cauchy problem $\partial_x u = -\varphi_x(\partial_t)u$ with $\varphi_x(s) = sF'(xs)$, on an explicit operator core. It identifies the scale evolution as a *general fractional* relaxation equation in the sense of [17] and [25], with scale-dilated kernel; the case $k(t) = \alpha t^{-\alpha}/\Gamma(1-\alpha)$ recovers the fractional scale-Cauchy problem of [8], Thm. 3.

Lemma A.5 (potential kernel)**.** *For $x > 0$ set $\varphi_x(s) := sF'(xs)$. Then φ_x is a nonzero Bernstein function, positive on $(0,\infty)$, and there is a unique positive, locally finite measure $\ell^{(x)}$ on $[0,\infty)$ — the* potential kernel *— with*

$$\widehat{\ell^{(x)}}(s) \;=\; \frac{1}{\varphi_x(s)} \;=\; \frac{1}{s\,F'(xs)}, \qquad s > 0.$$

Proof. With $B(s) := sF'(s) \in \mathrm{BF}\setminus\{0\}$ by Lemma 7.1(2), we have $\varphi_x(s) = B(xs)/x \in \mathrm{BF}\setminus\{0\}$, and $\varphi_x > 0$ on $(0,\infty)$ by Lemma 3.7.

For existence, *construct* the measure rather than representing it. Since φ_x is a Bernstein function without killing term it is the exponent of a subordinator: let $(\mu_\tau^{(x)})_{\tau\ge 0}$ be its laws, $\widehat{\mu_\tau^{(x)}}(s) = e^{-\tau\varphi_x(s)}$, and set

$$\ell^{(x)} \;:=\; \int_0^\infty \mu_\tau^{(x)}\, d\tau,$$

the subordinator's *potential measure*. By Tonelli its transform is $\int_0^\infty e^{-\tau\varphi_x(s)}\,d\tau = 1/\varphi_x(s)$ for every $s > 0$, which is finite, so $\ell^{(x)}$ is locally finite; and uniqueness is uniqueness of the Laplace transform for locally finite measures whose transform converges somewhere (Proposition 3.10).

Two remarks on this route. It uses no property of the CM class: neither $\mathrm{CM}\circ\mathrm{BF}\subseteq\mathrm{CM}$ nor Bernstein–Widder in its general form, both of which the obvious argument spends — at the cost only of the existence of the subordinator, which Theorem 7.3 has already used. And the integral $\int_0^\infty \mu_\tau^{(x)}\,d\tau$ is a measure only because $\tau\mapsto\mu_\tau^{(x)}$ is measurable, which is not automatic and is where the subordinator's increasing paths are actually used: $\mu_{\tau+\tau'}^{(x)} = \mu_\tau^{(x)} * \mu_{\tau'}^{(x)}$ with causality makes $\tau\mapsto\mu_\tau^{(x)}\big((-\infty,r]\big)$ nonincreasing, hence measurable, and a Dynkin argument on the generating π-system $\{(-\infty,r]\}$ extends this to every Borel set. □

Lemma A.6 (the potential kernel scales)**.** *For $x > 0$, $\ell^{(x)}$ is x times the pushforward of $\ell^{(1)}$ under $t\mapsto xt$; in density form, $\ell^{(x)}(t) = \ell^{(1)}(t/x)$.*

Proof. Both measures are causal and locally finite, and their transforms agree on $(0,\infty)$: $\widehat{\ell^{(x)}}(s) = x/B(xs) = x\,\widehat{\ell^{(1)}}(xs)$. Apply the uniqueness clause of Lemma A.5. □

Theorem A.7 (Sonine conservation)**.** *For every $x > 0$,*

$$\kappa^{(x)} * \ell^{(x)} \;=\; \mathrm{Leb}_{[0,\infty)}, \qquad \textit{i.e.} \qquad \int_{[0,t]} \ell^{(x)}\big([0,\,t-r]\big)\,\kappa^{(x)}(dr) \;=\; t \quad \textit{for all } t \ge 0.$$

Proof. Both sides are locally finite measures on $[0,\infty)$ with finite Laplace transforms, and those transforms agree:

$$\widehat{\kappa^{(x)}}(s)\,\widehat{\ell^{(x)}}(s) = \frac{\varphi_x(s)}{s}\cdot\frac{1}{\varphi_x(s)} = \frac{1}{s} = \widehat{\mathrm{Leb}}(s),$$

using Lemma A.1 for the first factor and Lemma A.5 for the second. Proposition 3.10 concludes. □

Note the scaling structure, on the absolutely continuous parts: the density of $\kappa^{(x)}$ dilates with L^1 normalization, $\frac{1}{x}k(\frac{t}{x})$, and $\ell^{(x)}$ with sup normalization, $\ell^{(x)} = x\,(\ell^{(1)} \circ (x\,\cdot)^{-1})$ (Lemma A.6). In the convolution the two normalizations cancel — substituting $t \mapsto xt$ carries $\kappa^{(x)} * \ell^{(x)}$ onto $\kappa^{(1)} * \ell^{(1)}$ scaled back to Lebesgue measure, the $\frac{1}{x}$ against the x — so the identity holds at every scale because it holds at one: the pair $(\kappa^{(x)}, \ell^{(x)})$ is itself scale-covariant. The drift atom is the exception that proves the rule: $b_0\delta_0$ does not dilate at all, and its counterpart in $\ell^{(x)}$ carries the compensating scale factor instead (Proposition A.9).

Corollary A.8 (exact invertibility; conservation of information)**.** *Define, on causal $v \in X$, the generalized derivative and integral*

$$D^{(x)}v := \partial_t\big(\kappa^{(x)} * v\big), \qquad I^{(x)}v := \ell^{(x)} * v.$$

Then $D^{(x)}I^{(x)}v = v$ in $\mathcal{D}'$, hence a.e.

Proof. By Theorem A.7 and associativity of convolution — Tonelli on causal supports — $\kappa^{(x)} * \ell^{(x)} * v = \mathrm{Leb} * v = \int_0^{\cdot} v(r)\,dr$, and $\partial_t \int_0^t v = v$. □

Corollary A.8 is the conservation content of the axioms in Volterra form, and it is worth being exact about what it proves: the generalized derivative $D^{(x)}$ has the explicit right inverse $I^{(x)}$ — the memory operator at each scale is exactly inverted by the potential kernel, in the one composition order stated. What this says about the cascade itself is indirect: the steps $\Phi_{x,y}$ are not the operators $D^{(x)}$, but the identity is what makes the memory-kernel form of the evolution (Proposition A.3) solvable for the input, consistently with (A5), which fixes the mass. Probabilistically, $\ell^{(x)}$ is the potential (renewal) measure of the subordinator with exponent φ_x, $\kappa^{(x)}$ is its drift-plus-Lévy-tail, and Theorem A.7 is the renewal identity balancing expected occupation against tail decay. In transform language the theorem is the tautology $(\varphi_x/s)\cdot(1/\varphi_x) = 1/s$; its content is entirely in the nature of the factors: the algebraic reciprocal of the memory operator's symbol is again the transform of a *causal positive* measure, so the inverse exists as a realizable system and not merely as a formula — and causality and positivity are exactly what the renewal construction of Lemma A.5 supplies.

Proposition A.9 (regularity of the pair)**.** *Write $\kappa^{(x)} = b_0\,\delta_0 + \kappa_c^{(x)}$, with $\kappa_c^{(x)}(dt) := \frac{1}{x}\,k\big(\frac{t}{x}\big)\,dt$ the absolutely continuous part of Lemma A.1 — an unconditional decomposition, k being nonincreasing. The two clauses below record when $\ell^{(x)}$ admits an analogous decomposition $\ell^{(x)} = c\,\delta_0 + \ell_c^{(x)}$, for a constant $c \geq 0$ not in general equal to b_0, and describe the regularity of $\kappa_c^{(x)}$ and $\ell_c^{(x)}$:*

1. *$\ell^{(x)}$ admits such a decomposition with $\ell_c^{(x)}$ a nonincreasing density if and only if $1/F'$ is a Bernstein function — equivalently, φ_x is a* special *Bernstein function; by dilation invariance the condition does not depend on x.*

2. *$\kappa_c^{(x)}$ is completely monotone iff k is completely monotone, which holds iff F' belongs to the Stieltjes class $\mathcal{S} := \big\{\, h(s) = \frac{a}{s} + b + \int_{(0,\infty)} \frac{\sigma(d\tau)}{s+\tau} \,\big\}$. In that case $\ell^{(x)}$ also admits the decomposition above with $\ell_c^{(x)}$ completely monotone, i.e. the absolutely continuous parts $(\kappa_c^{(x)}, \ell_c^{(x)})$ form a classical Sonine pair of CM kernels; moreover k completely monotone implies F is a complete Bernstein function.*

Proof. (1) is the characterization of the potential measures of special subordinators ([33], Thm. 11.3).

(2) Dilation preserves complete monotonicity, so the first equivalence reduces to k. If k is completely monotone, $k(t) = \int e^{-\tau t}\,\sigma(d\tau)$ for a positive measure σ, and Lemma 7.10 gives $F'(s) = b_0 + \int \sigma(d\tau)/(s+\tau) \in \mathcal{S}$. Conversely, an $\mathcal{S}$-representation of F' must have $a = 0$ —

an a/s term corresponds to an additive constant in k, contradicting $\int_1^\infty k(t)t^{-1}dt < \infty$ — and then $k(t) = \int e^{-\tau t}\sigma(d\tau)$ is completely monotone.

For the second assertion, $1/h \in \mathrm{CBF}$ for $h \in \mathcal{S}\backslash\{0\}$ ([33], Thm. 7.3), so $g := 1/F'(x\,\cdot) \in \mathrm{CBF}$ and $\widehat{\ell^{(x)}}(s) = g(s)/s$. Writing the Lévy–Khintchine triple of g as (a_g, b_g, ν_g), with completely monotone Lévy density because $g \in \mathrm{CBF}$ ([33], Thm. 6.2), gives $g(s)/s = \frac{a_g}{s} + b_g + \widehat{\bar\nu}_g(s)$, that is $\ell^{(x)} = b_g\,\delta_0 + \ell_c^{(x)}$ with $\ell_c^{(x)}(t) := a_g + \bar\nu_g(t)$; and $a_g + \bar\nu_g(t) = a_g + \int_t^\infty(\text{completely monotone})$ is completely monotone.

Finally, k completely monotone implies the Lévy density $k(t)/t$ of F is completely monotone, being a product of completely monotone functions, hence $F \in \mathrm{CBF}$ ([33], Thm. 6.2). □

The verified proof of the two equivalences in (2) never shows $a = 0$: it reads the a/s term as the atom $a\,\delta_0$ of the measure representing k, so that $k(t) = \int e^{-\tau t}(a\,\delta_0 + \sigma)(d\tau)$ is completely monotone whatever a is.

Part (2) delineates exactly where the classical Sonine calculus with completely monotone kernel pairs — the standing hypothesis of most of the general-fractional-calculus literature — applies: on the complete-Bernstein subfamily, which contains the local families classified in §10. For general admissible F the pair, and Theorem A.7, live at the level of measures.

Proposition A.10 (Volterra equation for the kernels)**.** *Let F be as in (7.1), let $x > 0$, and set $\mu_x := \mu_{0,x}$. Then, as measures on $[0,\infty)$,*

$$t\,\mu_x(dt) \;=\; \big(\theta_x * \mu_x\big)(dt), \qquad \theta_x(dt) := b_0\,x\,\delta_0(dt) + k\big(\tfrac{t}{x}\big)\,dt. \tag{V.1}$$

Proof. Taking Laplace transforms, $\mathcal{L}[t\,\mu_x](s) = -\frac{d}{ds}e^{-F(xs)} = xF'(xs)\,e^{-F(xs)}$, while by Lemma 7.10 with the substitution $\tau = t/x$, $\widehat{\theta_x}(s) = b_0 x + \int_0^\infty e^{-st}k(t/x)\,dt = xF'(xs)$. So both sides of (V.1) have the finite transform $xF'(xs)e^{-F(xs)}$ and coincide by Proposition 3.10. □

The measure θ_x is the drift atom together with the jump measure of μ_x weighted by jump size: its density $k(t/x)$ is t times the Lévy density $k(t/x)/t$. So (V.1) has a probabilistic reading: *size-biasing the total delay adjoins one independent extra jump, drawn from the length-weighted catalogue.* This is the classical half-line characterization of infinite divisibility ([36]) with the catalogue's role made explicit, and it is what makes the equation constructive: the kernel is assembled from its own delays.

Proposition A.11 (the Volterra equation determines the kernel)**.** *μ_x is the* unique *probability measure on $[0,\infty)$ satisfying (V.1).*

Proof. If ν is a probability measure on $[0,\infty)$ with $t\,\nu(dt) = \theta_x * \nu$, then $\hat\nu$ is differentiable on $(0,\infty)$ with $-\hat\nu'(s) = \mathcal{L}[t\nu](s) = xF'(xs)\,\hat\nu(s)$. Solving this linear ODE with $\hat\nu(0+) = 1$ gives $\hat\nu(s) = e^{-F(xs)}$, hence $\nu = \mu_x$ by Proposition 3.9. □

Proposition A.12 (the Volterra equation for the density)**.** *Let F be as in (7.1), let $x > 0$, and set $\mu_x := \mu_{0,x}$. If $k \not\equiv 0$ then μ_x is absolutely continuous, and its density ϕ_x solves the linear Volterra equation*

$$\big(t - b_0 x\big)\,\phi_x(t) \;=\; \int_0^t k\big(\tfrac{u}{x}\big)\,\phi_x(t-u)\,du \qquad \text{for a.e. } t > 0. \tag{V.2}$$

Away from its left endpoint $t = b_0 x$ — where the coefficient *$t - b_0 x$ vanishes; ϕ_x vanishes below it, the drift shifting every kernel by $b_0 x$ (Remark 7.5), but need not vanish at it — this is a Volterra equation of the second kind.*

Proof. When $k \not\equiv 0$, μ_x is a nondegenerate self-decomposable law by Remark 7.2, hence absolutely continuous ([32], Thm. 27.13). Restricting the identity (V.1) of Proposition A.10 to densities yields (V.2). □

Equation (V.2) (Proposition A.12) is the constructive counterpart of the inverse Laplace transform in Theorem 7.3: the kernels are *derived* from the memory kernel k by Volterra integration. At the level of the measure equation (V.1) the determination is proved (Proposition A.11); for the density equation we record here, as a heuristic and without a formal statement, why it determines ϕ_x rather than proving it: on $(b_0 x, \infty)$, where ϕ_x is supported, (V.2) is linear and homogeneous, and a *regular* equation of that form — continuous coefficient, bounded away from zero — has the trivial solution only. It is the vanishing of the coefficient $t - b_0 x$ exactly at the left endpoint of that interval that admits a nontrivial family instead, from which the unit-area axiom (A5) selects ϕ_x. (At that endpoint the *density* need not vanish: for the Gamma family it is positive at $\gamma = 1$ and divergent for $\gamma < 1$.) Positivity of that choice is the property Picard iteration of (V.2) preserves term by term, starting from a nonnegative seed, on the interior where the equation is regular — again a heuristic; the proved uniqueness lives at the measure equation (V.1).

Example A.13 (verification: the Gamma family)**.** For $F(s) = \gamma \log(1 + s)$ we have $b_0 = 0$, $k(u) = \gamma e^{-u}$, and $\phi_x(t) = t^{\gamma-1} e^{-t/x}/(\Gamma(\gamma) x^\gamma)$ by Proposition 8.9. Indeed, since $e^{-u/x} e^{-(t-u)/x} = e^{-t/x}$,

$$\int_0^t \gamma e^{-u/x} \, \frac{(t-u)^{\gamma-1} e^{-(t-u)/x}}{\Gamma(\gamma)\, x^\gamma} \, du \;=\; \frac{\gamma\, e^{-t/x}}{\Gamma(\gamma)\, x^\gamma} \cdot \frac{t^\gamma}{\gamma} \;=\; t\, \phi_x(t),$$

which is (V.2) for this family, as required.

The Sonine pair is equally explicit here. With $\varphi_x(s) = sF'(xs) = \gamma s/(1 + xs)$,

$$\kappa^{(x)}(dt) = \frac{\gamma}{x}\, e^{-t/x}\, dt, \qquad \ell^{(x)} = \frac{1}{\gamma}\, \mathrm{Leb} + \frac{x}{\gamma}\, \delta_0,$$

with transforms $\gamma/(1 + xs)$ and $(1 + xs)/(\gamma s)$, whose product is $1/s$: the conservation identity $\kappa^{(x)} * \ell^{(x)} = \mathrm{Leb}$ by inspection. In filter terms $D^{(x)}$ is the leaky differentiator $\gamma s/(1 + xs)$ and $I^{(x)}$ a proportional–integral element, and the conservation identity is their exact pole–zero cancellation — the general theorem asserting that such a causal, positive cancellation partner exists for every admissible family. The atom $\frac{x}{\gamma}\delta_0$ in the potential kernel is the b_g-atom of Proposition A.9, present although $b_0 = 0$ — the pair's two atoms are genuinely independent.

Remark A.14 (computation; relation to the 2005 series). For $F(s) = s^\alpha$, i.e. $k(u) = \alpha u^{-\alpha}/\Gamma(1 - \alpha)$, Picard iteration of (V.2) regenerates term by term the series expansion following Theorem 2 of [8]; the Volterra route thus contains the original series as its homogeneous special case. Numerically, (V.2) is a weakly singular Volterra equation of the second kind, amenable to standard product-integration quadrature, and it provides an inversion-free forward scheme for the kernels for arbitrary admissible k.

Remark A.15 (literature). Kernel pairs satisfying $\kappa * \ell \equiv 1$ go back to [34]; the operator calculus built on them is the general fractional calculus of [17] and [25], usually developed under completely monotone or Sonine-class kernel hypotheses, which by Proposition A.9(2) delineate the complete-Bernstein subfamily. In Volterra-equation theory the kernels $\kappa^{(x)}$ are the *completely positive* kernels of [4]; the resolvent theory of [30] then yields well-posedness and positivity of the evolution in Proposition A.3 by purely deterministic arguments. The integral equation (V.1) is the classical characterization of infinitely divisible laws on the half-line of [36].

A.2 The scale-Cauchy problem: Theorem 3′

This subsection upgrades Proposition A.3 from a distributional identity to a classical Cauchy problem in the scale variable, on the explicit operator core $\mathcal{D}$ of §3. The upgrade is worth its machinery. A distributional identity says the field *satisfies* the equation; a Cauchy problem with a uniqueness clause says the equation *determines* the field — the memory-line evolution becomes

a law of motion, the record at one scale fixing the record at every coarser one, with uniqueness in a natural class. It is the hemigroup counterpart of Theorem 3 of [8], which Remark A.23 recovers verbatim — hence the name Theorem 3′. And its pointwise form is what the delay-path corollary closing the appendix consumes. The plan: first the domain (Lemma A.16), then the operator (Definition A.17, the memory operator re-presented as a sum over delay events), then the law (Theorem 3′), with two codas — the generation-theory reading, recorded but never used, and the pointwise sign corollary.

Lemma A.16 (delay derivative)**.** *$\mathcal{D}$ is dense in X_0 and invariant under every τ_r and every $\Phi_{x,y}$, and for $f \in \mathcal{D}$,*

$$\lim_{h\downarrow 0} \tfrac{1}{h}\big(\tau_h f - f\big) = -\,f' \quad \text{in } X_0, \qquad \big\|\tau_r f - f\big\|_1 \le \min\big(2\|f\|_1,\ r\,\|f'\|_1\big).$$

Proof. Density is standard.

Invariance under τ_r. $\tau_r f$ is absolutely continuous, vanishes on $[0,r]$, and $(\tau_r f)' = \tau_r f'$.

Invariance under $\Phi_{x,y} = \mu_{x,y} * \cdot$. From $f = \mathbf{1}_{[0,\infty)} * f'$ we get $\mu * f = \mathbf{1}_{[0,\infty)} * (\mu * f')$, so $\mu * f$ is absolutely continuous with derivative $\mu * f' \in X_0$ and value 0 at 0.

The limit. On $[h,\infty)$, $\frac{1}{h}(f(t-h) - f(t)) + f'(t) = -\frac{1}{h}\int_{t-h}^{t}\big(f'(\rho) - f'(t)\big)\,d\rho$, whose L^1-norm tends to 0 by continuity of translation in L^1; on $[0,h)$ the contribution is $\frac{1}{h}\int_0^h |f| + \int_0^h |f'| \le \sup_{[0,h]}|f| + o(1) \to |f(0)| = 0$.

The estimate. $\tau_r f - f = -\int_0^r \tau_\rho f'\,d\rho$ as a Bochner integral — the integrated form of the limit — together with $\|T_\rho f'\|_1 = \|f'\|_1$ and the trivial bound $\|T_r f - f\|_1 \le 2\|f\|_1$. □

The memory-kernel operator $\varphi_x(\partial_t)$ has so far acted through transforms and convolutions; on the core $\mathcal{D}$ it also has a pointwise-defined presentation, and that is what the Cauchy problem needs. The two presentations read the function k in two ways. As a *memory weight* it smooths the record: the convolution form $\partial_t(\kappa^{(x)} * \cdot)$ of the previous subsection. Differentiated, it becomes an *event catalogue*: $\nu_1 = -dk$ assigns a rate to each delay depth r, the kernel $k(r) = \nu_1\big((r,\infty)\big)$ being the catalogue's tail, and the form below says that the infinitesimal scale step compares the record with its own delayed copies — one bracket $f - \tau_{xr} f$ per catalogue entry, at the catalogue's rates. Field reading and event reading of one operator; clause (5) of the lemma that follows is exactly the statement that they agree on the core. The following is the Phillips form of the generator.

Definition A.17 (per-scale generator; Phillips form)**.** Let ν_1 be the Lebesgue–Stieltjes measure $-dk$ on $(0,\infty)$, with k taken right-continuous and $k(\infty) = 0$, so that $\nu_1\big((r,\infty)\big) = k(r)$; and for $x > 0$ put $\nu_x := x^{-1}$ times the pushforward of ν_1 under $r \mapsto xr$, i.e. $\nu_x\big((r,\infty)\big) = k(r/x)/x$. For $f \in \mathcal{D}$ define

$$\varphi_x(\partial_t)\,f \;:=\; b_0\,f' \;+\; \int_0^\infty \big(f - \tau_r f\big)\,\nu_x(dr) \;=\; b_0\,f' \;+\; \frac{1}{x}\int_0^\infty \big(f - \tau_{xr} f\big)\,\nu_1(dr).$$

Lemma A.18 (properties of the generator)**.** *For $f \in \mathcal{D}$ and $x > 0$:*

1. *The integral converges absolutely in X_0, with*

$$\|\varphi_x(\partial_t) f\|_1 \;\le\; b_0\|f'\|_1 + \int_0^\infty \min\big(2\|f\|_1,\, xr\,\|f'\|_1\big)\,x^{-1}\nu_1(dr) \;<\; \infty.$$

2. *$\mathcal{L}\big[\varphi_x(\partial_t) f\big](s) = \varphi_x(s)\,\hat{f}(s)$ for $s > 0$, where $\varphi_x(s) = sF'(xs)$.*

3. *$\varphi_x(\partial_t)$ commutes with every $\Phi_{y,z}$ on $\mathcal{D}$.*

4. *$x \mapsto \varphi_x(\partial_t) f$ is continuous from $(0,\infty)$ to X_0.*

5. *$\kappa^{(x)} * f$ is a.e. equal to an absolutely continuous function with $\big(\kappa^{(x)} * f\big)' = \varphi_x(\partial_t) f$ a.e.; that is, the Phillips form coincides on $\mathcal{D}$ with the memory-kernel operator of Lemma A.1.*

Proof. (1) Combine Lemma A.16 with $\int (1 \wedge r)\, \nu_x(dr) < \infty$, which follows from $\int_0^1 k < \infty$ and $k(1) < \infty$ by integration by parts.

(2) Termwise: $\mathcal{L}[f'] = s\hat{f}$, using $f(0) = 0$, and $\mathcal{L}[f - T_r f] = (1 - e^{-sr})\hat{f}$. Fubini is justified by (1), and $b_0 s + \int (1 - e^{-sr})\nu_x(dr) = \varphi_x(s)$ by Lemma 7.10 — the two Bernstein representations of φ_x agree.

(3) Convolutions commute; pull $\Phi_{y,z}$ through the Bochner integral.

(4) In the dilated form of Definition A.17 the integrand $\frac{1}{x}(f - \tau_{xr} f)$ is, for x in a compact subset of $(0, \infty)$, continuous in x for each r by strong continuity of T, and dominated by $C \min(\|f\|_1,\, r\|f'\|_1)$, which is ν_1-integrable. Apply dominated convergence; the drift term is trivial.

(5) Both $\varphi_x(\partial_t) f \in L^1$ and $\kappa^{(x)} * f \in L^1_{\mathrm{loc}}$ are causal with finite Laplace transforms. By (2) and Lemma 7.10 the causal function $V(t) := \int_0^t \big(\varphi_x(\partial_t) f\big)(\rho)\, d\rho - (\kappa^{(x)} * f)(t)$ has transform $\varphi_x(s)\hat{f}(s)/s - F'(xs)\hat{f}(s) = 0$ for all $s > 0$, so $V = 0$ a.e. by Corollary 3.11. That is the claim; the value $(\kappa^{(x)} * f)(0+) = 0$ follows from $\kappa^{(x)}([0,t]) \to b_0$ and $\sup_{[0,t]} |f| \to 0$. □

Theorem A.19 (Theorem 3′: the scale-Cauchy problem)**.** *Let $(\Phi_{x,y})$ be as in Theorem 7.3, in the canonical gauge, and let $f \in \mathcal{D}$. Set $u(\cdot, x) := \Phi_{0,x} f$. Then:*

1. *$u(\cdot, x) \in \mathcal{D}$ for every $x \geq 0$;*
2. *$x \mapsto u(\cdot, x)$ belongs to $C\big([0,\infty); X_0\big) \cap C^1\big((0,\infty); X_0\big)$ and satisfies*

$$\partial_x u(\cdot, x) \;=\; -\,\varphi_x(\partial_t)\, u(\cdot, x) \quad (x > 0), \qquad u(\cdot, 0) = f,$$

   with ∂_x a norm-derivative in X_0;
3. *u is the* unique *solution in the class of $v \in C([0,\infty); X_0) \cap C^1((0,\infty); X_0)$ with $v(\cdot, x) \in \mathcal{D}$ for $x > 0$, satisfying the equation for $x > 0$ and $v(\cdot, 0) = f$.*

Proof. (1) is the Φ-invariance of $\mathcal{D}$ (Lemma A.16), and continuity at every $x \geq 0$ is (A7).

(2) Write $v(\cdot, x) := \varphi_x(\partial_t)\, u(\cdot, x)$. By Lemma A.18(3), $v(\cdot, x) = \Phi_{0,x}\, \varphi_x(\partial_t) f$, so by Lemma A.18(4), contractivity of $\Phi_{0,x}$, and (A7),

$$\|v(\cdot, x) - v(\cdot, x')\|_1 \;\leq\; \big\|\varphi_x(\partial_t) f - \varphi_{x'}(\partial_t) f\big\|_1 + \big\|(\Phi_{0,x} - \Phi_{0,x'})\, \varphi_{x'}(\partial_t) f\big\|_1 \;\longrightarrow\; 0,$$

so $x \mapsto v(\cdot, x)$ is continuous from $(0,\infty)$ to X_0.

By Lemma A.18(5) applied to $u(\cdot, x) \in \mathcal{D}$, each $\kappa^{(x)} * u(\cdot, x)$ is absolutely continuous with derivative $v(\cdot, x)$, vanishing at 0; hence for $0 < a \leq b$, Fubini gives

$$\int_a^b \big(\kappa^{(x)} * u(\cdot, x)\big)(t)\, dx \;=\; \int_0^t \int_a^b v(\rho, x)\, dx\, d\rho,$$

so the distributional ∂_t in Proposition A.3 evaluates to the Bochner integral $\int_a^b v(\cdot, x)\, dx$:

$$u(\cdot, b) - u(\cdot, a) \;=\; -\int_a^b v(\cdot, x)\, dx.$$

Since the integrand is continuous in x, the fundamental theorem of calculus for Bochner integrals yields $u \in C^1((0,\infty); X_0)$ with $\partial_x u = -v$, which is the equation.

(3) Let v be any solution in the stated class and fix $s > 0$. Then $\eta(x) := \hat{v}(s, x)$ is continuous on $[0,\infty)$ and C^1 on $(0,\infty)$, with

$$\eta'(x) = \mathcal{L}[\partial_x v](s) = -\mathcal{L}[\varphi_x(\partial_t) v](s) = -\varphi_x(s)\, \eta(x)$$

by Lemma A.18(2); differentiation and $\mathcal{L}$ exchange because $|\hat{g}(s)| \le \|g\|_1$. Solving this scalar linear ODE on $[\varepsilon, x]$ and letting $\varepsilon \downarrow 0$, using continuity at 0, gives $\eta(x) = \hat{f}(s)\, e^{-F(xs)}$. Thus $\hat{v}(s,x) = \hat{u}(s,x)$ for all $s > 0$, and $v = u$ in X_0 by Corollary 3.11 applied to their difference. □

Two things about the proof deserve saying. The uniqueness class is the natural one — norm C^1 solutions through the core — and what it excludes are solutions that leave the core or lose the boundary value. Within it, uniqueness is a *scalar* linear ODE per Laplace frequency, and no semigroup generation theory is invoked anywhere in the theorem — the preference of the methodological remark in §13 exercised once more, an argument about the object where a theory about a class of operators would also have served, at greater expense. Generation theory is nevertheless available, and Proposition A.20 below records what it gives: at each fixed scale the Phillips form generates the subordinate contraction semigroup, with $\mathcal{D}$ as a core — a reading of the same objects, not a dependency of the theorem.

Proposition A.20 (fixed-scale semigroups; core property)**.** *For fixed $x > 0$, the operator $-\varphi_x(\partial_t)$ with domain $\mathcal{D}$ is closable, and its closure generates the strongly continuous contraction semigroup on X_0 of convolutions by the probability measures with Laplace transforms $e^{-\tau\varphi_x(s)}$, $\tau \ge 0$ — the semigroup subordinate to the delay semigroup via φ_x. Moreover $\mathcal{D}$ is a core for that generator.*

Proof. φ_x is a Bernstein function without killing term by Lemma A.5, so Phillips' subordination theorem ([29], Thm. 4.3) applies to the delay semigroup, whose generator extends $f \mapsto -f'$ on $\mathcal{D}$ (Lemma A.16): the subordinate generator is the closure of the Phillips form on the generator's domain. That $\mathcal{D}$ is a core follows from the core criterion ([7], Prop. II.1.7), since $\mathcal{D}$ is dense and invariant under the subordinate semigroup, which consists of convolutions (Lemma A.16). Note that Theorem A.19 does *not* use this proposition — the Cauchy problem is solved by a scalar ODE, not by generation theory — so this result adds a reading rather than a dependency. □

The classical form now pays. Read pointwise, the scale-Cauchy equation becomes a difference form — crossing scale replaces the present value by catalogue-delayed past values — and at an extremum the sign of every term can be inspected. The corollary that closes the appendix is the delay-path companion of the counting form of §8.

Corollary A.21 (non-enhancement at past-dominating extrema)**.** *Let $f \in \mathcal{D}$ with f' bounded and continuous, and $u(t,x) := (\Phi_{0,x} f)(t)$. Then u, $\partial_t u$ and $\partial_x u$ are jointly continuous on $\mathbb{R} \times (0,\infty)$, the scale-Cauchy equation of Theorem A.19 holds pointwise,*

$$\partial_x u(t,x) \;=\; -\,b_0\,\partial_t u(t,x) \;-\; \frac{1}{x}\int_0^\infty \big[\,u(t,x) - u(t - xr,\,x)\,\big]\,\nu_1(dr),$$

the integral absolutely convergent, and at every past-dominating *point — a t with $u(t,x) \ge u(t',x)$ for all $t' \le t$ — at which additionally $b_0 = 0$ or $\partial_t u(t,x) = 0$,*

$$\partial_x u(t,x) \;\le\; 0,$$

with the dual inequality at past-dominated points. In particular, along any differentiable path $x \mapsto t_m(x)$ of temporal critical points, $\frac{d}{dx}\, u\big(t_m(x),x\big) = \partial_x u\big(t_m(x),x\big) \le 0$ wherever the tracked maximum dominates its past — whatever the path's speed.

Proof. Pointwise regularity. In the canonical gauge $u(t,x) = \mathbb{E}\, f(t - xT_1)$ and $\partial_t u(t,x) = \mathbb{E}\, f'(t - xT_1)$, the latter by dominated convergence (f' bounded); both are bounded by $\|f\|_\infty$, $\|f'\|_\infty$ and jointly continuous, again by dominated convergence, f and f' being bounded and continuous (f is absolutely continuous with $f' \in L^1$, and f' is continuous by hypothesis).

The pointwise identity. Set $g(t,x) := b_0\,\partial_t u(t,x) + \frac{1}{x}\int_0^\infty [u(t,x) - u(t-xr,x)]\,\nu_1(dr)$. The integral converges absolutely, the integrand being bounded by $\min\big(2\|f\|_\infty,\, xr\,\|f'\|_\infty\big)$ and $\int(1\wedge$

$r)\,\nu_1(dr)$ finite as in Lemma A.18(1); by the same domination, g is jointly continuous. Its Laplace transform is $\varphi_x(s)\,\hat{u}(s,x)$ — the computation of Lemma A.18(2), run pointwise — so $g(\cdot,x)$ is the continuous representative of $\varphi_x(\partial_t)\,u(\cdot,x)$. Theorem A.19(2) gives $u(\cdot,b)-u(\cdot,a) = -\int_a^b \varphi_x(\partial_t)u(\cdot,x)\,dx$ in X_0, hence $u(t,b)-u(t,a) = -\int_a^b g(t,x)\,dx$ for almost every t; both sides are continuous in t, so the identity holds for every t, and since g is continuous in x the fundamental theorem of calculus yields the pointwise derivative $\partial_x u(t,x) = -g(t,x)$, jointly continuous.

The sign. At a past-dominating point every bracket $u(t,x)-u(t-xr,x)$, $r>0$, is nonnegative, so the integral term of $-g$ is ≤ 0, and the drift term $-b_0\partial_t u$ vanishes under either proviso; the dual statement replaces $\ge$ by $\le$ throughout. The path clause is the chain rule — u is C^1 in each variable with jointly continuous partials, hence totally differentiable — with the ∂_t term killed by criticality, at whatever speed the path moves. □

The measure $\nu_1 = -dk$ is the delay-event catalogue of the dictionary of §3, and the identity says that crossing scale replaces the present value by catalogue-delayed past values; at a maximum that dominates its past, only smaller values can arrive, which is the sign. The proviso is sharp in both directions. A *secondary* maximum in the shadow of a larger earlier peak has brackets of both signs — the echo of the peak drains forward and can enhance it — which is the genuine physics of a delay line, not a defect of the formulation. The unconditional statement is the counting form of Proposition 8.13, and the two are complementary: counting non-creation from the signal, unconditional but restricted to the Pólya frequency members; pointwise non-enhancement along the feature's own delay path, conditional on past-domination but valid for every admissible family.

Example A.22 (the Gamma family: time-recursive form)**.** For $F(s) = \gamma\log(1+s)$ we have $\nu_x(d\rho) = \gamma x^{-2}e^{-\rho/x}\,d\rho$, an exponential memory, and $\varphi_x(s) = \gamma s/(1+xs)$, so

$$\varphi_x(\partial_t)\,u \;=\; \gamma\,\partial_t\big(1+x\,\partial_t\big)^{-1}u \;=\; \gamma\,w', \qquad \text{where } w \text{ solves } w + x\,w' = u,\quad w(0)=0.$$

The auxiliary variable w is computed by one causal first-order ODE, so the scale step is realized by a single recursive low-pass filter — the strongest possible form of the time-recursivity requirement of [8], §5: the observer embodies its past in one first-order state per scale.

Remark A.23 (recovering Theorem 3 of 2005). For $F(s) = s^\alpha$ with $0<\alpha<1$ we have $\nu_x\big((\rho,\infty)\big) = \alpha x^{\alpha-1}\rho^{-\alpha}/\Gamma(1-\alpha)$ and $b_0 = 0$, so $\varphi_x(\partial_t) = \alpha x^{\alpha-1}D^\alpha_{t,+}$ with $D^\alpha_{t,+}$ the Riemann–Liouville/Marchaud derivative, and Theorem A.19 reads $\partial_x u = -\alpha x^{\alpha-1}D^\alpha_{t,+}u$. Under the reparametrization $\tau := x^\alpha$ this is exactly

$$\partial_\tau u \;=\; -\,D^\alpha_{t,+}\,u,$$

Theorem 3 of [8] verbatim — identifying the scale parameter τ of that paper as the α-gauge of Remark 6.4 applied to the canonical scale coordinate.

Author contribution and use of AI

This article was written in close collaboration with generative AI, and that collaboration went well beyond editing assistance. The disclosure below is specific because a general one would misrepresent it.

What the author brought. The research programme, the questions it asks, and the point of departure are the author's. The semigroup theory of [8] is his own earlier work, and the decision to ask what survives of it when the semigroup axiom is weakened to a hemigroup is his. Every structural choice about what this article is, what it claims, and what it declines to claim was made by him.

How the work went. This material was to have formed part of a larger project, and for months before this article existed the author was reading, building tools, accumulating a research wiki, and writing that other article and its formalization — held up on a single point. The semigroup axiom plainly had to be weakened, and the right way to weaken it had not been found. The hemigroup formulation was that way, and it arrived on the evening of 5 August 2026. The substance of the working draft — the axioms, the characterization, and the shape of the argument that carries them — was written in that same extended conversation, overnight and into the following day. The article repository was opened two days later; the Lean 4 development was produced over the seven days after that. The remaining three weeks went on what the author takes to be the substance of authorship: understanding the mathematics well enough to stand behind it, replacing serviceable argument with motivated exposition, supplying what was missing, and scoping claims that had outrun their proofs.

What the AI did. The prose and the code were written by AI agents throughout, in dialogue with the author, who worked almost entirely from the conversation rather than from the editor. This includes the Lean 4 formalization essentially in full: the author's involvement there was slight, and none of the proof terms are his. It also includes mathematical content. A number of the results, the connections to the probabilistic literature, and several of the lines of argument were not in the author's plan and were learnt from the agents in the course of the work. They belong to the article, not merely to its presentation.

AI was also the first reviewer. Adversarial review was a working method throughout, not only a final gate: drafts were read blind, section by section, by reviewer agents that had no part in the writing, and the content revision was worked from their findings. Before the final revision the manuscript was then read by AI systems other than those that had written it: one review in journal style, returning a verdict of major revision, and one on presentation. Their findings drove the last pass — several headline claims were scoped back to what is actually proved, one proof was repaired, and one objection was answered with a new result. Both of these reviews are archived verbatim in the repository, under `notes/reviews/`, beside the plan that records the disposition of each finding. The author assessed each finding independently and accepted, narrowed, or declined it; no change was made because a reviewer asked for it alone. The author has read and revised the article throughout; beyond him, no human reader has yet reviewed it.

By the numbers. The sessions were archived as the work went on, so the division of labour can be stated rather than estimated. Between April and September 2026 the work ran to some 560 hours of recorded active session time. The author contributed about 106 000 words of direction and the agents about 1.87 million words of response, a ratio of roughly eighteen to one. The seven days that produced the Lean development wrote 24 700 lines of it, and took 205 turns and 9 710 words from the author — under four tenths of a word per line of proof.

These totals cover the work up to 27 September 2026, when version 1.1.0 was released; they are lower bounds, and they were computed on 2 October 2026. Both qualifications are load-bearing. A handful of sessions leave no transcript; some subagent traces were lost to context compaction before they could be archived; and reading on paper, thinking away from a keyboard, and work done under a separate professional obligation whose records cannot be exported leave no trace at all. The figures have also only ever moved in one direction as missing material was recovered — by roughly a fifth over the week the first version of this statement was written, and by about seven per cent more when they were computed again for this version over the same period — so they understate rather than overstate the agents' share.

Two conventions, because neither is self-evident. The line count is of lines of Lean *written* in those seven days, not of lines standing at the end of them: the development stood at 21 800 lines on 15 August 2026, having been edited as well as extended. And the author's word count

excludes the task briefs of unattended sessions, which this programme has dispatched from a work queue since 21 September 2026: those briefs are written by an agent, not typed by the author, and counting them as direction would have inflated it by some four fifths and made the ratio read as ten to one.

What this article would otherwise have been. The author would not, realistically, have written this article unaided — certainly not under the constraints of research pursued as a spare-time activity. Written alone it would have carried substantially less mathematical detail, and would have leaned on results imported from the probability literature at many of the points where it now proves them.

Responsibility. The author directed the work throughout, has satisfied himself of the mathematical content, and has checked every citation. Responsibility for this article, including for its errors, is his alone. What is machine-checked, and the axioms on which the verified development rests, are set out in §1.1; the validity of those results depends on the Lean kernel and not on how, or by whom, the proof terms were produced.

Version history

v1.1.0 (2026-09-27, DOI 10.5281/zenodo.22992450). The trust base is one cited fact. The existence of a subordinator with a given triple [33, Thm. 5.2], cited in v1.0.0 for the constructive direction of Theorem 7.3, is now proved, by a compound-Poisson construction; the constructive direction and the uniqueness rest on Lean's core axioms alone, and the analysis direction on self-decomposability [33, Prop. 5.17] alone. Newly machine-checked, with statements unchanged: Lemma 9.3, Lemma 9.4 (all four clauses), Lemma 9.20, and the two equivalences of Proposition A.9(2). Two statements corrected, neither affecting a result that uses them: the second display of Lemma 9.9 now assumes h continuous on $(0, \infty)$, which Mellin inversion needs and which its one use satisfies; and Lemma 10.4 now names its two readings of log-convexity, of which the locality theorem uses only the real one, and only where $z_* = \infty$. After the printed proofs of Lemmas 9.4 and 9.20 and of Proposition A.9, one sentence each says where the machine-checked proof takes another route. No result is renumbered, strengthened or withdrawn.

v1.0.0 (2026-09-02, DOI 10.5281/zenodo.22259187). First release.

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